

Newer Concepts in Diabetes, Cardio-Metabolism and Obesity

The Dr. Nehru Method

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This book is intended for educational purposes for medical professionals. It does not replace individual clinical judgement. The thresholds and staging frameworks described as the “Dr. Nehru Method” are proposed, India-specific operational tools and are identified as such throughout the text; they are not a substitute for established international consensus guidelines.

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Dedication

*To the patients of India, whose silent metabolic suffering inspired the search
for a better diagnostic paradigm; and to the clinicians willing to look
beneath the waterline.*

Foreword

[FOREWORD PLACEHOLDER — to be supplied by the invited author. The foreword should introduce the clinical context of the Indian diabetes epidemic and the rationale for a phenotype-specific approach.]

Preface

This book is written for the practising clinician who senses that the standard, glucose-centric model of diabetes care is failing the Indian patient — and who is looking for a coherent, evidence-informed framework to do better.

The Dr. Nehru Method is not a rejection of established medicine. It is an attempt to measure earlier, think upstream, and intervene before irreversible beta cell loss has occurred. Throughout these chapters, India-specific operational thresholds — a HOMA-IR above 2.3 and a TyG Index above 8.82 on the natural log scale — are used as practical decision points, validated in indexed publications from HOMA Healthcare Center, Hyderabad, in 2026.

Where a framework is proposed rather than established by international consensus, this is stated explicitly. The intent is honesty about the strength of evidence, so that the reader can apply clinical judgement with full awareness of what is proven and what is proposed.

Acknowledgements

[ACKNOWLEDGEMENTS PLACEHOLDER — to be supplied by the author, recognising co-investigators, the staff of HOMA Healthcare Center, and the patients who consented to anonymised case use.]

How to Use This Book

This book is organised into fourteen chapters that move from the failure of the standard paradigm (Chapters 1 and 2), through the mechanisms and measurable drivers of insulin resistance (Chapters 3 to 9), to the organ-level consequences (Chapters 10 and 11), the hormonal modifiers (Chapter 12), and finally to treatment — current and emerging (Chapters 13 and 14).

Each chapter is self-contained and ends with its own reference list, so it can be read in isolation or in sequence. Most chapters conclude with a short set of multiple-choice questions for self-assessment. A clean answer key is provided after each question set.

Recurring quantitative anchors — HOMA-IR above 2.3, TyG above 8.82 on the natural log scale, waist circumference above 90 cm in men and 80 cm in women — appear throughout and are intended as practical, India-specific operational thresholds rather than universal consensus cutoffs.

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Abbreviations

AWGS	Asian Working Group for Sarcopenia
BCAA	Branched-Chain Amino Acid
BIA	Bioelectrical Impedance Analysis
BMI	Body Mass Index
CVD	Cardiovascular Disease
DEXA	Dual-Energy X-ray Absorptiometry
FCPGR	Fasting C-Peptide to Glucose Ratio
FPG	Fasting Plasma Glucose
GLP-1	Glucagon-Like Peptide-1
HbA1c	Glycated Haemoglobin
HDL	High-Density Lipoprotein
HOMA-IR	Homeostatic Model Assessment of Insulin Resistance
IGF-1	Insulin-Like Growth Factor-1
IR	Insulin Resistance
LDL	Low-Density Lipoprotein
MASH	Metabolic Dysfunction-Associated Steatohepatitis
MASLD	Metabolic Dysfunction-Associated Steatotic Liver Disease
NAFLD	Non-Alcoholic Fatty Liver Disease
NASH	Non-Alcoholic Steatohepatitis
NEFA	Non-Esterified Fatty Acid

NWCO	Normal Weight Central Obesity
PCOS	Polycystic Ovary Syndrome
SAT	Subcutaneous Adipose Tissue
SHBG	Sex Hormone-Binding Globulin
T2DM	Type 2 Diabetes Mellitus
TOFI	Thin Outside, Fat Inside
TyG	Triglyceride-Glucose Index
VAT	Visceral Adipose Tissue
VLDL	Very Low-Density Lipoprotein
WC	Waist Circumference

CHAPTER 1

Why Standard Diabetes Care Is Failing India

The Illusion of Control

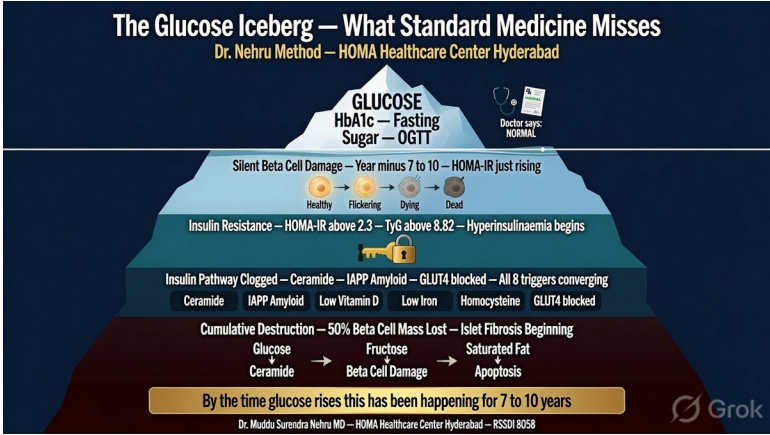


Figure 1.1 — The Glucose Iceberg: the visible tip of hyperglycaemia concealing a decade of subclinical metabolic destruction beneath the waterline. Source: Dr. Nehru Method, HOMA Healthcare Center, Hyderabad.

India carries the largest burden of type 2 diabetes of any nation on earth — over 101 million confirmed cases, with projections placing this figure at 134 million by 2045.[1,2] Yet the most disturbing statistic is not the prevalence. It is this: the majority of Indians who develop type 2 diabetes will spend a decade or more in metabolic deterioration before receiving a single correct diagnosis. They will be told their fasting glucose is borderline. Their HbA1c will be 5.8, 5.9, 6.0 — pre-diabetic, or more often, nothing to worry about yet. Their physician, trained on Western reference ranges and international guidelines designed for a different phenotype, will schedule a review in six months.

What the physician does not know — or does not measure — is that HOMA-IR in that patient is already 3.8. The insulin resistance cascade has been running for seven years.[3]

The HbA1c Trap

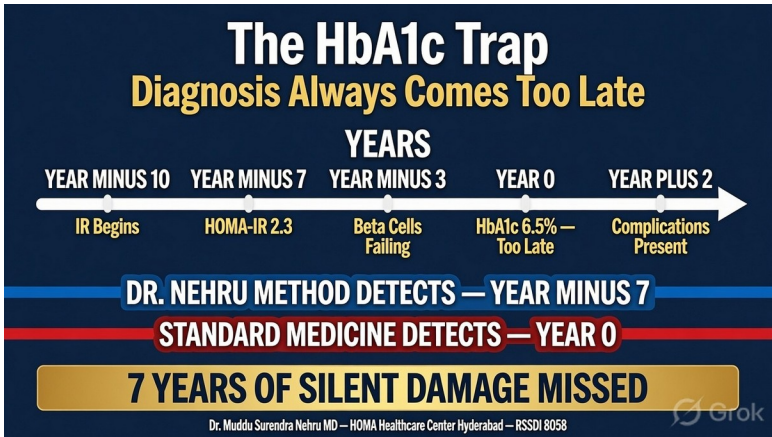


Figure 1.2 — Timeline of metabolic deterioration: HOMA-IR above 2.3 detects insulin resistance 7 years before HbA1c crosses the diagnostic threshold of 6.5%. Source: Matveyenko and Butler 2008; Nehru MS et al. 2026.

HbA1c became the diagnostic standard for type 2 diabetes because it is convenient, reproducible, and relatively inexpensive. A single number — 6.5 percent or above — defines the disease. Below that threshold, the patient is either normal or pre-diabetic, a category so loosely defined in routine clinical practice that it offers the physician no actionable urgency and the patient no real alarm.

The fundamental problem with HbA1c as a primary screening tool is temporal. By the time HbA1c crosses 6.5 percent, the pathophysiology it reflects is between seven and ten years old.[3] The islets of Langerhans have already begun to fail. Beta cell mass in Indian patients — already lower at baseline compared to Western populations[4] — has been eroding under sustained hyperinsulinaemia for most of the preceding decade. Between 40 and 65 percent of beta cell mass is already lost at the time of formal type 2 diabetes diagnosis — a finding that reframes diagnosis not as the beginning of disease but as confirmation of its advanced stage.[5] The microvascular bed has begun its slow compromise. The liver has been accumulating ectopic fat. The myocardium has been remodelling under the silent pressure of insulin-driven dyslipidaemia.

India has used a monitoring tool as a screening tool, and the gap between these two purposes has swallowed millions of patients.

The Glucose Iceberg

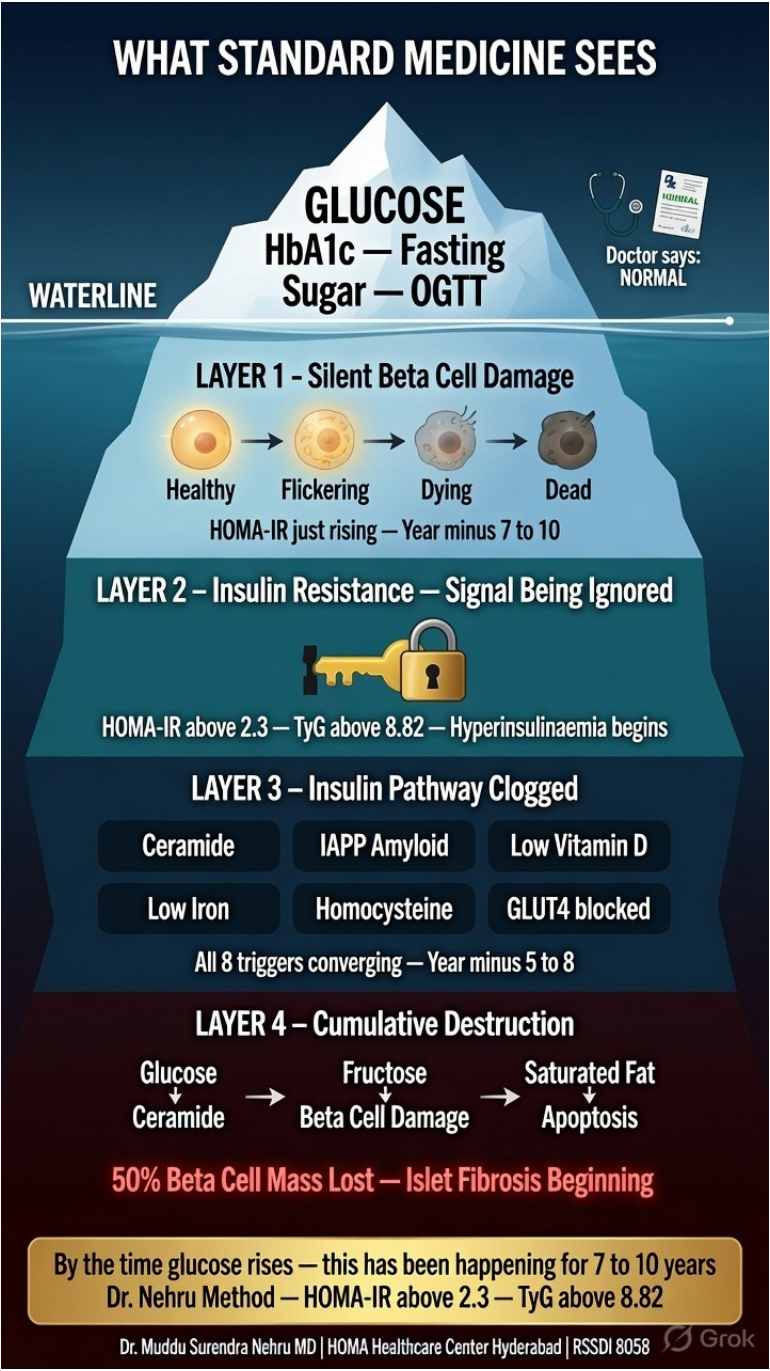


Figure 1.3 — Four layers of subclinical destruction beneath the waterline: silent beta cell damage, insulin resistance, blocked insulin pathway, and cumulative glucotoxicity, fructotoxicity and lipotoxicity. Source: Dr. Nehru Method.

Visualise the clinical presentation of type 2 diabetes as an iceberg. The visible portion — elevated fasting glucose, rising HbA1c, overt symptoms of hyperglycaemia — represents what standard care detects and treats. But the mass beneath the waterline is vastly larger: compensatory hyperinsulinaemia, progressive insulin resistance, visceral adiposity, atherogenic dyslipidaemia, systemic inflammation, endothelial dysfunction, mitochondrial impairment, and the slow collapse of beta cell reserve.

Every one of these processes begins and advances in the years — often the decade — before a single standard laboratory parameter crosses its diagnostic threshold.

The standard diabetes care pathway in India engages with the tip of this iceberg. A patient presents with fatigue and weight gain. Fasting glucose is 110 mg/dL. HbA1c is 6.1 percent. The physician advises dietary modification and returns the patient to follow-up in three months. What has not been measured is the HOMA-IR, which at that moment may be 4.5 — indicating Stage 3 insulin resistance on the Indian staging system.[6] What has not been evaluated is the TyG Index, which at 9.4 signals advanced cardiometabolic risk.[7,8] What has not been asked is whether this patient sleeps poorly, whether their waist circumference exceeds 90 centimetres at a body weight of 68 kilograms, whether they consume high glycaemic index carbohydrates at every meal, whether their serum Vitamin D is 9 ng/mL, or whether they have lost four kilograms of muscle mass over the preceding three years without any apparent change in body weight.

The Indian Phenotype Problem

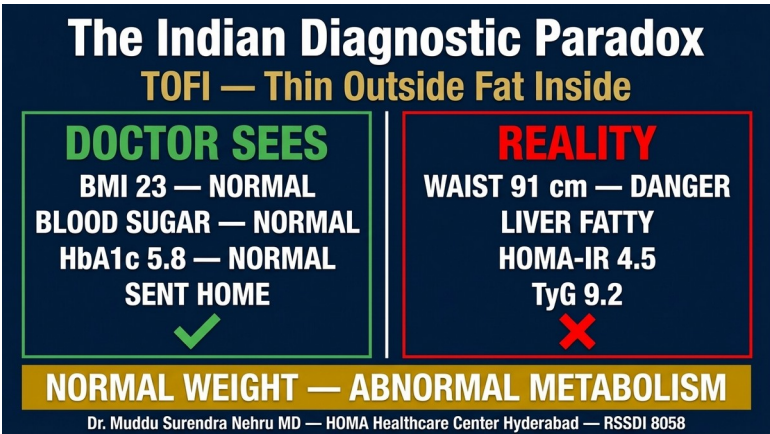


Figure 1.4 — The Indian Diagnostic Paradox: TOFI (Thin Outside Fat Inside) and NWCO (Normal Weight Central Obesity). Normal BMI masking visceral adiposity, fatty liver, and Stage 3 insulin resistance. Source: Narayan and Kanaya, *Diabetologia* 2020.

The failure is compounded by a biological reality that international guidelines have persistently failed to account for: the Indian metabolic phenotype is fundamentally different from the European population on which most diagnostic thresholds were derived.[4]

The Indian thin-fat phenotype — characterised by disproportionately high visceral and intramuscular fat at body weights that would be considered normal or even lean by Western standards — creates a diagnostic paradox. A man weighing 68 kilograms with a BMI of 23.4 will not trigger obesity-related metabolic screening in most Indian clinics, because the BMI cutoff of 25 imported from Western guidelines classifies him as normal weight.

He is not. His waist circumference of 91 centimetres places him above the Asian-specific abdominal obesity threshold. His visceral fat index is elevated. His liver is fatty. His skeletal muscle is insulin-resistant. He is what this framework designates as NWCO — Normal Weight Central Obesity — or in the more evocative terminology now entering the literature: TOFI — Thin Outside, Fat Inside.

This phenotype is not a clinical curiosity. It is the dominant metabolic pattern in urban India. Studies from Hyderabad, Chennai, Pune, and Delhi have repeatedly demonstrated that Indians develop insulin resistance and type 2

diabetes at lower BMI values, lower visceral fat volumes, and younger ages than their Western counterparts.[4] Early beta cell exhaustion — a phenomenon where the Indian pancreas reaches functional decompensation sooner and at lower degrees of metabolic stress — further accelerates the clinical timeline.[3,5]

The Missing Numbers: HOMA-IR and TyG Index

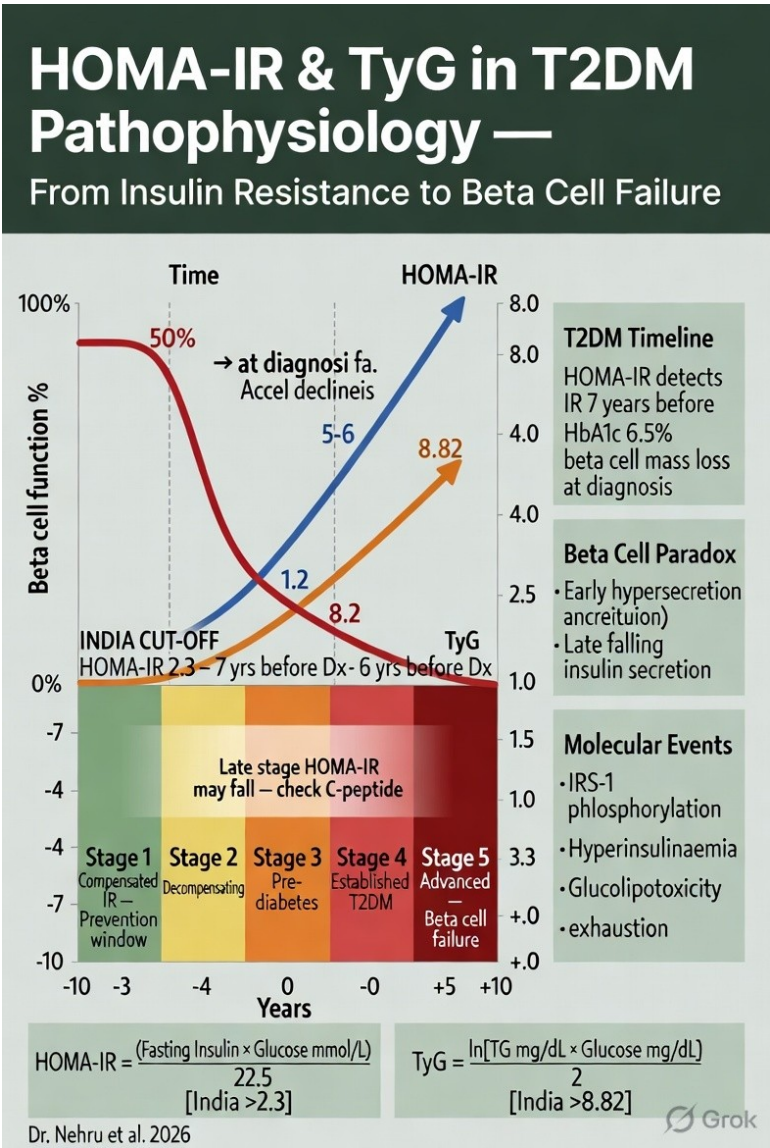


Figure 1.5 — HOMA-IR and TyG Index in T2DM pathophysiology: five-stage India-specific classification from Stage 1 (1.0-2.3) to Stage 5 (above 10.0), with TyG cutoff above 8.82 validated in Indian adults. Source: Nehru MS et al. Int J Med 2026.

There are two numbers that should appear on every metabolic workup in India. They do not.

HOMA-IR — the Homeostatic Model Assessment of Insulin Resistance — is calculated from fasting insulin and fasting glucose. It requires no special equipment, no complicated protocol, only a fasting blood draw and a validated formula. Yet it remains absent from the standard diabetes screening panel in virtually every primary care setting in this country.

The reason is circular: HOMA-IR is not on the standard panel because no guideline mandates it, and no Indian guideline mandated it because no validated Indian cutoff existed — until now. Three indexed publications from HOMA Healthcare Center, Hyderabad, published in 2026, have established that the appropriate HOMA-IR cutoff for detecting clinically significant insulin resistance in Indian adults is above 2.3.[6,7,8] The widely cited Western cutoff of 2.5 to 3.0 misclassifies a substantial proportion of Indian patients whose insulin resistance is already driving cardiovascular and metabolic pathology at lower values.

The staging system derived from this research defines five clinically actionable stages: Stage 1 from 1.0 to 2.3 representing early compensatory insulin resistance; Stage 2 from 2.3 to 4.0 indicating established resistance requiring active intervention; Stage 3 from 4.0 to 6.0 marking advanced resistance with high cardiovascular risk; Stage 4 from 6.0 to 10.0 representing critical metabolic failure; and Stage 5 above 10.0 indicating severe hyperinsulinaemic exhaustion. This is not a theoretical construct. Each stage maps to a clinical trajectory and a treatment imperative.

The Triglyceride-Glucose Index — TyG — provides a parallel and complementary measure. Calculated as the natural logarithm of the product of fasting triglycerides in mg/dL and fasting glucose in mg/dL, divided by two, the TyG Index is now recognised internationally as a reliable, inexpensive surrogate for insulin resistance with additional cardiovascular predictive value independent of conventional lipid fractions.[9] While this threshold of 8.82 was first identified in South Korean population data, three indexed Indian publications from 2026 have validated its applicability in the

Indian metabolic context, confirming equivalent predictive accuracy for insulin resistance and cardiovascular risk in Indian adults.[6,7,8] The validated Indian cutoff is above 8.82, within the natural log scale range of 7.5 to 10.5.

A Patient Who Should Have Been Caught Earlier

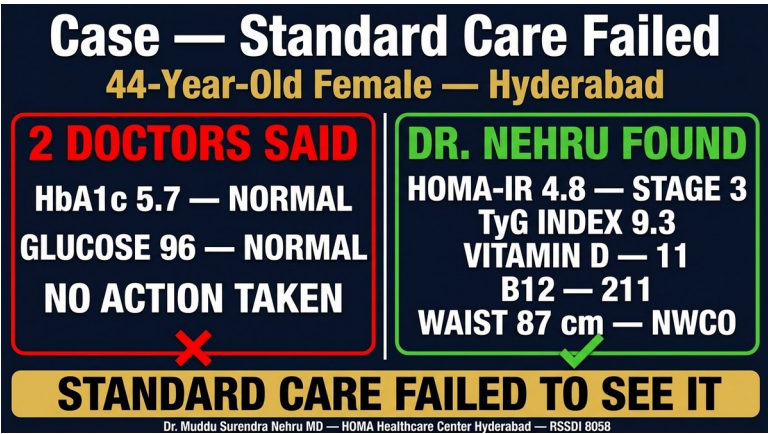


Figure 1.6a — Anonymised case: 44-year-old female, two prior consultations with normal investigations, HOMA-IR 4.8 Stage 3 and TyG 9.3 detected at first Dr. Nehru Method assessment. Standard care had failed to see the disease.

A 44-year-old woman, a software engineer in Hyderabad, presented with complaints of persistent fatigue, poor sleep quality, and cognitive slowing — what she described as brain fog. She had been investigated twice in the preceding two years at reputable clinics. Her fasting glucose was 96 mg/dL. Her HbA1c was 5.7 percent. Both times she was reassured and discharged without further action.

Her HOMA-IR, calculated for the first time at this consultation, was 4.8 — Stage 3 insulin resistance.[6] Her TyG Index was 9.3.[7] Her serum Vitamin D was 11 ng/mL. Her B12 was 211 pg/mL. Her waist circumference was 87 centimetres at a body weight of 61 kilograms — a textbook NWCO presentation.[4] She had quietly withdrawn from physical activity over two years. Her body weight had not changed, but her body composition had shifted silently toward fat and away from muscle.

Beyond Prevention — Secondary and Tertiary Care

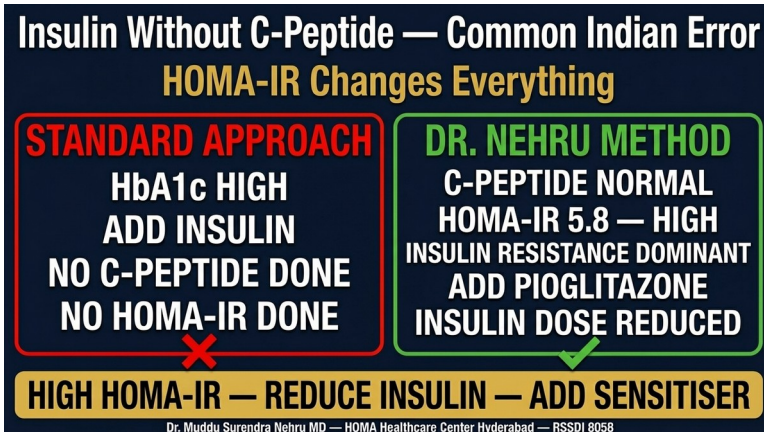


Figure 1.6b — Insulin prescribed without C-peptide or HOMA-IR assessment: a common and correctable error in Indian secondary care. When HOMA-IR is high and C-peptide is normal, insulin resistance dominates — insulin sensitiser reduces insulin requirement. Source: Dr. Nehru Method, HOMA Healthcare Center, Hyderabad.

The Dr. Nehru Method is not a framework for prevention alone. Its impact is equally significant — and arguably more immediately visible — in secondary and tertiary care, where patients are already on multiple medications and still deteriorating.

Consider the patient already on insulin. In Indian clinical practice, insulin is frequently initiated on the basis of a high HbA1c without prior C-peptide measurement and without HOMA-IR assessment.[6] When C-peptide returns normal and HOMA-IR is found to be 5.8 or above, the clinical picture changes entirely: this is not a patient with beta cell failure requiring exogenous insulin. This is a patient with dominant insulin resistance in whom the addition of a sensitiser — pioglitazone or saroglitazar — will reduce the insulin requirement substantially, often allowing dose reduction within eight to twelve weeks.

The second pattern is equally common and equally avoidable: the patient on four or five oral agents whose HbA1c remains above 9 percent. In the absence of HOMA-IR measurement, the clinical instinct is to add another drug. When HOMA-IR is measured and found to be 7.4, and TyG Index is 9.8, the picture clarifies: the dominant driver is unaddressed insulin resistance. The therapeutic response is not addition but substitution —

replacing agents that act downstream of the resistance with agents that address the resistance itself.

The Eight Triggers — A Framework Introduced

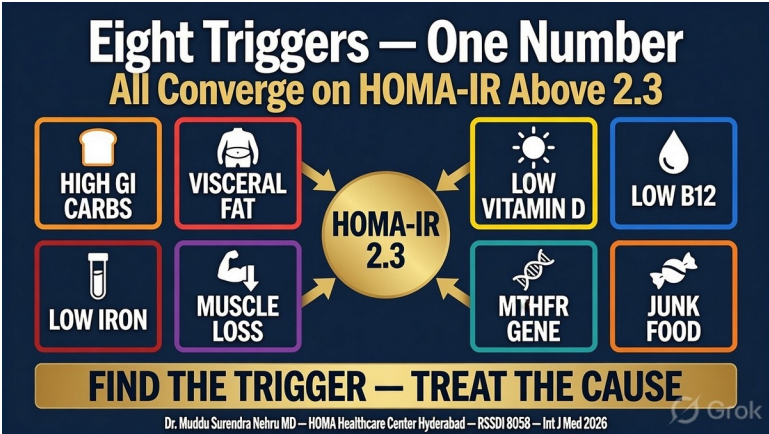


Figure 1.7 — The Indian Health Constitution: eight metabolic triggers all converging on HOMA-IR above 2.3. Each trigger is independently measurable and specifically treatable. Source: Nehru MS et al. Int J Med 2026.

Insulin resistance does not arise from a single cause. In the Indian clinical context, it arises from the convergence of multiple metabolic triggers, each capable of independently driving HOMA-IR above 2.3, and collectively capable of accelerating it to catastrophic levels.[6,7,8]

The Indian Health Constitution framework identifies eight primary triggers: high glycaemic index carbohydrate consumption, visceral obesity in its TOFI and NWCO forms, Vitamin D deficiency, B12 deficiency, iron deficiency, sarcopenia, the MTHFR C677T gene variant disrupting methylation and homocysteine metabolism, and ultra-processed junk food consumption with its attendant inflammatory and endocrine burden.

Why a New Framework Is Not Optional

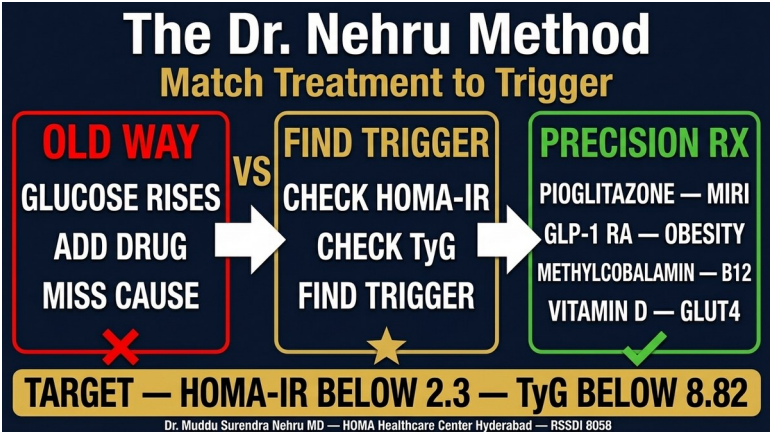


Figure 1.8 — The Dr. Nehru Method: precision treatment matched to dominant trigger. Target HOMA-IR below 2.3, TyG below 8.82. Replace drug addition with trigger identification. Source: HOMA Healthcare Center, Hyderabad, RSSDI 8058.

India cannot treat its way out of a diabetes epidemic using a diagnostic framework designed to detect that epidemic too late.[1,2]

The medical community in India has at its disposal the data, the tools, and now the validated cutoffs to identify insulin resistance a decade before HbA1c rises.[6,7,8] The eight triggers that drive that resistance are identifiable, measurable, and in most cases reversible. The infrastructure required to implement this — a fasting insulin assay, a triglyceride level, a careful clinical assessment of waist circumference and body composition — exists at every tertiary hospital and most secondary care centres in this country.

What has been missing is not technology. It is framework. A coherent, clinically validated, India-specific system for identifying metabolic risk early, staging it accurately, and intervening before the beta cell begins its irreversible decline.

The chapters that follow do not offer incremental modifications to existing guidelines. They present a reconceptualisation of what diabetes care in India must become — built not on the convenience of imported thresholds, but on the reality of the Indian metabolic phenotype, and on evidence generated from Indian patients, in Indian clinics, by Indian researchers.

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CHAPTER 2

The Glucose Iceberg

The Glucose Iceberg — What Lies Beneath Standard Screening

The Iceberg Metaphor — Why It Is Clinically Exact

Figure 2.1 — The Glucose Iceberg: four layers of subclinical destruction beneath the waterline, invisible to standard screening for 7 to 10 years before HbA1c rises. Source: Dr. Nehru Method, HOMA Healthcare Center Hyderabad.

The iceberg is not a metaphor chosen for its elegance. It is chosen for its precision. What a ship's navigator sees above the waterline — and what destroys the vessel — is a fraction of the structure that lies beneath. In type 2 diabetes, what the physician measures — fasting glucose, HbA1c, the occasional lipid panel — is the visible fraction. The mass that causes the damage has been building beneath the surface for a decade before a single standard parameter crosses its threshold.

[IMAGE PLACEHOLDER — Figure 2.1]

Figure 2.1 — The Glucose Iceberg — visible hyperglycaemia above the waterline and the four hidden layers of subclinical metabolic destruction beneath.

Beta cell dysfunction begins its progressive, irreversible course ten to twelve years before the clinical diagnosis of type 2 diabetes. By the time HbA1c crosses 6.5 percent, approximately 50 percent of beta cell function is already lost.[1] This is not a new finding. The UKPDS cohort established it in 1995. What is remarkable is that despite this knowledge being available for three decades, the diagnostic framework used in Indian primary care today still relies on the same HbA1c threshold — a number that, by definition, confirms the disease only after half the functional beta cell mass has been surrendered.

[IMAGE PLACEHOLDER — Figure 2.2]

Figure 2.2 — Layered progression of beta cell damage preceding diagnosis.

Layer One — Silent Beta Cell Damage

Figure 2.2 — Layer 1: Silent beta cell damage from Year minus 7 to 10. Progressive loss from healthy to flickering to stressed to dying — patient feels normal, doctor says normal, HOMA-IR is rising. Source: Matveyenko and Butler 2008.

The first and deepest layer of the iceberg is the one furthest from clinical awareness. In the years between Year minus ten and Year minus seven — a decade before the HbA1c crosses any diagnostic threshold — individual beta cells within the islets of Langerhans are beginning to fail. Some are dedifferentiating, reverting toward progenitor states. Some are entering stress pathways. Some are dying via ceramide-mediated apoptosis.[3]

Islet amyloid polypeptide — IAPP — begins to aggregate into amyloid deposits within the islets during this phase. Butler and colleagues demonstrated in an autopsy study of 124 subjects that islet amyloid deposition and a significant deficit in beta cell mass are already present in the prediabetic state — before any clinical diagnosis of type 2 diabetes has been made.[3] The islet architecture is being remodelled by a process that leaves no footprint on the standard blood panel.

The patient during this phase feels entirely normal. Their fasting glucose is 92 mg/dL. Their HbA1c is 5.4 percent. Their physician has no reason to investigate further. What is rising silently — and what HOMA-IR would detect if it were measured — is the earliest signal of compensatory hyperinsulinaemia: the beta cell working harder to maintain euglycaemia against a peripheral tissue resistance that has not yet declared itself on any standard test.[2,12]

Layer Two — Compensatory Hyperinsulinaemia and Insulin Resistance

Figure 2.3 — Layer 2: Insulin resistance — the signal arrives but the door is locked. HOMA-IR above 2.3, TyG above 8.82, hyperinsulinaemia beginning. Muscle, liver, and adipose tissue all resistant. Source: Dr. Nehru Method.

The Whitehall II study — a longitudinal cohort of 6,538 civil servants followed over years — provides the clearest human data on what happens beneath the waterline in the years before diabetes diagnosis. Tabák and colleagues demonstrated that fasting insulin and HOMA beta cell function rise compensatorily between four and three years before T2DM diagnosis,

followed by a steep decline, while HOMA insulin sensitivity drops steeply up to five years prior to diagnosis.[2] The pancreas is fighting a losing battle — overproducing insulin to compensate for a resistance it cannot overcome.

During this layer, three tissue beds are simultaneously developing resistance: skeletal muscle, the liver, and visceral adipose tissue. The insulin signal arrives at the cell membrane with full force. The problem is not the signal — it is the receptor response and the downstream cascade.

In Asian Indians, this layer has a distinctive and dangerous character. Raji and colleagues demonstrated that non-obese Asian Indian men — matched for the same normal body mass index as Caucasians — have significantly higher visceral adipose tissue and exhibit a 66 percent prevalence of insulin resistance.[6] The Indian patient in Layer Two may have a BMI of 23, a waist circumference of 89 centimetres, a fasting glucose of 98 mg/dL — and a HOMA-IR of 3.6, indicating Stage 2 insulin resistance on the India-specific staging system. Standard medicine sees nothing. The iceberg is growing.

Simultaneously, atherogenic dyslipidaemia is establishing itself. The Kailuan cohort study — 43,360 participants — demonstrated that the atherogenic index of plasma rises significantly and predicts the onset of T2DM over a four to eight year longitudinal period before overt diagnosis.[7] TyG Index above 8.82, on the natural log scale, signals this phase with precision that no single lipid fraction can match.[13]

Layer Three — The Blocked Molecular Pathway

Figure 2.4 — Layer 3: Six molecular blockers converging — ceramide blocks Akt, IAPP amyloid deposits, low Vitamin D impairs GLUT4, low iron activates HIF-1alpha, homocysteine damages IRS-1, GLUT4 vesicle translocation fails. Year minus 5 to 8. Source: Dr. Nehru Method.

Layer Three is where the molecular machinery of insulin signalling begins to fail at identifiable, specific, treatable points. This is the layer where the eight triggers of the Indian Health Constitution converge — not metaphorically but biochemically — on the same pathway.[12,13,14]

Ceramide accumulates in skeletal muscle and acts as a direct inhibitor of insulin signalling. Chavez and Summers demonstrated that ceramide blocks insulin-stimulated Akt phosphorylation — the critical signalling node that activates GLUT4 translocation to the cell surface.[4] Without Akt

phosphorylation, the downstream cascade fails. Glucose cannot enter the cell. The GLUT4 storage vesicles remain sequestered in intracellular compartments.

Garvey and colleagues confirmed in human skeletal muscle biopsies that insulin resistance is characterised by severely depleted GLUT4 storage vesicles at the plasma membrane and a failure of GLUT4 to translocate upon insulin stimulation.[5] This is not a consequence of long-standing diabetes. It is present in insulin-resistant individuals years before diagnosis — when fasting glucose is still normal.

Petersen and Shulman demonstrated that severe insulin resistance in skeletal muscle is linked to an 80 percent accumulation of intramyocellular lipid and a 30 percent reduction in mitochondrial oxidative phosphorylation.[9] The mitochondria — the energy factories of the muscle cell — are fragmented and dysfunctional. This is not merely a consequence of metabolic stress. It is a driver of it, creating a self-amplifying cycle of lipid accumulation, ceramide generation, and further insulin resistance.

Vitamin D deficiency — present in 70 to 80 percent of urban Indians — impairs beta cell insulin secretion and reduces GLUT4 expression independently. B12 deficiency from vegetarian diet raises homocysteine, which damages insulin receptor tyrosine kinase and accumulates methylmalonic acid. Iron deficiency activates HIF-1alpha, raising hepatic triglycerides and impairing GLUT4 translocation. Each of these is a separate, measurable, correctable trigger — and each is absent from the standard workup.

Layer Four — Cumulative Destruction

Figure 2.5 — Layer 4: Cumulative destruction — glucotoxicity, fructotoxicity, and lipotoxicity converging on beta cell failure. 50% beta cell mass lost, islet fibrosis beginning, IAPP amyloid deposited. By the time glucose rises, this has been happening for 7 to 10 years.

Source: Dr. Nehru Method.

Layer Four is where the three toxic pathways converge: glucotoxicity, fructotoxicity, and lipotoxicity. At this depth of the iceberg — Year minus three to Year minus one — the cumulative damage to beta cell mass has crossed the threshold of metabolic reserve.[1]

Repeated glucose spikes activate SREBP-1c — the master transcription factor for hepatic de novo lipogenesis. Shimano and colleagues demonstrated that hyperinsulinaemia and high glucose upregulate SREBP-1c, driving lipid accumulation and subsequent hepatic insulin resistance.[11] The liver, already accumulating ectopic fat, begins generating VLDL at an accelerated rate. TyG Index continues its rise. The atherogenic risk compounds.

Fructose — from ultra-processed food and high-fructose beverages increasingly prevalent in urban India — directly drives hepatic de novo lipogenesis through a pathway that bypasses the normal regulatory checkpoints of glucose metabolism. Unlike glucose, fructose does not stimulate insulin secretion and does not trigger satiety signals — making it the most metabolically dangerous sugar for the Indian urban diet.

Thejashwini and colleagues found that NAFLD has a prevalence of 22 percent in normoglycaemic, non-diabetic first-degree relatives of T2DM patients, with an odds ratio above 13 for association with insulin resistance. [10] This is Layer Four made visible: the liver is already fatty, the beta cell is already compromised, and the standard blood panel still reads normal.

The Intervention Window — What Standard Medicine Misses

Figure 2.6 — The intervention window: HOMA-IR 2.3 to 4.9 — 90% reversal possible if dominant trigger identified and treated. Standard medicine enters at Year 0. Dr. Nehru Method enters at Year minus 7. Source: Dr. Nehru Method, HOMA Healthcare Center Hyderabad.

Between Layer Two and Layer Three — when HOMA-IR is between 2.3 and 4.9, when TyG is between 8.82 and 9.5, when the beta cell is compensating but has not yet crossed into decompensation — lies the intervention window. [12,13] This is the clinical opportunity that standard medicine systematically misses, because standard medicine does not measure what defines it.

In this window, the dominant trigger can be identified. Vitamin D deficiency corrected. B12 replaced — always with methylcobalamin, never with cyanocobalamin, which fails to cross the blood-brain barrier and has no role in neurological protection. Iron deficiency treated. High glycaemic index carbohydrate replaced. Sarcopenia addressed with resistance training and protein. The MTHFR C677T variant managed with methylfolate.

The expected clinical response when the dominant trigger is correctly identified and treated is a fall in HOMA-IR of 0.5 to 1.5 units at three months.[12,13,14] This is not a theoretical projection. It is the observed clinical outcome in the patient cohort that generated the three indexed publications from HOMA Healthcare Center, Hyderabad.

The tragedy of standard diabetes care in India is not that this window does not exist. It is that standard medicine does not look for it — because the tools used to look, HOMA-IR and TyG Index, are not on the standard panel. The iceberg grows. The window closes. By the time the physician acts, the patient is no longer in the window. They are at the tip.

Four Molecular Steps — From Trigger to Beta Cell Failure

Figure 2.7 — Four molecular steps from trigger to beta cell failure: Step 1 trigger raises HOMA-IR, Step 2 hyperinsulinaemia activates SREBP-1c, Step 3 glucotoxicity plus lipotoxicity activates ER stress, Step 4 beta cell mass falls and C-peptide drops. Source: Dr. Nehru Method.

Understanding the iceberg requires understanding the molecular sequence that connects trigger to beta cell failure. The Dr. Nehru Method maps this as four steps — a clinical teaching framework that transforms an abstract pathophysiology into an actionable clinical roadmap.[12,13,14]

Step One: Any trigger — high glycaemic index carbohydrate, visceral obesity, Vitamin D deficiency, B12 deficiency, iron deficiency, sarcopenia, MTHFR C677T, junk food — raises fasting insulin and fasting glucose. HOMA-IR crosses 2.3. This is Stage 2 insulin resistance. The peripheral tissues — muscle, liver, fat — are no longer responding normally to insulin. The beta cell begins compensatory hypersecretion.

Step Two: Chronic hyperinsulinaemia activates SREBP-1c in the hepatocyte. [11] Hepatic de novo lipogenesis begins. TyG Index rises above 8.82. Ectopic fat accumulates in the liver and in skeletal muscle. Lipotoxicity begins at the islet level — palmitate and saturated fatty acids drive ceramide synthesis, which feeds back to block Akt phosphorylation further, worsening insulin resistance. HOMA-IR rises toward Stage 3.

Step Three: Glucotoxicity — repeated glucose spikes — activates the unfolded protein response in the beta cell endoplasmic reticulum. ER stress

drives pro-apoptotic signalling. Beta cell mass begins to fall. If all eight triggers are converging simultaneously — as they do in the typical urban Indian patient with poor diet, Vitamin D deficiency, sedentary lifestyle, and MTHFR variant — this step is accelerated dramatically.[4,9]

Step Four: Beta cell mass falls below the critical threshold for maintaining euglycaemia. Insulin secretory reserve is exhausted. Fasting glucose crosses 126 mg/dL. HbA1c crosses 6.5 percent. The physician diagnoses type 2 diabetes. The patient receives their first prescription. But 50 percent of the beta cell mass is already gone,[1] IAPP amyloid is deposited throughout the islets,[3] mitochondria are fragmented,[9] and the liver is fatty.[10] This is Year zero. This is where standard medicine begins. This is where the Dr. Nehru Method has been working for seven years already.

What Standard Medicine Sees — and What It Misses

Figure 2.8 — What standard medicine sees vs what is actually happening: standard medicine checks HbA1c and fasting glucose, says normal, sends the patient home — while HOMA-IR is rising, beta cells are dying, TyG is above 8.82, and 7 years are passing. Source: Dr. Nehru Method.

The clinical consequence of this iceberg structure is a systematic mismatch between the disease and its detection. Standard medicine uses tools optimised for Year zero — the moment of overt hyperglycaemia — and applies them to a disease that begins at Year minus ten.

Endothelial dysfunction — measured by impaired flow-mediated dilation — is present in normoglycaemic individuals with insulin resistance and prediabetes, years before HbA1c crosses 6.5 percent.[8] The cardiovascular risk is accumulating silently. The microvascular bed is compromising. The physician, seeing a normal HbA1c, reassures the patient and schedules a review in six months.

This is not negligence. It is the logical consequence of a framework that defines the disease by its most downstream measurable marker and builds the entire clinical response around that marker. The HbA1c was designed to monitor diabetes — not to prevent it. Applying it as a prevention tool is a category error with consequences measured in millions of patients and billions of rupees of avoidable healthcare expenditure.

The Indian Health Constitution framework — built on HOMA-IR above 2.3 as the single detection number — addresses this category error directly. It enters the iceberg where treatment changes outcomes.[12,13,14] It maps the four layers. It identifies the dominant trigger. It treats the cause, not the consequence. It measures the response — HOMA-IR fall of 0.5 to 1.5 units at three months — not the downstream marker.

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CHAPTER 3

The Eight Triggers of Insulin Resistance

Chapter 3 — The Eight Triggers of Insulin Resistance

The Indian Health Constitution Framework

This chapter presents the Indian Health Constitution Framework as a practical clinical lens for understanding why insulin resistance develops early, silently, and often in combination in Asian Indian populations. The chapter is written as an operational framework, not as universal international consensus, and it uses proposed thresholds from published Indian cohorts — HOMA-IR above 2.3 and TyG above 8.82 natural log — to identify metabolic dysfunction before overt diabetes appears.¹ Each of the eight triggers described below has independent supporting evidence in the literature, though the framework that links them together as a unified clinical staging system is a proposed Dr. Nehru Method construct rather than an internationally validated diagnostic protocol.



Figure 3.1 — The Eight Triggers of Insulin Resistance in the Indian phenotype — conceptual map.

3.1 — High GI Carbohydrates — The Chronic Hyperinsulinaemia Loop

For many Indian patients, insulin resistance begins with repeated exposure to high glycaemic load meals built around polished rice, refined wheat, bakery foods, sweets, and frequent snacking. These foods are rapidly digested and absorbed, creating sharp postprandial glucose excursions that force the pancreas to release more insulin again and again. Over time, the body adapts to this environment by sustaining higher fasting insulin levels — the hallmark of chronic hyperinsulinaemia.

This trigger is especially important because it is culturally embedded and often normalised as ordinary daily diet. The clinical issue is not occasional

carbohydrate intake but the cumulative metabolic stress of repeated glucose spikes. In the Indian phenotype, this pattern may appear long before frank hyperglycaemia, while HOMA-IR gradually rises above the proposed operational threshold of 2.3.

A practical clinical interpretation is that insulin resistance is often not a single disease event but a dietary rhythm problem. When refined carbohydrates dominate most meals, the pancreas is forced into a compensatory state that can last for years. This is why carbohydrate quality, portion structure, meal timing, and food processing level matter as much as total calories within this framework.

3.2 — Visceral Obesity — TOFI and Ectopic Fat

Visceral obesity is a major trigger of insulin resistance even when BMI appears only mildly elevated or entirely normal. The TOFI phenotype — thin outside, fat inside — describes individuals who carry disproportionate visceral and ectopic fat despite an apparently lean body shape. In this setting, fat is deposited in the liver, pancreas, and abdominal cavity, where it behaves metabolically rather than merely as stored energy.

A patient with normal weight may still carry high waist circumference, raised triglycerides, fatty liver, and elevated insulin — the scale tells a reassuring story the body is not actually living.

The central problem is portal fatty acid flux. Visceral adipose tissue releases free fatty acids directly into the portal circulation, increasing hepatic fat delivery and worsening hepatic insulin resistance. As the liver becomes more insulin resistant, it overproduces glucose and triglyceride-rich lipoproteins, which can drive the TyG index upward. This trigger matters because BMI can miss the real pathology entirely. In the Indian phenotype, this hidden adiposity often explains why metabolic disease occurs at lower body mass than in Western cohorts.

Clinically, TOFI should be considered a high-risk phenotype whenever abdominal obesity, fatty liver, raised triglycerides, or family history of diabetes are present. The framework treats ectopic fat not as a cosmetic issue but as a source of inflammatory and metabolic signalling that steadily worsens systemic insulin resistance.

3.3 — Vitamin D Deficiency — A Proposed Insulin Signalling Modifier

Vitamin D deficiency is common in India, and many studies report high biochemical prevalence when global cutoffs are applied. However, its role in insulin resistance should be framed carefully — the mechanism is proposed, not definitively proven. The proposed pathways include vitamin D receptor-mediated effects on pancreatic beta cell function, altered calcium signalling, and modulation of peripheral insulin sensitivity.²



Figure 3.2 — Micronutrient triggers: vitamin D and B12 deficiency in the Indian metabolic context.

The Indian literature suggests prevalence estimates can be very high, with many cohorts reporting deficiency rates in the broad range of 55 to 91 percent, depending on cutoff and population studied. At the same time, recent Indian review evidence argues that global thresholds may overstate disease burden and that outcome-linked, population-specific reference standards may be more appropriate for Indian populations.²

Within this framework, vitamin D deficiency is best understood as a metabolic amplifier rather than a standalone cause of insulin resistance. It may worsen insulin secretion, immune balance, and muscular glucose handling, particularly when combined with obesity, low physical activity, and poor diet. Because the mechanism is still debated, it is described here as proposed rather than definitive. The clinical message is simple — identify vitamin D deficiency as a potentially modifiable cofactor, but avoid claiming it is the sole driver of insulin resistance. Screening and correction remain reasonable and low-cost, while universal thresholds may not map perfectly onto Indian physiology.

3.4 — Vitamin B12 Deficiency — The Metformin Double Hit

Vitamin B12 deficiency is highly relevant in India because vegetarian dietary patterns and metformin exposure can act together as a double hit. In Indian cohorts, B12 deficiency has been reported as widespread, including estimates

around 47 percent in north Indian populations, with substantial prevalence also documented in diabetes and endocrine practice settings. Metformin further increases risk, making this deficiency especially relevant in patients with prediabetes and type 2 diabetes.³

The biochemical consequence is impaired one-carbon metabolism, disturbed methylation, and accumulation of metabolites linked to neuropathy and metabolic inefficiency. B12 deficiency should not be treated as a minor laboratory abnormality — it interacts with homocysteine metabolism, nerve function, and energy pathways, thereby amplifying the metabolic burden of insulin resistance.

For clinical supplementation within this framework, methylcobalamin is the preferred form, as the active bioavailable form supporting the methylation cycle directly without requiring hepatic conversion. A useful clinical framing is that B12 deficiency can worsen fatigue, neuropathic symptoms, and metabolic recovery even when glucose numbers appear acceptable. In a patient already struggling with insulin resistance, untreated B12 deficiency may conceal or aggravate the larger syndrome.

3.5 — Iron Deficiency — Ferritin and Cellular Energy Failure

Iron deficiency is an underappreciated trigger of insulin resistance, especially when ferritin falls below 30 ng/mL. Iron is essential for mitochondrial enzymes, oxygen handling, and ATP production, so deficiency can impair cellular energy generation long before overt anaemia becomes apparent. When tissues do not produce ATP efficiently, glucose transport and utilisation become less effective, contributing to insulin resistance.

The framework treats this as an energy problem as much as a haematological one. Muscle, liver, and pancreatic beta cells all depend on adequate mitochondrial function. If iron deficiency persists, the body may require greater insulin output to achieve the same metabolic result, deepening compensatory hyperinsulinaemia. A practical clinical mistake is to overlook ferritin because haemoglobin remains within range — this misses early iron depletion and delays intervention. Ferritin below 30 ng/mL should be treated as a meaningful metabolic red flag, especially when fatigue, hair loss, reduced exercise tolerance, or poor glycaemic control coexist.

Iron deficiency also intersects with diet quality, menstrual blood loss, gastrointestinal disease, and vegetarian food patterns. For this reason it should be evaluated as a systemic contributor rather than an isolated lab value. Correcting iron deficiency is part of restoring cellular energy and reducing the insulin burden on already stressed tissues.

3.6 — Sarcopenia — Loss of the Glucose Sink

Sarcopenia is a powerful trigger because skeletal muscle is the body's largest glucose sink. Healthy muscle stores and disposes of glucose efficiently under insulin's guidance, so loss of muscle mass reduces the body's capacity to handle carbohydrate loads. The result is higher circulating glucose, greater insulin demand, and progressive insulin resistance.

A patient can carry central adiposity and low muscle mass simultaneously — sarcopenic obesity — making BMI not just unhelpful but actively misleading about true metabolic risk.

The relationship is bidirectional. Insulin resistance promotes inflammation, inactivity, and protein catabolism, which accelerate muscle loss. In turn, sarcopenia worsens insulin resistance by reducing insulin-mediated glucose uptake and by limiting the metabolic reserve needed to buffer dietary carbohydrate exposure. This creates a self-reinforcing cycle especially important in ageing adults and in individuals with low protein intake or physical inactivity. Muscle preservation is therefore not just about strength or mobility — it is a core insulin-sensitising strategy. Resistance exercise, adequate protein intake, and correction of nutrient deficits all matter because they restore the muscle compartment as an effective glucose sink.

3.7 — MTHFR C677T — Methylation Impairment and Homocysteine

The MTHFR C677T variant is an important genetic modifier in insulin resistance, but it must be framed cautiously. In India, prevalence appears variable across subgroups, and estimates around 25 percent should be presented as approximate rather than universal. The mutation affects folate-dependent methylation pathways, which can impair homocysteine remethylation and disturb cellular methylation balance.⁴

The proposed mechanism is that impaired methylation may contribute to endothelial dysfunction, oxidative stress, and altered cellular repair. These effects can worsen metabolic resilience, especially when combined with poor diet, B12 deficiency, and obesity. The result is not a standalone disease but a biological susceptibility that may magnify insulin resistance under real-world conditions. Homocysteine becomes clinically relevant because it reflects disruption of one-carbon metabolism — when elevated, it may indicate a need to look more closely at folate status, vitamin B12 status, and related nutritional factors. In patients who are metabolically compliant but still insulin resistant, this genetic-metabolic pathway may help explain why progress is slower than expected. The most defensible framing is that MTHFR C677T may contribute to methylation impairment and homocysteine elevation in susceptible Indian subgroups, acting as a trigger or amplifier of insulin resistance rather than a sole determinant.

3.8 — Ultra-Processed Junk Food — Dysbiosis and Inflammation

Ultra-processed junk food is a major modern trigger because it combines refined starches, added sugars, industrial fats, excess salt, and low fibre density. This pattern alters the gut microbiome, promotes dysbiosis, and weakens the intestinal barrier. When gut permeability rises, lipopolysaccharide exposure may increase, feeding systemic inflammation and worsening insulin resistance.

The gut-liver axis is central here. Inflammatory signals from the intestine can reach the liver through the portal circulation and intensify hepatic insulin resistance. At the same time, high-fructose and high-calorie processed foods increase de novo lipogenesis, raising triglycerides and pushing the TyG index upward. This trigger is especially relevant in urban India, where convenience food, packaged snacks, sugary beverages, and restaurant meals are increasingly common. The problem is not sugar alone but the full matrix of processing, low fibre, and repeated hyperpalatable exposure. These foods encourage overeating and suppress metabolic satiety signals, making them a long-term driver of weight gain and insulin resistance. Ultra-processed food is not simply unhealthy — it is metabolically disruptive, worsening inflammation, disturbing the microbiome, and reinforcing the same insulin

resistance loop already being driven by diet, visceral fat, and nutrient deficiencies.

3.9 — Convergence — The Final Common Pathway

These eight triggers do not act independently — they converge on the same final metabolic pathway. High GI diets, visceral fat, micronutrient deficiencies, sarcopenia, genetic susceptibility, and ultra-processed food exposure all intensify hyperinsulinaemia until the pancreas can no longer compensate. The proposed operational thresholds used throughout this framework — HOMA-IR above 2.3 and TyG above 8.82 natural log, drawn from published Indian cohorts — are labelled the Dr. Nehru Method for practical clinical staging rather than universal consensus.

The convergence point is beta cell stress and eventual beta cell dysfunction. Once compensatory insulin secretion can no longer match the demand created by insulin resistance, glucose levels rise and the clinical phase of dysglycaemia becomes visible. This is why the framework emphasises early identification of these triggers, before the transition from compensation to decompensation becomes difficult to reverse.

The operational value of this approach is that it creates a structured clinical language for Indian patients. Instead of waiting for overt diabetes, clinicians can recognise the full cluster of risks and intervene at the level of diet, body composition, nutrition, genetics, and inflammation. The final common pathway is not simply hyperglycaemia — it is the progressive loss of beta cell reserve, occurring silently across multiple converging pathways at once.

Clinical Summary — Chapter 3

- Eight triggers converge on HOMA-IR above 2.3 and TyG above 8.82 — proposed Dr. Nehru Method operational thresholds, not universal consensus.
- High GI carbohydrates — repeated postprandial glucose spikes drive chronic compensatory hyperinsulinaemia.
- TOFI phenotype — visceral and ectopic fat at normal BMI — BMI alone is an unreliable screening tool.
- Vitamin D deficiency — proposed, not proven, insulin signalling modifier — 55 to 91% prevalence reported in Indian cohorts.

- Vitamin B12 deficiency — metformin double hit — methylcobalamin preferred form for replacement.
- Iron deficiency — ferritin below 30 ng/mL — mitochondrial ATP impairment — metabolic red flag.
- Sarcopenia — bidirectional with insulin resistance — sarcopenic obesity can mask true metabolic risk.
- MTHFR C677T — approximately 25% prevalence in India — variable across subgroups — amplifier not sole cause.
- Ultra-processed food — gut dysbiosis — LPS — systemic inflammation — TyG elevation via de novo lipogenesis.

Multiple Choice Questions

1. Which carbohydrate pattern most strongly promotes chronic hyperinsulinaemia in this framework?
A. Low-fibre refined food pattern B. High-protein pattern C. High GI rice and wheat dominance D. Intermittent fasting alone
2. What best describes the TOFI phenotype?
A. High BMI with low visceral fat B. Thin outside, fat inside C. Normal waist with no ectopic fat D. Lean body mass with high muscle
3. Ferritin below which level is treated as metabolically significant in this chapter?
A. 10 ng/mL B. 20 ng/mL C. 30 ng/mL D. 50 ng/mL
4. Which vitamin deficiency is linked here with the metformin double hit?
A. Vitamin A B. Vitamin C C. Vitamin B12 D. Vitamin K
5. The proposed operational TyG threshold in this framework is:
A. 7.0 B. 8.82 C. 9.5 D. 10.5

Answers: 1. C 2. B 3. C 4. C 5. B

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CHAPTER 4

Hyperinsulinaemia — The Invisible Driver

Chapter 4: Hyperinsulinaemia — The Invisible Driver

The Silent Pre-Clinical Phase

In the conventional paradigm of metabolic medicine, the diagnosis of type 2 diabetes mellitus is predicated upon the elevation of plasma glucose. However, glucose is merely the final common pathway—a late-stage manifestation of a metabolic derangement that may have been evolving silently for years to over a decade. The true initiating event, the invisible driver of this cardiometabolic cascade, is hyperinsulinaemia.

[IMAGE PLACEHOLDER — Figure 4.1]

Figure 4.1 — The decade of silent damage: hyperinsulinaemia preceding hyperglycaemia.

The Dr. Nehru Method posits that compensatory hyperinsulinaemia is not simply an innocent bystander or a benign physiological response to insulin resistance. Rather, it is a primary pathogenic force. Chronic hyperinsulinaemia actively drives visceral adiposity, promotes non-alcoholic fatty liver disease (NAFLD), exacerbates polycystic ovary syndrome (PCOS), accelerates atherogenesis, and stimulates cellular proliferation via the insulin-like growth factor-1 (IGF-1) axis [1] [2]. This chapter explores the prolonged pre-clinical phase preceding clinical hyperglycaemia, advocating for the routine measurement of fasting insulin and the early staging of insulin resistance using the Homeostatic Model Assessment for Insulin Resistance (HOMA-IR).

The Prolonged Pre-Clinical Window

The natural history of type 2 diabetes is characterized by a prolonged pre-clinical phase. Landmark prospective data, including trajectory analyses from the Whitehall II study, have demonstrated that insulin sensitivity begins to decline and compensatory insulin secretion begins to rise years before the onset of impaired fasting glucose [3]. Available evidence suggests a

prolonged pre-clinical phase of years to over a decade before clinical hyperglycaemia, although the exact timeline varies by individual. Within the clinical framework of the Dr. Nehru Method, we recognize this period as a window of "silent damage."

During this prolonged phase, the pancreatic beta cells hyper-secrete insulin to overcome peripheral tissue resistance, maintaining euglycaemia at the expense of hyperinsulinaemia. The patient appears healthy on standard screening panels; fasting plasma glucose (FPG) and glycated haemoglobin (HbA1c) remain within normal limits. However, the continuous exposure of the endothelium, liver, and adipose tissue to supra-physiologic insulin concentrations initiates a cascade of structural and functional alterations [4].

Fasting Insulin: The Neglected Biomarker

Despite its critical role in early pathogenesis, fasting insulin is rarely ordered in routine primary care. This omission is largely driven by assay standardisation challenges as well as cost and access issues: insulin assays are more expensive than standard glucose tests, lack universal standardization, and do not fit neatly into current guideline-driven reimbursement models that remain glucocentric.

However, fasting insulin provides an invaluable window into beta cell stress. A fasting insulin level above 12 mIU/L should be considered an early marker of metabolic concern. While the precise threshold for hyperinsulinaemia remains debated globally, levels exceeding 12 mIU/L—even in the presence of normal glucose—indicate that the beta cells are working overtime to maintain homeostasis.

HOMA-IR: A Clinical Tool for Early Detection

To quantify the degree of insulin resistance in a clinical setting, the Homeostatic Model Assessment for Insulin Resistance (HOMA-IR) is the most practical and validated tool. Originally developed as a mathematical model of the glucose-insulin feedback loop, it provides a robust surrogate for the gold-standard hyperinsulinaemic-euglycaemic clamp described by DeFronzo [5].

The HOMA-IR is calculated using the following formula:
$$\text{HOMA-IR} = (\text{Fasting Plasma Glucose in mg/dL} \times \text{Fasting Insulin in mIU/L}) / 405$$

[IMAGE PLACEHOLDER — Figure 4.2]

Figure 4.2 — HOMA-IR staging — the five-stage Indian classification from compensatory resistance to hyperinsulinaemic exhaustion.

In the Dr. Nehru Method, we propose a population-specific operational threshold for the Indian phenotype. Based on our validated published data from a tertiary metabolic clinic in urban India, a HOMA-IR value above 2.3 is considered indicative of significant insulin resistance [6]. While we acknowledge that other Indian studies may report different values, the cutoff of 2.3 has been validated in our cohorts as a highly sensitive marker for identifying central obesity-linked dyslipidemia.

The Pathogenic Consequences of Hyperinsulinaemia

The proposition that hyperinsulinaemia is a central driver of disease, rather than just a marker, is supported by a robust body of evidence linking high insulin levels to diverse systemic pathologies.

Visceral Adiposity, NAFLD, and PCOS

Insulin is a potent anabolic hormone. Chronic hyperinsulinaemia actively promotes lipogenesis and inhibits lipolysis, particularly in the visceral adipose depot. This creates a vicious cycle: expanding visceral fat releases free fatty acids and adipocytokines that further worsen peripheral insulin resistance, demanding even more insulin secretion.

In the liver, hyperinsulinaemia drives de novo lipogenesis, leading to hepatic steatosis or NAFLD. This ectopic fat deposition impairs hepatic insulin clearance, further elevating systemic insulin levels. In the ovaries, hyperinsulinaemia acts synergistically with luteinizing hormone to stimulate androgen production, a core mechanism in the pathogenesis of Polycystic Ovary Syndrome (PCOS).

Cardiovascular Disease: An Independent Risk Factor

The relationship between hyperinsulinaemia and cardiovascular disease (CVD) is well established. Early reviews by Stout highlighted the atherogenic potential of insulin, demonstrating its ability to stimulate smooth muscle cell proliferation and lipid synthesis within the arterial wall [7].

Crucially, hyperinsulinaemia is an independent risk factor for ischaemic heart disease, separate from its association with obesity or dyslipidaemia. The landmark Québec Cardiovascular Study demonstrated that high fasting insulin levels prospectively predicted ischaemic heart disease in men, establishing insulin itself as an atherogenic driver [8]. Furthermore, as demonstrated in our own clinical research, waist-integrated indices of insulin resistance (such as TyG-WC) strongly correlate with atherogenic dyslipidaemia (elevated triglycerides and low HDL-C) in the Indian population [9]. The Dr. Nehru Method emphasizes a locked cutoff of TyG-WC (natural log > 8.82) as an indicator of this risk.

The IGF-1 Axis and Cancer Risk

Beyond metabolic and cardiovascular consequences, hyperinsulinaemia is increasingly recognized as a driver of oncogenesis. Insulin and insulin-like growth factor-1 (IGF-1) share significant structural homology. Chronic hyperinsulinaemia decreases the hepatic production of IGF-binding proteins, leading to increased levels of free, bioactive IGF-1 [10].

This hyperactivation of the IGF-1 axis may stimulate cellular proliferation and inhibit apoptosis. Large prospective cohorts, such as the American Cancer Society study by Calle et al., have established an epidemiological association between obesity and increased mortality from multiple cancers, including breast, colon, and prostate [11]. It is important to emphasize that this represents an epidemiological association consistent with a biologically plausible mechanism, rather than proven direct causation.

Beta Cell Stress and the Proinsulin Ratio

As insulin resistance progresses and the demand for insulin escalates, the beta cells undergo significant stress. Initially, this manifests as compensatory hyperplasia and hypersecretion. However, over time, the beta cell machinery becomes overwhelmed.

A key marker of this stress is an alteration in proinsulin processing. Proinsulin is the precursor molecule cleaved into equimolar amounts of mature insulin and C-peptide. Under conditions of chronic hyperstimulation, the beta cells prematurely release disproportionate amounts of unprocessed or partially processed proinsulin into the circulation. An elevated ratio of fasting proinsulin to mature insulin is a sensitive indicator of beta cell dysfunction and impending failure [12].

Furthermore, measuring C-peptide provides a reliable assessment of endogenous insulin secretion and residual beta cell function, as it is co-secreted with insulin but not subject to the same rapid hepatic clearance.

Elaborating on the Mechanisms of Insulin-Driven Pathology

To fully appreciate why the Dr. Nehru Method places such profound emphasis on early hyperinsulinaemia, it is essential to delve deeper into the specific molecular and physiological mechanisms by which excess insulin inflicts silent damage over this prolonged pre-clinical window. The transition from a state of physiological compensation to one of pathological alteration is gradual, multifaceted, and insidious.

The Endothelial Dysfunction Cascade

The vascular endothelium is not merely an inert barrier; it is a highly active endocrine organ responsible for maintaining vascular tone, regulating inflammation, and preventing thrombosis. In a healthy state, insulin acts on the endothelium to stimulate the production of nitric oxide (NO) via the PI3K/Akt pathway, promoting vasodilation and enhancing blood flow to skeletal muscle, which in turn facilitates glucose disposal.

However, in the setting of insulin resistance and compensatory hyperinsulinaemia, a critical pathway divergence occurs. The PI3K/Akt pathway becomes blunted due to the accumulation of intracellular lipid metabolites (such as diacylglycerols and ceramides) and inflammatory cytokines. Conversely, the MAPK (mitogen-activated protein kinase) pathway, which mediates insulin's growth-promoting and mitogenic effects, remains sensitive or even becomes hyperactive.

This selective insulin resistance means that while the vasodilatory and protective effects of insulin are lost, its pro-atherogenic effects are amplified. The hyperactive MAPK pathway stimulates the proliferation and migration of vascular smooth muscle cells, increases the expression of vascular cell adhesion molecules (VCAM-1), and promotes the secretion of pro-coagulant factors like Plasminogen Activator Inhibitor-1 (PAI-1). The net result is a shift toward a pro-inflammatory, pro-thrombotic, and vasoconstrictive endothelial phenotype, and early vascular risk may begin during this pre-clinical phase, often years before clinical diabetes is diagnosed.

The Hepatic Burden: De Novo Lipogenesis and VLDL Secretion

The liver is central to the metabolic derangements driven by hyperinsulinaemia. In the fasting state, a healthy liver suppresses glucose production (gluconeogenesis) in response to basal insulin levels. In the insulin-resistant state, the liver becomes "deaf" to insulin's suppressive signal on gluconeogenesis, contributing to fasting hyperglycaemia in later stages.

However, paradoxically, the liver remains highly sensitive to insulin's lipogenic signals. The transcription factor Sterol Regulatory Element-Binding Protein 1c (SREBP-1c), which drives de novo lipogenesis (the conversion of excess carbohydrates into fatty acids), is potently upregulated by hyperinsulinaemia. This results in a massive influx of newly synthesized fatty acids into the hepatic parenchyma.

To manage this lipid burden, the liver packages these fatty acids into Very Low-Density Lipoproteins (VLDL) and secretes them into the circulation. This overproduction of VLDL is the primary driver of hypertriglyceridaemia, the hallmark of atherogenic dyslipidaemia. Furthermore, the excess VLDL particles exchange their triglycerides for cholesterol esters with High-Density Lipoproteins (HDL) and Low-Density Lipoproteins (LDL), mediated by Cholesteryl Ester Transfer Protein (CETP). This exchange results in the formation of small, dense LDL particles (which are highly atherogenic) and triglyceride-enriched HDL particles (which are rapidly cleared by the kidneys, leading to low HDL-C levels). This complex interplay perfectly explains the strong correlation we observed in our clinic between the TyG-WC index and the TG/HDL-C ratio.

Adipose Tissue Dysfunction: The Ectopic Fat Overflow

Adipose tissue is the primary storage site for excess energy. Under normal conditions, insulin suppresses lipolysis (the breakdown of stored fat) and promotes triglyceride synthesis. However, as visceral adipose tissue expands and becomes inflamed (often infiltrated by macrophages forming "crown-like structures"), it becomes severely insulin resistant.

The failure of insulin to suppress lipolysis in visceral fat leads to a continuous, uncontrolled release of Non-Esterified Fatty Acids (NEFAs) directly into the portal circulation, inundating the liver. This portal "fat flux" exacerbates hepatic insulin resistance and drives further VLDL production.

Moreover, when the subcutaneous adipose tissue reaches its maximal expansibility limit—a threshold that is notably lower in the South Asian "thin-fat" phenotype—the excess energy spills over into ectopic sites, including the liver, skeletal muscle, and the pancreas itself. The accumulation of fat within the pancreatic islets (pancreatic steatosis) directly impairs beta cell function, accelerating the progression from Stage 3 (Beta Cell Stress) to Stage 4 (Beta Cell Exhaustion) in the Dr. Nehru Method HOMA-IR staging system.

The Neurological Implications and the Role of Methylcobalamin

While the cardiovascular and hepatic consequences of hyperinsulinaemia are well-documented, the neurological impact of this prolonged pre-clinical phase is increasingly recognized. Chronic hyperinsulinaemia and the accompanying microvascular endothelial dysfunction contribute to early nerve damage, often presenting as subclinical peripheral neuropathy long before overt hyperglycaemia.

In managing these early neuropathic changes, particularly in patients who may later require metformin therapy (which is known to interfere with Vitamin B12 absorption), proactive nutritional support is essential. Within the Dr. Nehru Method, we emphasize that methylcobalamin is the preferred form of Vitamin B12 supplementation. As the active, bioavailable form, methylcobalamin bypasses the need for enzymatic conversion in the liver, making it more effective for neurological repair. While cyanocobalamin is

not universally contraindicated, it is not preferred due to its requirement for metabolic activation and the trace release of cyanide during its conversion.

The Dr. Nehru Method: HOMA-IR Staging

To operationalize the management of insulin resistance, the Dr. Nehru Method proposes a clinical staging system based on HOMA-IR. This framework allows for targeted interventions before irreversible beta cell dysfunction occurs.

Stage 0 (Optimal Sensitivity): HOMA-IR < 1.5. Normal beta cell function, no significant metabolic risk.

Stage 1 (Early Resistance): HOMA-IR 1.5 – 2.3. Onset of compensatory hyperinsulinaemia. Fasting insulin may be elevated (>12 mIU/L) with normal glucose.

Stage 2 (Established Resistance): HOMA-IR 2.3 – 3.5. Exceeds the validated Indian cutoff [6]. Often accompanied by central obesity, early NAFLD, or dyslipidaemia.

Stage 3 (Beta Cell Stress): HOMA-IR 3.5 – 5.0. Impaired glucose tolerance may emerge. Elevated proinsulin ratio.

Stage 4 (Beta Cell Exhaustion): HOMA-IR > 5.0. Progressive beta cell dysfunction which may become difficult to reverse at advanced stages. Clinical diabetes is often present.

Stage 5 (Beta Cell Failure): Declining HOMA-IR and C-peptide despite hyperglycaemia, indicating profound loss of secretory capacity requiring exogenous insulin.

Conclusion

Hyperinsulinaemia is the invisible engine driving the modern epidemic of cardiometabolic disease. By shifting our clinical focus from late-stage hyperglycaemia to early-stage hyperinsulinaemia, we can identify patients during this prolonged pre-clinical window. Through the routine use of fasting insulin, the application of HOMA-IR staging, and the recognition of insulin's direct pathogenic role in vascular and cellular biology, we can intervene early, during the phase when metabolic risk is still potentially modifiable.

Multiple Choice Questions

1. According to the Dr. Nehru Method, what is the proposed operational threshold for HOMA-IR indicative of insulin resistance in the Indian phenotype?
 - A. 1.5
 - B. 2.3
 - C. 3.5
 - D. 5.0
2. Which of the following is a primary reason fasting insulin is not routinely ordered in primary care?
 - A. It is less sensitive than HbA1c for diagnosing diabetes.
 - B. Commercial reasons and lack of universal standardization.
 - C. It cannot be measured accurately in obese patients.
 - D. It fluctuates too rapidly to provide useful clinical data.
3. The landmark Québec Cardiovascular Study (Després et al., 1996) demonstrated that hyperinsulinaemia is an independent risk factor for which condition?
 - A. Ischaemic heart disease
 - B. Colon cancer
 - C. Polycystic Ovary Syndrome
 - D. Non-alcoholic fatty liver disease
4. An elevated ratio of fasting proinsulin to mature insulin primarily indicates:
 - A. Enhanced hepatic insulin clearance
 - B. Optimal beta cell function
 - C. Beta cell stress and dysfunction
 - D. Severe hypoglycaemia
5. How does chronic hyperinsulinaemia influence cancer risk?
 - A. By directly mutating cellular DNA
 - B. By decreasing the production of IGF-binding proteins, leading to increased free IGF-1
 - C. By suppressing the immune system's ability to detect tumors
 - D. By increasing the rate of cellular apoptosis

Answers: 1. B 2. B 3. A 4. C 5. B

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CHAPTER 5

Visceral Obesity, TOFI, and the Indian Paradox

Why BMI Fails the Indian Patient

For decades, the global medical community has relied upon the Body Mass Index (BMI) as the primary anthropometric tool to screen for obesity and assess cardiometabolic risk. Derived from the simple ratio of weight to height squared, BMI offers an easily calculable metric for large-scale epidemiological studies. However, when applied to the Indian population, BMI is not merely imprecise—it is fundamentally misleading.

The Indian patient presents a unique metabolic challenge, often characterized by severe insulin resistance, atherogenic dyslipidaemia, and a high risk of cardiovascular disease despite having a BMI that falls within the “normal” range by Western standards. This phenomenon, often referred to as the Indian Paradox, necessitates a radical departure from BMI-centric screening. The Dr. Nehru Method advocates for a deeper understanding of body composition, specifically the critical distinction between subcutaneous and visceral adipose tissue, to accurately stratify and manage cardiometabolic risk.

The Inadequacy of BMI for the Indian Population

The standard World Health Organization (WHO) BMI classifications—defining overweight as a BMI of 25 kg/m² or higher, and obesity as 30 kg/m² or higher—were derived predominantly from Caucasian populations. These cutoffs assume a universal relationship between total body weight, adiposity, and disease risk.

However, South Asians possess a distinct body composition. At any given BMI, an Indian individual typically has a significantly higher percentage of total body fat, less lean muscle mass, and a greater concentration of fat deposited in the abdominal region compared to a Caucasian individual [1]. This was famously illustrated by the “Y-Y paradox,” comparing Dr. Chittaranjan Yajnik to his British colleague Dr. John Yudkin. Despite having

an identical BMI of 22.3 kg/m², dual-energy X-ray absorptiometry (DEXA) scans revealed that Dr. Yajnik had nearly double the body fat percentage (21.2% vs. 9.1%) [1].



Figure 5.1 — The Y-Y Paradox — identical BMI, divergent body fat percentage (Yajnik vs Yudkin).

Recognizing this disparity, a WHO expert consultation proposed lowered BMI action points for Asian populations, identifying increased risk beginning at a BMI of 23 kg/m² and high risk at 27.5 kg/m² [2]. Subsequent Indian consensus guidelines, led by experts such as Misra et al., further refined these recommendations, defining overweight in Asian Indians as a BMI of 23–24.9 kg/m² and obesity as a BMI $\geq$ 25 kg/m² [3]. Despite these lowered thresholds, BMI remains an incomplete metric because it cannot distinguish between muscle and fat, nor can it identify where the fat is stored.

The TOFI Phenotype: Thin Outside, Fat Inside

The failure of BMI to capture true metabolic risk is most evident in the TOFI phenotype: “Thin Outside, Fat Inside.” A TOFI individual appears lean or normal-weight outwardly, possessing a BMI within the acceptable range, yet harbors a dangerous accumulation of visceral adipose tissue (VAT) internally.

In the Dr. Nehru Method, we emphasize that TOFI is not merely a cosmetic curiosity; it is a state of profound clinical importance. The TOFI patient is often lulled into a false sense of security by their “normal” weight, and primary care physicians relying solely on the weighing scale may miss the opportunity for early intervention. These individuals frequently present with Normal Weight Central Obesity (NWCO), exhibiting the classic metabolic derangements of the metabolic syndrome—insulin resistance, hypertriglyceridaemia, and low HDL cholesterol—long before they develop overt hyperglycaemia.

The Personal Fat Threshold

To understand why a lean-appearing Indian patient develops type 2 diabetes while a heavily obese Caucasian patient may remain metabolically healthy,

we must look to the “Personal Fat Threshold” hypothesis proposed by Taylor and Holman, building on foundational metabolic concepts [4].

This hypothesis posits that every individual has a genetically and phenotypically determined capacity to store fat safely in subcutaneous adipose depots. The subcutaneous fat acts as a metabolic “sink,” safely sequestering excess energy. However, once this personal threshold is exceeded, the subcutaneous depot can no longer expand. The excess energy must go somewhere, and it spills over into ectopic sites—primarily the visceral cavity, the liver, the skeletal muscle, and the pancreas.

Crucially, the personal fat threshold for the Indian population is remarkably low. Due to evolutionary, genetic, and epigenetic factors, the South Asian subcutaneous adipose tissue loses its expansibility much earlier than in other ethnicities. Consequently, the overflow of fat into visceral and ectopic depots occurs at much lower total body weights, explaining the high prevalence of metabolic disease in non-obese Indians.

Subcutaneous vs. Visceral Fat: A Metabolic Difference

Not all fat is created equal. The anatomical location of adipose tissue dictates its physiological behavior and its impact on systemic metabolism.

Subcutaneous adipose tissue (SAT), located just beneath the skin, is relatively benign when functioning normally. It is highly sensitive to insulin, effectively suppresses lipolysis, and secretes beneficial adipokines such as adiponectin, which enhances peripheral insulin sensitivity and exerts anti-inflammatory effects.

Visceral adipose tissue (VAT), located deep within the abdominal cavity and surrounding the internal organs, is a highly active, pathogenic endocrine organ. VAT is metabolically rigid and severely insulin resistant. It is characterized by a high rate of basal lipolysis, constantly releasing non-esterified fatty acids (NEFAs) directly into the portal vein, inundating the liver.

Furthermore, expanded VAT becomes heavily infiltrated by macrophages, transforming into a chronic source of systemic inflammation. It secretes a host of pathogenic adipokines and cytokines, including resistin, tumor necrosis factor-alpha (TNF-alpha), and interleukin-6 (IL-6), while

simultaneously downregulating the production of protective adiponectin [5]. This inflammatory, lipotoxic milieu is the primary driver of systemic insulin resistance.

The Ectopic Fat Overflow: Liver, Muscle, and Pancreas

When the visceral depot expands, the resulting portal “fat flux” directly impacts the liver. Hepatic steatosis (fatty liver) develops, impairing the liver’s ability to suppress glucose production and promoting the overproduction of Very Low-Density Lipoproteins (VLDL), leading to atherogenic dyslipidaemia.

Ectopic fat also accumulates in skeletal muscle (intramyocellular lipids), interfering with the insulin signaling cascade (specifically the PI3K/Akt pathway) and blunting glucose uptake. Fat deposition around the heart (epicardial and pericardial fat) exerts direct lipotoxic and inflammatory effects on the myocardium and coronary arteries, independently increasing cardiovascular risk.

Perhaps most critically for the pathogenesis of type 2 diabetes is the accumulation of fat within the pancreas itself. Pancreatic steatosis exposes the delicate beta cells to high local concentrations of free fatty acids and inflammatory cytokines. This beta cell lipotoxicity impairs insulin secretion, accelerates beta cell apoptosis, and drives the progression from compensatory hyperinsulinaemia to absolute insulin deficiency.



Figure 5.2 — Ectopic fat overflow into liver, muscle, heart, and pancreas (TOFI).

Measuring the Invisible: Waist Circumference and Beyond

Given the pathogenic nature of visceral fat, clinical assessment must move beyond the weighing scale. Waist circumference (WC) is the most practical, cost-effective surrogate marker for visceral adiposity in a routine clinical setting. Landmark work by Lean et al. established waist circumference as a powerful predictor of cardiovascular risk, independent of BMI [6].

[IMAGE PLACEHOLDER — Figure 5.3]

Figure 5.3 — Waist circumference measurement — the Asian Indian cutoffs (>90 cm men, >80 cm women).

For the Asian Indian population, the cutoffs for central obesity are significantly lower than international standards. Based on consensus guidelines, central obesity is defined as a waist circumference greater than 90 cm in men and greater than 80 cm in women [3]. These measurements should be a mandatory vital sign in every clinical encounter.

For a more precise assessment of body composition, advanced modalities such as Dual-Energy X-ray Absorptiometry (DEXA) and Bioelectrical Impedance Analysis (BIA) are increasingly utilized in specialized metabolic clinics. While not yet universally standardized for all populations, some BIA reference ranges propose that a Visceral Adipose Tissue (VAT) area exceeding 100 cm² indicates a significantly elevated risk of metabolic complications. We present this >100 cm² threshold as a useful clinical reference point, though we acknowledge it is not a universal consensus guideline.

The Indian Reality: The ICMR-INDIAB Data

The urgency of shifting our focus to visceral obesity is underscored by national epidemiological data. The landmark ICMR-INDIAB study revealed a staggering prevalence of central obesity across India [7]. The study demonstrated that while generalized obesity (measured by BMI) was prevalent, the rates of abdominal obesity (measured by waist circumference) were substantially higher, affecting over half of the urban population and a significant proportion of the rural population.

This high prevalence of central obesity directly correlates with the rising tide of insulin resistance and dyslipidaemia in the country. The metabolic consequences of this visceral adiposity are profound, and they are best quantified using integrated indices.

The Dr. Nehru Method: Integrating Anthropometry and Biochemistry

In the Dr. Nehru Method, we bridge the gap between anthropometry (where the fat is) and biochemistry (what the fat is doing). Visceral obesity is the anatomical driver; insulin resistance and atherogenic dyslipidaemia are the biochemical consequences.

To capture this relationship, we utilize the Homeostatic Model Assessment for Insulin Resistance (HOMA-IR) and the Triglyceride-Glucose (TyG) index. As established in our published clinical cohorts, we propose population-specific thresholds for the Indian phenotype: a HOMA-IR above 2.3 and a TyG index (natural log) above 8.82 [8] [9]. These metrics, particularly when integrated with waist circumference (as in the TyG-WC index), provide a highly sensitive, low-cost method for identifying the high-risk, centrally obese patient long before clinical diabetes ensues.

(Note regarding B12 supplementation: When addressing the metabolic and neurological needs of these patients, methylcobalamin is preferred in the Dr. Nehru Method as the active bioavailable form; cyanocobalamin is not universally contraindicated but is not preferred.)

Elaborating on the Clinical Implications of TOFI

The “Thin Outside, Fat Inside” (TOFI) phenotype is not merely a descriptive classification; it represents a significant clinical blind spot in standard medical practice. When a patient presents with a normal BMI, the clinical index of suspicion for metabolic disease drops precipitously. Routine lipid panels or fasting glucose tests may be delayed or omitted entirely, under the false assumption that the patient's weight protects them from cardiometabolic risk.

However, the TOFI patient is actively undergoing the same pathogenic processes as the visibly obese patient. The visceral fat is actively secreting inflammatory cytokines and non-esterified fatty acids, driving hepatic insulin resistance and beta cell lipotoxicity. Because these patients lack the subcutaneous “sink” to store excess energy safely, their metabolic deterioration can be surprisingly rapid. By the time fasting glucose levels rise enough to trigger a diabetes diagnosis, the TOFI patient may already have

advanced endothelial dysfunction, established atherogenic dyslipidaemia, and significant beta cell loss.

Therefore, the Dr. Nehru Method insists that clinical vigilance must not be dictated by the weighing scale. A normal BMI should never be a reason to withhold a waist circumference measurement or a thorough metabolic evaluation, particularly in the South Asian population where the TOFI phenotype is highly prevalent.

Deep Dive into the ICMR-INDIAB Findings

The ICMR-INDIAB study is one of the most comprehensive epidemiological efforts to map the metabolic health of the Indian subcontinent, and its findings regarding abdominal obesity are sobering. The data revealed that the prevalence of abdominal obesity was not only alarmingly high in urban centers but was also rapidly penetrating rural populations, blurring the traditional urban-rural divide in metabolic risk.

Specifically, the study highlighted a stark contrast between generalized obesity (measured by BMI) and abdominal obesity (measured by waist circumference). In many regions, the prevalence of abdominal obesity was more than double that of generalized obesity. This means that for every Indian patient identified as obese by BMI, there is at least one other patient—and often more—who is entirely missed by BMI screening but carries a dangerous burden of visceral fat.

Furthermore, the ICMR-INDIAB data demonstrated a strong, independent correlation between abdominal obesity and the prevalence of both pre-diabetes and overt type 2 diabetes. This large-scale, population-level evidence firmly validates the core premise of the Dr. Nehru Method: visceral adiposity, not total body weight, is the primary driver of the Indian metabolic epidemic. The study serves as a clarion call for a paradigm shift in public health screening, emphasizing waist circumference and advanced metabolic indices over simple height-weight ratios.

The Path Forward: A Call for Precision Anthropometry

To effectively combat the rising tide of cardiometabolic disease in India, we must adopt a more precise approach to anthropometry and risk stratification.

The reliance on BMI as a standalone screening tool must be abandoned. Instead, clinical practice must incorporate a multi-tiered assessment strategy.

First, waist circumference must become a mandatory measurement for every adult patient, regardless of their BMI. The Asian Indian cutoffs (>90 cm for men, >80 cm for women) should trigger immediate metabolic evaluation, including fasting insulin, lipid profile, and HOMA-IR calculation.

Second, clinicians must be educated to recognize the TOFI phenotype and the concept of the personal fat threshold. Understanding that a lean-appearing patient may have exhausted their subcutaneous storage capacity is crucial for early intervention.

Finally, we must integrate these anthropometric findings with robust biochemical markers. The TyG-WC index, which combines the anatomical risk of waist circumference with the biochemical reality of insulin resistance (triglycerides and fasting glucose), represents the future of accessible, high-precision metabolic screening. By embracing these newer concepts, we can move beyond the limitations of BMI and provide the targeted, proactive care that the Indian patient urgently requires.

Conclusion

The Indian Paradox—the high prevalence of cardiometabolic disease at low body weights—is only a paradox if we continue to view the body through the flawed lens of BMI. By recognizing the unique “thin-fat” phenotype, understanding the low personal fat threshold of the South Asian population, and appreciating the profound pathogenic difference between subcutaneous and visceral fat, we can identify the true drivers of disease. The TOFI patient is in imminent danger, and the tape measure, rather than the weighing scale, is our most vital tool for early detection.

Multiple Choice Questions

1. According to the WHO expert consultation for Asian populations, at what Body Mass Index (BMI) does “high risk” begin, serving as a public health action point for obesity?

- A. 25.0 kg/m²
- B. 27.5 kg/m²
- C. 30.0 kg/m²

D. 32.5 kg/m²

2. The “Y-Y Paradox” (Yajnik and Yudkin) best illustrates which clinical concept regarding the Indian phenotype?

- A. Indians have a higher basal metabolic rate than Caucasians.
- B. Indians have a lower risk of cardiovascular disease at a normal BMI.
- C. Indians have a significantly higher body fat percentage at the same BMI compared to Caucasians.
- D. Indians store fat primarily in subcutaneous rather than visceral depots.

3. In the context of the “Personal Fat Threshold” hypothesis, what occurs when an individual's subcutaneous adipose tissue capacity is exceeded?

- A. The body increases its basal metabolic rate to burn the excess fat.
- B. Fat safely accumulates in the subcutaneous tissue of the lower extremities.
- C. Excess energy spills over into ectopic sites such as the liver, muscle, and pancreas.
- D. The liver decreases its production of Very Low-Density Lipoproteins (VLDL).

4. Which of the following adipokines, secreted by healthy subcutaneous fat, exerts protective anti-inflammatory effects and enhances insulin sensitivity?

- A. Resistin
- B. Tumor Necrosis Factor-alpha (TNF-alpha)
- C. Interleukin-6 (IL-6)
- D. Adiponectin

5. Based on consensus guidelines for Asian Indians, what are the waist circumference cutoffs defining central obesity?

- A. > 102 cm in men; > 88 cm in women
- B. > 94 cm in men; > 80 cm in women
- C. > 90 cm in men; > 80 cm in women
- D. > 85 cm in men; > 75 cm in women

Answers: 1. B 2. C 3. C 4. D 5. C

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CHAPTER 6

Proinsulin, C-Peptide and Beta Cell Reserve

Chapter 6 — Proinsulin, C-Peptide and Beta Cell Reserve

Biomarkers of Beta Cell Stress in the Indian Diabetic

6.1 — Introduction

Standard diabetes diagnosis relies on late-stage glucose markers — fasting plasma glucose, two-hour OGTT glucose, and HbA1c. These markers reflect a relatively advanced stage of metabolic dysfunction. By the time HbA1c crosses 6.5 percent, beta cell reserve has already been substantially depleted. Emerging evidence highlights the critical importance of earlier biomarkers — hyperinsulinaemia, proinsulin, and C-peptide — which can indicate beta cell stress and insulin resistance years before glucose dysregulation becomes apparent. In the Dr. Nehru Method, these biomarkers are integrated with the India-specific operational thresholds of HOMA-IR above 2.3 and TyG above 8.82 to provide a complete picture of beta cell health.¹²



Figure 6.1 — Proinsulin processing and C-peptide co-secretion — markers of beta cell reserve.

6.2 — Fasting Hyperinsulinaemia — The Earliest Detectable Signal

Fasting hyperinsulinaemia — characterised by elevated circulating insulin levels in the fasting state — is increasingly recognised as the primary detectable defect preceding insulin resistance by years or even decades. Longitudinal studies have demonstrated that insulin resistance can precede the development of type 2 diabetes by 10 to 13 years, with fasting insulin levels capturing this compensatory elevation early in the disease trajectory.³ The condition is self-perpetuating — pre-exposure to hyperinsulinaemia

stimulates a greater insulin-induced secretory response, independently of insulin sensitivity.

Fasting insulin above 12 mIU/L indicates the beta cell is working overtime to maintain glucose homeostasis — even when fasting glucose is completely normal.

In the Dr. Nehru Method, a fasting insulin above 12 mIU/L is an early marker of metabolic concern. The HOMA-IR formula — fasting glucose in mg/dL multiplied by fasting insulin in mIU/L divided by 405 — provides a quantitative measure of this compensatory hyperinsulinaemia. The India-specific operational cutoff of above 2.3 defines the threshold for active clinical intervention in the Indian phenotype.¹

Beyond glucose metabolism, hyperinsulinaemia is an independent predictor of cardiovascular disease risk. The Quebec Cardiovascular Study of 2,103 men demonstrated that the odds ratio for ischaemic heart disease was 1.69 for each standard deviation increase in insulin concentration, independent of traditional cardiovascular risk factors.⁴ Hyperinsulinaemia also drives visceral fat accumulation through its potent lipogenic effects, activating SREBP-1c and promoting hepatic de novo lipogenesis — creating the atherogenic dyslipidaemia pattern characteristic of the Indian metabolic phenotype.⁵

6.3 — Proinsulin — Definition, Cutoffs, and Clinical Significance

Proinsulin is the single-chain precursor molecule to insulin, synthesised in the pancreatic beta cells. It undergoes proteolytic cleavage in the secretory granules to yield equimolar amounts of mature insulin and C-peptide. In healthy individuals, the ratio of proinsulin to insulin is typically low — less than 5 percent of circulating insulin — reflecting efficient processing by the prohormone convertases PC1/3 and PC2. A normal fasting intact proinsulin level is generally below 11 pmol/L, though some references cite the upper limit at 10 pmol/L.⁶

Proinsulin above 11 pmol/L signals that the beta cell Golgi apparatus is overwhelmed — it is releasing immature secretory granules because demand exceeds processing capacity.

Elevated proinsulin levels are a sensitive indicator of beta cell stress and dysfunction, rising significantly before HbA1c reaches diabetic levels. Studies confirm that proinsulin levels are elevated early in prediabetes, reflecting impaired proinsulin processing capacity.⁷ The proinsulin to C-peptide ratio serves as a marker of beta cell endoplasmic reticulum stress and impaired proinsulin conversion — an elevated ratio indicates impending beta cell dysfunction.⁸

South Asian populations consistently exhibit higher fasting proinsulin levels compared to white Caucasians, even when matched for BMI.⁹ This suggests a greater degree of beta cell stress and impaired prohormone processing in Indian patients, contributing to their higher susceptibility to type 2 diabetes at lower BMI values and younger ages. Asian Indians exhibit higher fasting insulin and proinsulin levels compared to other ethnicities at similar BMI — a direct reflection of the thin-fat phenotype where metabolic stress accumulates invisibly within a normal-appearing body.

Elevated proinsulin is not merely a marker of beta cell stress — it is an independent predictor of cardiovascular events. The Quebec Cardiovascular Study and the Insulin Resistance Atherosclerosis Study confirmed this association, establishing proinsulin as a prognostic marker for cardiovascular disease in prediabetic individuals that operates independently of conventional lipid and glucose parameters.¹⁰

6.4 — C-Peptide — The True Measure of Beta Cell Reserve

C-peptide — the connecting peptide — is co-secreted with insulin in equimolar amounts from the pancreatic beta cells. Unlike insulin, which undergoes 50 to 80 percent first-pass hepatic extraction and has a half-life of 3 to 5 minutes, C-peptide undergoes negligible hepatic extraction and is cleared primarily by the kidneys with a half-life of approximately 30 minutes. This makes C-peptide a far more stable and reliable indicator of endogenous insulin secretion than peripheral insulin measurement.

Normal fasting C-peptide ranges from 0.5 to 2.0 ng/mL. A fasting C-peptide above 2.0 ng/mL indicates significant hyperinsulinaemia and insulin resistance — the beta cell is hypersecreting to overcome peripheral resistance.

A fasting C-peptide below 0.5 ng/mL indicates severe depletion of beta cell mass, suggesting absolute insulin deficiency.¹¹

C-peptide trajectory maps the complete beta cell journey — rising during compensation, peaking under stress, then falling as reserve depletes. It is the most honest measure of how much beta cell the patient has left.

In the early stages of prediabetes, C-peptide rises as beta cells compensate for increasing insulin resistance. As beta cell dysfunction progresses, C-peptide eventually declines, reflecting reduction in endogenous insulin production. Longitudinal studies demonstrate an exponential fall in C-peptide over approximately 5 years following the onset of type 1 or type 2 diabetes. Asian Indian cohorts with prediabetes frequently exhibit high fasting C-peptide levels — indicating significant insulin resistance even at lower BMI — before eventual decline.¹²

6.5 — The FCPGR — Fasting C-Peptide to Creatinine Ratio

Because C-peptide is renally cleared, its serum concentration can be falsely elevated in patients with impaired glomerular filtration. The Fasting C-Peptide to Creatinine Ratio — FCPGR — corrects for renal function and provides a more reliable metric of residual beta cell function. The operational thresholds in clinical practice are as follows. An FCPGR above 0.020 nmol/mmol indicates substantial residual beta cell function — these patients may respond to insulin sensitisers and beta cell-preserving therapies. An FCPGR below 0.010 nmol/mmol indicates critical beta cell failure — exogenous insulin is clinically mandatory. These thresholds should be interpreted as operational clinical guides rather than universally validated cutoffs, as the evidence base is primarily from specialised diabetes populations.¹³

6.6 — India Versus China Versus West — Beta Cell Comparison

Comparative studies reveal significant ethnic differences in beta cell function. Asian Indians have a lower genetically determined beta cell function — as measured by HOMA-B — compared to white Europeans. East

Asian populations show lower HOMA-B values but maintain a similar beta cell response relative to insulin resistance when compared to Caucasians. These differences reflect both genetic and environmental influences on beta cell health.¹⁴

The age of type 2 diabetes onset differs significantly by ethnicity. Indians and Chinese typically develop type 2 diabetes at younger ages — 40 to 50 years — compared to Western populations — 50 to 60 years. BMI at diagnosis is also lower — approximately 25 kg/m² for Asian Indians versus 30 kg/m² for Western populations. This earlier onset at lower BMI directly reflects the lower personal fat threshold and lower baseline beta cell reserve that characterises the South Asian phenotype.¹⁵

First-phase insulin response — a crucial indicator of beta cell function — is significantly lower in Asian Indians compared to Caucasians even when matched for age and BMI. This reduced capacity to secrete insulin rapidly in response to glucose explains the heightened susceptibility to type 2 diabetes in South Asian populations and underscores why the India-specific HOMA-IR cutoff of above 2.3 is appropriate — beta cell stress manifests at lower absolute insulin resistance in this phenotype.¹

6.7 — The Dr. Nehru Method Beta Cell Staging — Proposed Framework

Integrating HOMA-IR, fasting insulin, proinsulin, and C-peptide into a coherent staging framework allows the clinician to map where each patient sits on the beta cell continuum. This is the Dr. Nehru Method proposed clinical staging — not an internationally standardised system but a clinically useful operational framework.

Stage 1 — Compensated insulin resistance — HOMA-IR above 2.3, fasting insulin rising, C-peptide normal or rising, proinsulin normal. The beta cell is hypersecreting but processing efficiently. Stage 2 — Beta cell stress — HOMA-IR persisting above 2.3, proinsulin above 11 pmol/L, proinsulin to insulin ratio above 0.20, C-peptide peaking. The processing machinery is beginning to fail. Stage 3 — Beta cell exhaustion — C-peptide falling from peak, FCPGR approaching 0.010, HbA1c crossing 6.5 percent. Stage 4 — Beta cell failure — C-peptide below 0.5 ng/mL, FCPGR below 0.010, insulin mandatory.

Figure 6.2 — Proinsulin-to-insulin ratio rising with beta cell endoplasmic reticulum stress.

High C-peptide with high HOMA-IR means the beta cell is still working — treat the insulin resistance. Low C-peptide with high glucose means the beta cell has failed — provide insulin without delay.

Clinical Summary — Chapter 6

1. Fasting hyperinsulinaemia above 12 mIU/L is the earliest detectable signal of metabolic dysfunction — present years before glucose rises.
2. HOMA-IR above 2.3 — India operational threshold — Nehru MS Int J Med 2026.
3. Proinsulin above 11 pmol/L indicates beta cell Golgi stress — processing overwhelmed.
4. Proinsulin to insulin ratio above 0.20 indicates impaired conversion — beta cell decompensation.
5. C-peptide 0.5 to 2.0 ng/mL normal range — above 2.0 indicates IR — below 0.5 indicates failure.
6. FCPGR above 0.020 — residual function present — below 0.010 — insulin mandatory.
7. Indian patients show higher proinsulin at same BMI than Europeans — greater beta cell stress.
8. C-peptide trajectory — rises then falls — maps complete beta cell journey.

Multiple Choice Questions

1. Fasting intact proinsulin above which level indicates significant beta cell Golgi stress?
A. 5 pmol/L B. 8 pmol/L C. 11 pmol/L D. 20 pmol/L
2. FCPGR below 0.010 nmol/mmol indicates:
A. Robust beta cell function B. Critical beta cell failure — insulin mandatory

C. Normal insulin sensitivity D. Hypoglycaemia risk

3. Why is C-peptide superior to peripheral insulin for measuring beta cell output?

A. C-peptide stimulates insulin secretion B. C-peptide undergoes negligible hepatic extraction

C. C-peptide lowers blood glucose D. C-peptide is secreted by alpha cells

4. Which population characteristic explains higher proinsulin at same BMI in Indians?

A. Higher subcutaneous fat B. Greater beta cell reserve

C. Lower beta cell function and higher IR at lower BMI D. Higher renal clearance

5. The proinsulin to insulin ratio above 0.20 indicates:

A. Optimal beta cell function B. Beta cell endoplasmic reticulum stress

C. Normal glucose tolerance D. Absence of insulin resistance

Answers: 1. C 2. B 3. B 4. C 5. B

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CHAPTER 7

TyG Index and Beta Cell Failure

Chapter 7 — TyG Index and Beta Cell Failure

Beyond a Number — The Triglyceride-Glucose Index as a Metabolic Distress Signal

7.1 — What the TyG Index Actually Measures

The triglyceride-glucose index — TyG — is calculated as the natural logarithm of fasting triglycerides in mg/dL multiplied by fasting glucose in mg/dL divided by two. This simple formula, requiring no additional cost beyond a standard fasting lipid and glucose measurement, captures something that neither triglycerides nor glucose can capture alone — the integrated metabolic disturbance of insulin resistance. Triglycerides rise when hepatic de novo lipogenesis accelerates under the influence of insulin resistance and SREBP-1c activation. Fasting glucose rises when hepatic glucose output increases as peripheral insulin signalling fails. TyG captures both numerators rising simultaneously, encoding the convergence of hepatic fat metabolism and glucose dysregulation into a single computable number. In the natural log scale, the India-specific cutoff of above 8.82 — validated in three indexed publications from HOMA Healthcare Center Hyderabad in 2026 — identifies patients at risk for progressive beta cell damage years before HbA1c crosses the diagnostic threshold.¹²³

TyG above 8.82 on the natural log scale is the earliest computable signal of atherogenic insulin resistance in Indian adults. It costs nothing beyond a standard lipid profile and fasting glucose.

The TyG Index must always be stated in natural log scale values — ranging from 7.5 to 10.5 in clinical populations — and must never be confused with the simple linear ratio of triglycerides to glucose that some international studies report as values of 4.x or 5.x. These are different calculations and different scales. The India-specific cutoff of 8.82 applies exclusively to the natural log formula. Clinicians using the linear ratio scale and comparing to 8.82 will systematically misclassify patients.

7.2 — TyG Predicts Type 2 Diabetes Before Glucose Rises

The landmark evidence for TyG as an early diabetes predictor comes from Tirosh and colleagues, who studied 13,936 apparently healthy young adults in a prospective cohort over five years. Rising fasting triglycerides over the observation period independently predicted incident type 2 diabetes — regardless of baseline BMI and fasting glucose. This finding established that triglyceride elevation — captured efficiently by the TyG Index — is not merely a consequence of insulin resistance but a measurable precursor to glycaemic failure, detectable years before glucose crosses the diagnostic threshold.⁴ The clinical implication is direct — a patient with TyG above 8.82 and normal HbA1c is not metabolically safe. They are on a measurable trajectory toward beta cell exhaustion, and the TyG elevation is the earliest visible signal of that journey.

Rising triglycerides independently predicted type 2 diabetes over five years, regardless of BMI and fasting glucose — Tirosh et al., Diabetes Care, 2008, N=13,936.

Guo and colleagues extended this understanding in 914 patients across normal, impaired, and diabetic glucose tolerance states, demonstrating that the TyG Index is negatively correlated with early and late-phase pancreatic beta cell insulin secretion assessed during oral glucose tolerance testing. As TyG rises, C-peptide secretory capacity falls.⁵ Song and colleagues confirmed in a further cohort that TyG inversely correlates with HOMA-B — the homeostatic model assessment of beta cell function — establishing that TyG tracks not just insulin resistance but the functional reserve of the beta cell itself.⁶ These are not surrogate findings. They establish TyG as a direct mirror of beta cell health, measurable from a standard laboratory panel at zero additional cost.

7.3 — The Five-Stage TyG Beta Cell Journey

The progression from metabolic health to beta cell exhaustion can be mapped through five stages using the TyG Index alongside HOMA-IR and C-peptide. This framework — the Dr. Nehru Method proposed staging system — is not an internationally standardised diagnostic classification but a clinically

actionable conceptual model for identifying where each patient sits on the continuum and what intervention is appropriate at each stage.

[IMAGE PLACEHOLDER — Figure 7.1]

Figure 7.1 — The five-stage TyG beta cell journey from metabolic health to exhaustion.

In Stage 1 — the metabolically healthy state — TyG is below 8.82, HOMA-IR is below 2.3, and C-peptide is normal. The beta cell is uncompromised. In Stage 2 — the compensating state — TyG crosses 8.82 and HOMA-IR crosses 2.3, but C-peptide remains normal or rises as the beta cell increases insulin output to overcome peripheral resistance. HbA1c is still normal. This is the critical intervention window. In Stage 3 — the stressed state — TyG approaches 9.5 to 10.0, HOMA-IR rises to 4.0 to 6.0, and the proinsulin ratio begins to rise above 0.20. C-peptide is peaking. The beta cell is working at maximum capacity and beginning to show evidence of lipotoxic stress. In Stage 4 — the failing state — C-peptide begins to fall, proinsulin ratio rises above 0.30, HbA1c enters the prediabetic range of 5.7 to 6.4 percent, and TyG exceeds 10.0. In Stage 5 — the exhausted state — C-peptide is very low, HbA1c has crossed the diagnostic threshold, and the beta cell has been depleted of functional reserve. Insulin is mandatory. This staging framework should be framed clearly as the Dr. Nehru Method clinical protocol — not as published international consensus — and should be applied with clinical judgment informed by individual patient context.

7.4 — The Ceramide Connection — From Triglycerides to Beta Cell Damage

The biological mechanism linking elevated triglycerides to beta cell damage operates primarily through the ceramide lipotoxicity pathway. This is a proposed and mechanistically well-supported pathway — not a proven universal linear sequence in all patients — and should be understood as such when teaching or citing. Severe hypertriglyceridaemia may increase lipotoxic stress through increased circulating saturated fatty acid exposure. Saturated fatty acids — particularly palmitate — can promote ceramide formation via the de novo ceramide synthesis pathway. Maedler and colleagues demonstrated in isolated human pancreatic islets from organ donors that prolonged palmitate exposure significantly blunts glucose-stimulated insulin

secretion and promotes beta cell apoptosis — an effect not seen with monounsaturated fatty acids.⁷ This human islet evidence — not merely cell line data — gives the ceramide pathway genuine clinical relevance.

[IMAGE PLACEHOLDER — Figure 7.2]

Figure 7.2 — The ceramide lipotoxicity pathway linking triglycerides to beta cell damage.

Palmitate exposure in isolated human pancreatic islets blunts glucose-stimulated insulin secretion and promotes apoptosis — Maedler et al., Diabetes, 2001.

Lu and colleagues provided clinical human evidence for ceramide involvement in beta cell dysfunction in 1,974 subjects, demonstrating that elevated plasma ceramide C18:0 is strongly associated with decreased insulin sensitivity, reduced beta cell function measured by disposition index, and higher risk of incident type 2 diabetes.⁸ Ceramide impairs insulin signalling through Akt and PKB pathway inhibition in experimental systems, and contributes to beta cell apoptosis through caspase-3 activation — a recognised apoptosis pathway in multiple cell models.⁹ Chronic fatty acid exposure can reduce proinsulin gene expression and beta cell function.¹⁰ The VLDL-related lipid flux contributes to lipotoxicity through the release of saturated fatty acids — VLDL particles are not intrinsically toxic in isolation but become harmful through the lipotoxic microenvironment created when lipolysis releases their fatty acid cargo near beta cells.¹¹

The clinical implication is that TyG above 8.82 — indicating elevated fasting triglycerides and fasting glucose — represents not merely a cardiovascular risk marker but a signal of ongoing lipotoxic beta cell stress. The correct clinical response is not a fibrate prescription. It is HOMA-IR measurement, TyG calculation, and identification of which of the eight metabolic triggers is driving the insulin resistance and triglyceride elevation.

7.5 — Hypertriglyceridaemia and Pancreatitis — The Beta Cell Destroyer

Severe hypertriglyceridaemia carries a risk of acute pancreatitis that rises steeply with escalating triglyceride levels. Pedersen and colleagues demonstrated in a Danish population cohort of 116,550 subjects that the

prevalence of acute pancreatitis is approximately 10 percent in individuals with triglycerides between 1,000 and 1,999 mg/dL, and rises to approximately 20 percent when triglycerides exceed 2,000 mg/dL.¹² Pancreatitis does not merely inflame the exocrine pancreas — it destroys endocrine tissue simultaneously. Every episode of hypertriglyceridaemia-induced pancreatitis carries the risk of significant and irreversible beta cell mass loss, the extent of which depends on the degree of pancreatic necrosis.

Post-pancreatitis diabetes — classified as Type 3c diabetes — is a distinct entity from type 2 diabetes, characterised by combined exocrine and endocrine insufficiency, early insulin dependence, and a metabolic phenotype that responds poorly to standard oral hypoglycaemic agents designed for type 2 disease. Ewald and colleagues established that Type 3c diabetes accounts for approximately 5 to 9 percent of all diabetes cases globally.¹³ Indian data from Mohan and colleagues documents the unique subset of fibrocalculous pancreatic diabetes and tropical chronic pancreatitis — conditions driven by chronic nutritional deficiency and cassava toxicity in some populations — that produce Type 3c diabetes with early insulin dependence and distinctive metabolic characteristics.¹⁴

Type 3c diabetes is not type 2 diabetes. It requires different management. Every case of hypertriglyceridaemia-induced pancreatitis risks creating it.

Chylomicronaemia syndrome — characterised clinically by eruptive xanthomas, lipaemia retinalis, hepatosplenomegaly, and severe pancreatitis risk — occurs when fasting triglycerides exceed 1,000 mg/dL and frequently exceed 2,000 mg/dL. It is pathogenetically linked to defects in chylomicron clearance through mutations affecting LPL, APOC2, APOC3, APOA5, or GPIHBP1. The prevalence of familial chylomicronaemia syndrome is approximately 1 in 300,000, though multifactorial chylomicronaemia is considerably more common.¹⁵

7.6 — Pharmacological Approaches to TG-Driven Beta Cell Stress

GLP-1 receptor agonists reduce fasting and postprandial triglycerides by approximately 12 to 22 percent through mechanisms that include delayed gastric emptying, reduced hepatic VLDL and apoB-48 production, and

decreased intestinal chylomicron secretion — effects that extend beyond weight loss alone.¹⁶ In the context of TyG-driven beta cell stress, the GLP-1 receptor agonist class occupies a unique position — simultaneously reducing triglycerides, improving insulin resistance, reducing body weight, and exerting direct beta cell protective effects in experimental systems.

Fenofibrate reduces triglycerides by 30 to 50 percent through PPAR α agonism. Li and colleagues observed in 1,974 patients with hypertriglyceridaemia and normal glucose tolerance that a fenofibrate-treated subgroup of 92 patients showed reduction in HOMA-IR and HOMA-B — suggesting reduced beta cell secretory burden.¹⁷ However, no clinical randomised controlled trial has established that fenofibrate directly protects beta cell mass or prevents beta cell apoptosis in humans. The FIELD and ACCORD Lipid trials both failed to show cardiovascular benefit in general populations, confirming that triglyceride reduction without concurrent apoB particle reduction does not reduce cardiovascular events — the mechanism being that TG is a marker of atherogenic particle burden rather than a direct causal cardiovascular target in isolation.

Lee and colleagues demonstrated that EPA and DHA inhibit NLRP3 inflammasome activation and downstream IL-1 β secretion in experimental work, providing a mechanistic basis for the anti-inflammatory benefit of omega-3 supplementation in lipotoxic states.¹⁸ This evidence is primarily preclinical. No clinical randomised controlled trial has demonstrated that omega-3 supplementation directly reduces ceramide accumulation or prevents beta cell apoptosis in humans with hypertriglyceridaemia. Omega-3 supplementation at 1 to 4g daily produces 15 to 30 percent triglyceride reduction and is a clinically reasonable adjunct in patients with elevated TyG — framed as triglyceride reduction and anti-inflammatory support rather than proven beta cell preservation.

Treat the insulin resistance driving TyG elevation — not the TyG number in isolation. The number is the signal. The root is the target.

7.7 — The Indian Clinical Imperative

The Indian clinical population carries specific characteristics that make TyG monitoring particularly urgent. The ICMR-INDIAB study of 16,607 subjects

established that dyslipidaemia prevalence in India is approximately 81 percent, driven predominantly by low HDL in 67 to 72 percent and elevated triglycerides in 30 percent of the population.¹⁹ The atherogenic dyslipidaemia pattern — high TG, low HDL, small dense LDL — is the dominant Indian lipid phenotype, not hypercholesterolaemia. This means that standard lipid panels focused on LDL and total cholesterol systematically underestimate cardiovascular and metabolic risk in Indian patients.

Jin and colleagues demonstrated in 4,321 patients in an Asian cardiovascular cohort that the TyG Index independently predicts major adverse cardiovascular events — outperforming both LDL-C and fasting glucose for identifying high-risk patients.²⁰ The Mendelian randomisation data of Ference and colleagues in 654,783 subjects established that triglyceride lowering reduces cardiovascular events only in proportion to the concurrent reduction in apoB-containing atherogenic particles — not through triglyceride reduction itself.²¹ These two findings together create the framework for Indian clinical practice: TyG predicts cardiovascular events in Asian populations better than LDL, but the correct therapeutic target is not TG in isolation — it is the insulin resistance that drives TG elevation and the atherogenic particle burden it creates.

The Dr. Nehru Method operational protocol for TyG above 8.82 is therefore: measure HOMA-IR to confirm insulin resistance, identify which of the eight metabolic triggers is driving the resistance, treat the trigger, recheck TyG and HOMA-IR at three months, and expect measurable falls in both indices as the root cause is addressed. This is not a guideline-mandated protocol but a clinically rational and evidence-informed approach that integrates the India-specific validated cutoffs of HOMA-IR above 2.3 and TyG above 8.82 with systematic trigger identification.

7.8 — What This Means for Indian Clinical Practice

The TyG Index should be calculated for every patient who receives a standard fasting lipid profile and fasting glucose measurement. It requires no additional test, no additional cost, and no additional blood draw. The natural log formula — $\ln$ of fasting triglycerides in mg/dL multiplied by fasting glucose in mg/dL divided by two — can be calculated in seconds on any clinical calculator. The India-specific cutoff of above 8.82 on the natural log scale identifies patients at risk for progressive beta cell damage and

cardiovascular disease in the Indian population, validated in three indexed international publications from 2026.

When TyG exceeds 8.82, the clinical response is a metabolic workup — not a fibrate prescription. HOMA-IR, C-peptide, fasting insulin, Vitamin D, B12, ferritin, TSH, and a body composition assessment should follow. The trigger driving the insulin resistance should be identified and treated. When the trigger is corrected, TyG falls — not because TG was targeted directly, but because the metabolic fire driving both TG and glucose elevation has been reduced. This is the fundamental difference between managing a number and treating a patient.

The TyG Index is not a lipid number. It is a metabolic distress signal. The correct response is a metabolic workup — not a triglyceride-lowering drug.

Clinical Summary — Chapter 7

1. TyG Index is calculated as $\ln(\text{TG mg/dL} \times \text{Glucose mg/dL} \div 2)$. India-specific cutoff above 8.82 on natural log scale. Never express as 4.x or 5.x — that is a different formula and different scale.
2. TyG above 8.82 predicts type 2 diabetes before HbA1c rises — Tirosh 2008, N=13,936.
3. TyG inversely correlates with C-peptide secretion and HOMA-B — Guo 2022 N=914, Song 2024.
4. Ceramide lipotoxicity pathway — palmitate from hypertriglyceridaemia — beta cell apoptosis — proposed mechanism, well-supported experimentally and in human islet studies.
5. Ceramide C18:0 clinically associated with reduced beta cell function in humans — Lu et al., PLOS Medicine, 2020, N=1,974.
6. Pancreatitis risk — approximately 10 percent at TG 1,000 to 1,999 mg/dL — doubles above 2,000 mg/dL — Pedersen 2016, N=116,550.
7. Post-pancreatitis diabetes is Type 3c — distinct from T2DM — manage differently — Mohan V, Indian data confirmed.
8. Semaglutide reduces TG 12 to 22 percent — mechanism confirmed — Hjerpsted 2018, N=30 RCT.

9. TyG predicts MACE in Asian cohorts better than LDL-C and fasting glucose — Jin 2018, N=4,321.
10. Correct response to TyG above 8.82 is metabolic workup and trigger treatment — not fibrate alone.

Multiple Choice Questions

1. The TyG Index is calculated using which formula on natural log scale?
 - A. $\ln(\text{TG} + \text{Glucose} \div 2)$
 - B. $\ln(\text{TG} \times \text{Glucose} \div 2)$
 - C. $\text{TG} \times \text{Glucose} \div 405$
 - D. $\ln(\text{TG} \div \text{Glucose} \times 2)$
2. Which study with N=13,936 demonstrated that rising TG independently predicts T2DM before HbA1c rises?
 - A. UKPDS
 - B. Whitehall II
 - C. Tirosh et al. Diabetes Care 2008
 - D. CHARLS cohort
3. The India-specific TyG cutoff validated by Nehru MS et al. 2026 on the natural log scale is:
 - A. Above 4.5
 - B. Above 5.0
 - C. Above 8.82
 - D. Above 10.0
4. In the ceramide lipotoxicity pathway, which saturated fatty acid primarily promotes ceramide formation in beta cells?
 - A. Oleic acid
 - B. Palmitate
 - C. DHA
 - D. Linoleic acid
5. Post-pancreatitis diabetes is classified as:
 - A. Type 1 diabetes
 - B. Type 2 diabetes
 - C. Type 3c diabetes
 - D. LADA

Answers: 1. B 2. C 3. C 4. B 5. C

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CHAPTER 8

Sarcopenia and Metabolic Disease

Chapter 8 — Sarcopenia and Metabolic Disease

The Muscle-Insulin Resistance Connection in Indian Clinical Practice

8.1 — Why Sarcopenia Belongs in a Diabetes Book

Skeletal muscle is the largest insulin-sensitive tissue in the human body, accounting for approximately 70 to 80 percent of insulin-stimulated glucose disposal under euglycaemic clamp conditions. When muscle mass falls — through sarcopenia — this disposal capacity falls proportionally, and the metabolic burden on the remaining beta cell reserve increases. The relationship between sarcopenia and insulin resistance is therefore not incidental. It is mechanistic, bidirectional, and clinically urgent. Insulin resistance impairs muscle protein synthesis through disruption of the mTORC1 anabolic signalling pathway, while reduced muscle mass worsens insulin resistance by reducing the peripheral glucose sink. Each condition accelerates the other. In the Indian clinical population — where sarcopenia occurs at lower body weight and lower BMI than in Western populations, where dietary protein intake is frequently below recommended levels, and where HOMA-IR above 2.3 is already prevalent — this bidirectional relationship creates a metabolic spiral that standard diabetes management does not address.

Skeletal muscle accounts for 70 to 80 percent of insulin-stimulated glucose disposal. When muscle falls, the beta cell compensates alone — until it cannot.

The Asian Working Group for Sarcopenia — AWGS 2019 — provides the diagnostic framework appropriate for Indian patients. Sarcopenia is defined by the presence of low muscle mass confirmed by measurement, alongside either low muscle strength or low physical performance. Low muscle mass alone — without functional impairment — is termed possible sarcopenia or pre-sarcopenia. Confirmed sarcopenia requires both low mass and low

strength or performance. Severe sarcopenia requires all three — low mass, low strength, and low physical performance. These distinctions matter clinically because the intervention priority and prognosis differ at each stage.

8.2 — Diagnostic Criteria and Cutoff Values

The AWGS 2019 criteria use skeletal muscle index — SMI — as the primary measure of muscle mass. SMI is calculated by dividing appendicular lean mass in kilograms by height in metres squared. The appendicular lean mass represents the sum of lean tissue in both arms and both legs, excluding trunk, and is obtained from either dual-energy X-ray absorptiometry or bioimpedance analysis. The cutoff values differ by measurement method and must not be interchanged. For DEXA, low muscle mass is defined as SMI below 7.0 kg per metre squared in men and below 5.4 kg per metre squared in women. For BIA, the cutoff in women is 5.7 kg per metre squared — not 5.4 — reflecting the different measurement characteristics of the two modalities. Using 5.7 for DEXA or 5.4 for BIA constitutes a methodological error that will systematically misclassify patients.¹

Muscle strength is assessed by handgrip dynamometry. The AWGS 2019 cutoffs are below 28 kg in men and below 18 kg in women. The Indian Sarcopenia Consensus — developed through a modified Delphi method with expert panels across India and published in 2025 — recommends a male grip cutoff of below 27.5 kg, acknowledging that Indian men have slightly lower baseline grip than East Asian populations on which the AWGS thresholds were calibrated.² Both cutoffs are clinically acceptable. The reference used should be stated when documenting the diagnosis.

[IMAGE PLACEHOLDER — Figure 8.1]

Figure 8.1 — Grip strength and muscle staging in the diagnosis of sarcopenia.

Physical performance assessment uses the Short Physical Performance Battery — SPPB — with a score below 9 out of 12 indicating poor performance, the six-minute walk test with below 400 metres indicating severe functional impairment, and the Timed Up and Go test with above 12 seconds indicating fall risk and severe sarcopenia. Gait speed below 0.8 metres per second on a four-metre flat corridor walk is an additional performance marker used in some centres. These functional tests require

minimal equipment — a corridor, a stopwatch, and a chair — and should be performed at every metabolic clinic visit in patients above 40 years with HOMA-IR above 2.3, as the operational threshold for metabolic risk screening in Indian adults.

DEXA women cutoff is 5.4 kg per metre squared. BIA women cutoff is 5.7. These are not interchangeable. The method determines the threshold.

8.3 — The Five-Stage Sarcopenia-Metabolic Journey

The progression from metabolic health through sarcopenia to severe sarcopenic obesity can be mapped through five stages that integrate muscle mass, grip strength, insulin resistance markers, and beta cell biomarkers. This staging framework represents the Dr. Nehru Method proposed clinical classification and should be understood as an operational tool for clinical practice rather than an internationally standardised diagnostic system.

In Stage 0 — the pre-sarcopenic metabolic stage — SMI remains within normal range and grip strength is preserved, but HOMA-IR has already crossed above 2.3 and TyG Index has crossed above 8.82. Irisin — the exercise-responsive myokine confirmed to be lower in sarcopenic patients in human observational studies and a 2025 meta-analysis — may be beginning to fall, and myostatin — the well-established negative regulator of skeletal muscle mass — may be beginning to rise. C-peptide is normal or rising as the beta cell compensates. HbA1c is normal. Standard medicine declares this patient healthy. The Dr. Nehru Method identifies this as the critical intervention window — when muscle is fully recoverable and beta cell reserve is intact.³⁴

In Stage 1 — early sarcopenia — SMI approaches the cutoff threshold and grip may be borderline. HOMA-IR is typically between 3.0 and 5.0, TyG between 8.82 and 9.5, and C-peptide is at or near its peak as beta cell compensation reaches maximum. The patient reports fatigue on exertion and reduced exercise tolerance. Intervention at this stage produces excellent outcomes. In Stage 2 — confirmed sarcopenia — SMI falls below the AWGS cutoff, grip falls below threshold, and at least one physical performance measure is impaired. HOMA-IR is typically between 4.0 and 6.0, C-peptide is beginning to fall from its peak, and HbA1c may be entering

the prediabetic range. In Stage 3 — severe sarcopenia — all three AWGS criteria are met, HOMA-IR exceeds 6.0, C-peptide is significantly reduced, and HbA1c has often crossed the diagnostic threshold for type 2 diabetes. In Stage 4 — sarcopenic obesity — the phenotype is characterised by low muscle mass concealed by preserved or elevated total body weight, high visceral fat area above 100 cm squared on BIA, and HOMA-IR frequently exceeding 8.0. BMI may appear normal or only mildly elevated, making standard BMI-based screening entirely inadequate for identifying this high-risk phenotype.

8.4 — The SMI Calculation

The skeletal muscle index is derived from the DEXA or BIA report as follows. Appendicular lean mass equals the sum of right arm lean mass plus left arm lean mass plus right leg lean mass plus left leg lean mass. Trunk lean mass is excluded. SMI equals appendicular lean mass in kilograms divided by height in metres squared. As a practical example — a patient with right arm lean mass of 3.2 kg, left arm lean mass of 3.1 kg, right leg lean mass of 7.8 kg, and left leg lean mass of 7.6 kg has an appendicular lean mass of 21.7 kg. At a height of 1.70 metres, height squared equals 2.89, and SMI equals 21.7 divided by 2.89, giving 7.51 kg per metre squared — normal by AWGS 2019 criteria for men. SMI is the only value requiring manual calculation from the DEXA or BIA report. All other body composition values — total body fat percentage, visceral fat area, appendicular lean mass absolute value, android and gynoid fat distribution, bone mineral density, T-score, and Z-score — are reported directly by the machine, though some outputs such as visceral fat area and fat mass index depend on the specific manufacturer and software version in use and are not universally available across all systems.

8.5 — Protein, Muscle, and the BCAA Paradox

Adequate dietary protein is the substrate requirement for muscle protein synthesis, and its insufficiency is one of the most consistent and correctable contributors to sarcopenia in Indian patients. The Indian Council of Medical Research recommended dietary allowance of 0.8 grams of protein per kilogram of body weight per day represents the minimum for nitrogen balance in healthy sedentary adults. It is not an adequate target for patients with established sarcopenia, insulin resistance, or metabolic disease.

Evidence from sarcopenia and resistance training literature supports protein targets of 1.2 to 1.6 grams per kilogram per day for patients with insulin resistance and borderline muscle mass, and 1.6 to 2.0 grams per kilogram per day for patients with confirmed sarcopenia or those actively attempting muscle mass recovery. These are proposed clinical targets based on resistance training and sarcopenia intervention literature and should be applied with individual patient context rather than presented as universally mandated thresholds.

Leucine — one of the three branched-chain amino acids alongside isoleucine and valine — is the primary activator of mTORC1-mediated muscle protein synthesis. A leucine threshold of approximately 2 to 3 grams per meal is required to trigger the mTORC1 anabolic signal in muscle tissue. Plant-based protein sources commonly consumed in Indian diets — particularly cooked lentils and rice — provide lower leucine density per gram of protein than animal-based sources such as egg white, chicken, and whey protein. This does not make plant protein valueless, but it means that Indian vegetarian and vegan patients may need deliberately higher total protein intake and deliberate food combining to reliably achieve the per-meal leucine threshold for mTORC1 activation. Soy protein — the highest-quality plant protein in terms of essential amino acid profile — is available in Indian markets as tofu, tempeh, and soy milk and should be specifically recommended for vegetarian patients with sarcopenia risk.

High serum BCAAs are a marker of insulin resistance — not a cause of sarcopenia. BCAAs with exercise build muscle. BCAAs without exercise in an insulin-resistant patient may worsen insulin resistance.

The branched-chain amino acid paradox is one of the most important and most frequently misunderstood concepts in metabolic nutrition. Wang and colleagues demonstrated in a metabolomic profiling study of 2,422 subjects that elevated fasting serum BCAAs — isoleucine, leucine, and valine — are strongly and independently associated with insulin resistance and predict incident type 2 diabetes up to twelve years before diagnosis.⁵ This finding appears to contradict the recommendation to increase protein and BCAA intake for muscle building. The resolution of this apparent contradiction is mechanistic. In healthy individuals with normal insulin sensitivity, dietary BCAAs are efficiently taken up by skeletal muscle for protein synthesis,

keeping circulating levels normal. In insulin-resistant individuals, impaired muscle glucose and amino acid uptake causes BCAAs to accumulate in the circulation — not because more are consumed, but because muscle cannot clear them effectively. Elevated serum BCAAs are therefore a marker of the insulin resistance impairing muscle uptake — not an independent cause of that resistance. The correct clinical response is to treat insulin resistance first, restoring muscle's capacity to utilise dietary protein effectively, and to combine protein intake with resistance exercise, which independently enhances muscle amino acid uptake through AMPK-mediated GLUT4 and amino acid transporter upregulation.

[IMAGE PLACEHOLDER — Figure 8.2]

Figure 8.2 — Branched-chain amino acids as a marker of impaired muscle amino acid uptake.

8.6 — Myokines — A Conceptual Framework

Skeletal muscle functions as an endocrine organ, secreting a class of signalling molecules termed myokines during and after muscular contraction. Irisin — derived from cleavage of the FNDC5 protein — is an exercise-responsive myokine whose circulating levels are confirmed to be lower in patients with sarcopenia in human observational studies and a 2025 meta-analysis, with positive correlations to muscle mass and grip strength.³ Whether irisin directly promotes beta cell survival and insulin secretion — as suggested by experimental and limited human islet data — has not been confirmed in human intervention trials, and this beta cell protective role should be described as a proposed mechanism with experimental support rather than an established clinical effect.

Myostatin is a well-established negative regulator of skeletal muscle mass, confirmed in both human and experimental systems. Its inhibitory role on satellite cell activation and muscle protein synthesis is the mechanistic basis for the anti-myostatin drug class currently in clinical development.⁴ In sarcopenic states, myostatin rises as muscle mass falls, further suppressing muscle regeneration in a self-reinforcing cycle. No clinically validated serum myostatin cutoff has been established for sarcopenia diagnosis — studies remain exploratory and assay-dependent — and myostatin measurement is a research tool rather than a routine clinical test in Indian laboratories as of 2026.

Interleukin-6 exhibits context-dependent biology that is essential to understand in the sarcopenia-metabolic disease relationship. Acute IL-6 released from contracting skeletal muscle during exercise is widely accepted as part of an anti-inflammatory physiological response — stimulating fat oxidation and contributing to the metabolic benefits of physical activity. Chronic IL-6 elevation in visceral obesity and progressive muscle loss represents a different biological state — pro-inflammatory, contributing to hepatic insulin resistance through SOCS3 activation and IRS-1 pathway impairment. Sarcopenia shifts this balance from the acute beneficial exercise-derived IL-6 response toward the chronic harmful inflammatory state, because less functional muscle means less exercise-derived acute IL-6 while more visceral fat produces more chronic inflammatory IL-6.⁶

8.7 — Exercise Prescription for Sarcopenia

Resistance training is the only intervention that simultaneously increases muscle mass and reduces HOMA-IR through a single mechanism — enhanced muscle glucose uptake via AMPK-mediated GLUT4 upregulation. A randomised controlled trial demonstrated significant HOMA-IR improvement with resistance training — though the specific mechanism through irisin was not confirmed in that study, with irisin response being variable across exercise trials.⁷ The prescription for sarcopenia management in patients with insulin resistance above 2.3 on HOMA-IR should include resistance training three times per week at approximately 70 percent of maximum effort, performing eight to twelve repetitions per set, with progressive resistance increase every two weeks as tolerated. A minimum of eight weeks of consistent resistance training should be completed before repeat SMI assessment, as body composition changes are not reliably detectable earlier by standard BIA.

Aerobic exercise complements but does not replace resistance training for sarcopenia management. A combination of 150 minutes of moderate-intensity aerobic activity per week alongside three resistance training sessions produces better metabolic and body composition outcomes than either modality alone. For patients with severe sarcopenia and very low physical performance scores, exercise should begin with supervised chair-based resistance exercises and progressive walking before advancing to gym-based resistance training. The six-minute walk test result provides a useful

baseline for monitoring functional improvement alongside SMI and grip strength.

Resistance training three times per week combined with adequate protein is the only intervention that simultaneously rebuilds muscle and reduces HOMA-IR.

8.8 — Sarcopenic Obesity — The Most Missed Phenotype

Sarcopenic obesity is the simultaneous presence of excess adiposity and low muscle mass — a combination that produces worse metabolic outcomes than either condition alone. In Indian clinical practice, this phenotype is frequently missed because BMI — which reflects total body mass relative to height — cannot distinguish between fat mass and lean mass. A patient with BMI of 24 to 26 may have visceral fat area above 100 cm squared and SMI below the AWGS cutoff simultaneously — meeting criteria for both central obesity and sarcopenia while appearing clinically normal by standard anthropometric assessment. This phenotype — sometimes called TOFI, thin outside fat inside — is particularly prevalent in Indian adults who are physically inactive, have chronic insulin resistance, and have inadequate protein intake.

Body composition assessment by BIA provides the most practical initial screening for sarcopenic obesity in Indian clinical settings. The combination of SMI below the AWGS cutoff, visceral fat area above 100 cm squared, and phase angle below 5 degrees — where phase angle below 5 is used as an adjunct clinical marker, not a core AWGS diagnostic criterion — identifies patients who require further evaluation and intervention. When BIA results are borderline or when the clinical picture suggests body composition abnormality despite normal BIA findings, DEXA provides definitive assessment of appendicular lean mass, android fat distribution, android to gynoid ratio, and bone mineral density simultaneously. DEXA also allows calculation of fat mass index — fat mass in kilograms divided by height in metres squared — which carries independent metabolic risk significance above 5.5 kg per metre squared, though this threshold comes from European population data and should be applied cautiously to Indian patients pending India-specific normative data.

The treatment of sarcopenic obesity requires simultaneous attention to both components — fat reduction and muscle preservation. Standard calorie restriction and weight loss without resistance training and adequate protein will reduce both fat mass and lean mass, potentially worsening the sarcopenic component while improving the obesity component. Bimagrumab — an investigational monoclonal antibody targeting activin type II receptors, which block myostatin and related negative muscle regulators — has demonstrated in a 75-person metabolic trial that simultaneous fat mass reduction and lean mass increase is pharmacologically achievable. The body composition profile produced by bimagrumab — increased lean mass alongside reduced fat mass — is unusual among current obesity therapies, and combination studies with GLP-1 receptor agonists are underway. Bimagrumab remains investigational and is not yet available for clinical use in India as of 2026.⁸

8.9 — The Complete Muscle Assessment Panel

The Dr. Nehru Method muscle assessment panel for patients above 40 years with HOMA-IR above 2.3 proceeds in a stepwise sequence from zero-cost functional tests to laboratory investigations to body composition imaging. Grip strength using a Jamar or equivalent dynamometer and the Timed Up and Go test require only standard clinic equipment and produce immediate results at effectively zero per-test cost after device acquisition. These two tests should be performed at every metabolic clinic visit and serve as the most sensitive early indicators of functional muscle decline. BIA using an InBody or equivalent bioimpedance device provides SMI, visceral fat area, and phase angle at a cost of approximately Rs 500 to 2,000 at diagnostic centres in Hyderabad and should be performed every three months in patients under active sarcopenia management. DEXA should be used to confirm sarcopenia diagnosis when BIA results are borderline, to provide definitive appendicular lean mass measurement, and for annual assessment in confirmed sarcopenic patients. The laboratory panel for sarcopenia evaluation should include HOMA-IR and TyG as primary metabolic markers, C-peptide to assess beta cell reserve, 25-hydroxyvitamin D as Vitamin D deficiency directly impairs muscle function through VDR-mediated mechanisms, B12 as methylcobalamin deficiency causes neuromuscular dysfunction and is particularly prevalent in Indian patients on long-term metformin, ferritin as iron deficiency impairs mitochondrial

function and exercise capacity, and morning testosterone in men above 40 years as hypogonadism is a treatable hormonal driver of sarcopenia that is systematically underinvestigated in Indian metabolic practice.

Clinical Summary — Chapter 8

- Sarcopenia is defined by AWGS 2019 as low muscle mass plus low muscle strength or low physical performance. SMI below 7.0 men and below 5.4 women by DEXA. SMI below 5.7 women by BIA. Grip below 27.5 men by Indian consensus 2025.
- SMI is calculated as appendicular lean mass divided by height squared. DEXA and BIA cutoffs differ for women and must not be interchanged.
- HOMA-IR above 2.3 impairs muscle protein synthesis through mTORC1 disruption. Sarcopenia worsens insulin resistance by reducing peripheral glucose disposal. The relationship is bidirectional.
- Protein target 1.2 to 1.6g per kg per day for IR patients. 1.6 to 2.0g for confirmed sarcopenia. These are proposed clinical targets — not ICMR mandated thresholds.
- High serum BCAAs are a marker of insulin resistance — not a dietary excess. Treat insulin resistance first. Combine protein with resistance exercise.
- Irisin is confirmed lower in sarcopenia — human meta-analysis 2025. Myostatin inhibits muscle growth — confirmed. Direct beta cell effects of both are experimental only.
- Resistance training three times weekly reduces HOMA-IR — RCT confirmed. Minimum eight weeks before reassessing SMI.
- Sarcopenic obesity — low muscle plus high fat at normal BMI — TOFI phenotype. BIA identifies it. DEXA confirms it. Standard BMI misses it entirely.
- Bimagrumab — investigational anti-myostatin antibody — increased lean mass alongside fat reduction in N=75 metabolic trial — not yet available India 2026.

Multiple Choice Questions

1. The AWGS 2019 SMI cutoff for women using DEXA is:
 - A. Below 5.7 kg/m²
 - B. Below 5.4 kg/m²
 - C. Below 6.0 kg/m²
 - D. Below 4.9 kg/m²
2. Which myokine is confirmed lower in sarcopenia by human observational data and a 2025 meta-analysis?
 - A. Myostatin
 - B. IL-6
 - C. Irisin
 - D. Apelin
3. High serum BCAAs in insulin-resistant patients represent:
 - A. Excessive dietary protein intake
 - B. A marker of impaired muscle amino acid uptake
 - C. Direct cause of beta cell toxicity
 - D. A sign of normal metabolic health
4. The Indian Sarcopenia Consensus 2025 grip cutoff for men is:
 - A. Below 30 kg
 - B. Below 28 kg
 - C. Below 27.5 kg
 - D. Below 25 kg
5. Sarcopenic obesity in a patient with normal BMI is best identified by:
 - A. BMI and waist circumference alone
 - B. BIA or DEXA body composition assessment
 - C. Serum myostatin level
 - D. HbA1c and fasting glucose

Answers: 1. B 2. C 3. B 4. C 5. B

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CHAPTER 9

The Gut-Liver-Pancreas Axis

The Gut–Liver–Pancreas Axis: A Major Upstream Regulator of Metabolic Health

Section 1 — Introduction

Clinicians trained in the classical model of type 2 diabetes mellitus view the pancreatic beta cell as the central protagonist of metabolic failure. Insulin deficiency, progressive beta cell exhaustion, and glucose toxicity have dominated the teaching curriculum for decades. While these mechanisms remain fundamentally important, they represent a downstream consequence rather than the initiating event in the majority of patients presenting in clinical practice today.

The gut–liver–pancreas axis has emerged as one of the most significant upstream regulators of whole-body metabolic health. This is not a simple linear pathway. It is an integrated, bidirectional network involving endocrine signalling, bacterial metabolite production, immune activation, portal nutrient sensing, and hepatic metabolic programming. Understanding this axis is not an academic luxury — it is essential clinical knowledge for any practitioner managing insulin resistance, type 2 diabetes, obesity, or non-alcoholic fatty liver disease.



Figure 9.1 — The gut-liver-pancreas axis and portal circulation.

A critical framing principle must be stated clearly at the outset: the gut–liver–pancreas axis is a major upstream regulator — it is not the sole cause of diabetes or metabolic syndrome. The pancreas is not merely a passive victim of gut dysfunction. It is a dynamic organ that participates actively in metabolic regulation and deteriorates under sustained metabolic stress regardless of whether gut signals are the primary driver. Nor is this a single pathway. Multiple mechanisms operate simultaneously, reinforcing one another in a web of interaction that must be appreciated in its totality.

The Dr. Nehru Method of clinical metabolic assessment situates the gut–liver–pancreas axis as one of the eight major upstream triggers in the Indian Health Constitution framework — specifically converging on the pathways of insulin resistance as quantified by HOMA-IR and the Triglyceride-Glucose Index. This chapter delineates the anatomical, endocrine, microbial, and nutritional dimensions of this axis and their relevance to the clinical management of metabolic disease in the Indian population.

Section 2 — Portal Circulation Anatomy: Why the Liver Sees Everything First

To appreciate the primacy of the liver in metabolic regulation, one must begin with anatomy. The venous drainage of the gastrointestinal tract — from the stomach, small intestine, large intestine, pancreas, and spleen — converges into the portal vein. This single venous conduit delivers the entirety of gut-absorbed nutrients, gut-derived hormones, bacterial metabolites, and microbial translocation products directly to the hepatic sinusoids before they enter the systemic circulation.

The liver is therefore the first major filter for gut-derived signals. Every meal consumed by a patient results in a portal surge — a bolus delivery of glucose, amino acids, short-chain fatty acids, gut hormones, bacterial lipopolysaccharide fragments, and other bioactive molecules arriving at hepatic parenchymal cells before any other organ is exposed. The liver must process, respond to, and often detoxify these signals within minutes.

There is one important anatomical exception that is frequently overlooked in clinical teaching. Dietary lipids — triglycerides, phospholipids, and fat-soluble vitamins absorbed in the small intestine — are not transported via the portal vein. They are packaged into chylomicrons within intestinal epithelial cells and secreted into the mesenteric lymphatics. These chylomicrons travel through the thoracic duct and enter the systemic circulation via the left subclavian vein, bypassing hepatic first-pass metabolism. This has important clinical implications: postprandial hypertriglyceridaemia and chylomicronaemia arise from lymphatic rather than portal lipid transport. The liver encounters these lipids only after lipoprotein lipase has partially hydrolysed the chylomicrons to form chylomicron remnants.

Understanding this anatomical architecture explains why the liver occupies its central position in the metabolic axis. Hepatic insulin resistance — the inability of insulin to suppress hepatic glucose production — is one of the earliest measurable metabolic defects in patients with insulin resistance. The TyG Index above 8.82 on the natural logarithm scale, as validated in published Indian cohorts by Nehru MS et al. in IJCPR 2026, serves as an operational proxy marker of hepatic and systemic insulin resistance in clinical practice.

Clinical Insight: Dietary fats travel via lymphatics — not portal vein. Postprandial hypertriglyceridaemia reflects chylomicron load, not portal lipid delivery. This distinction matters when counselling patients on meal composition and postprandial glucose versus lipid management.

Section 3 — The Incretin System: Gut-Derived Amplification of Insulin Secretion

The incretin effect is one of the most clinically important and therapeutically exploited phenomena in modern diabetology. The term describes the observation that oral glucose elicits a substantially greater insulin secretory response than the same quantity of glucose administered intravenously. The difference — attributable to gut-derived hormones released in response to nutrient ingestion — accounts for approximately half to two-thirds of total postprandial insulin secretion in healthy individuals. This is high-grade evidence, well established through decades of metabolic physiology research.



Figure 9.2 — Incretin signalling and the gut microbiome in metabolic regulation.

Two hormones are principally responsible for the incretin effect:

Glucagon-Like Peptide-1 (GLP-1) is secreted by L cells located predominantly in the distal ileum and colon, although L cells are distributed throughout the gut. GLP-1 is released within minutes of nutrient exposure — particularly in response to carbohydrates and fats — and acts on pancreatic beta cells to augment glucose-dependent insulin secretion. The critical word is glucose-dependent: GLP-1 enhances insulin release only when blood glucose is elevated, providing an intrinsic safety mechanism against

hypoglycaemia. GLP-1 additionally suppresses glucagon secretion, delays gastric emptying, and promotes satiety through central nervous system signalling.

Glucose-Dependent Insulinotropic Peptide (GIP) is secreted by K cells located in the duodenum and jejunum. GIP was the first incretin hormone identified and shares with GLP-1 the property of glucose-dependent insulin stimulation. However, in individuals with established type 2 diabetes, the GIP response is often preserved while the GLP-1 response is attenuated — a finding that partly explains why GLP-1 receptor agonists have proven more effective as therapeutic agents than GIP agonists alone in this population.

The clinical importance of the incretin system extends beyond pharmacology. When gut anatomy is disrupted — as in bariatric surgery, small intestinal disease, surgical resection, or severe dysbiosis — incretin secretion is altered. Patients with rapid gastric emptying may have exaggerated GLP-1 responses and postprandial hypoglycaemia. Those with proximal gut disease may have impaired GIP secretion. These nuances are directly relevant to the management of diabetic patients with concurrent gastrointestinal pathology.

From a pharmacological standpoint, GLP-1 receptor agonists — including liraglutide, semaglutide, and dulaglutide — represent the most direct translation of incretin physiology into clinical therapeutics. Their efficacy in reducing HbA1c, body weight, and cardiovascular events is now supported by robust trial evidence. The mechanism of this drug class is not pharmacological novelty — it is restoration and amplification of a physiological gut-derived signal that is diminished in the setting of type 2 diabetes.

Section 4 — Gut Microbiome and Insulin Resistance: Signal, Mechanism, and Evidence Grade

The human gut harbours an estimated 38 trillion microorganisms — a community collectively referred to as the gut microbiome — whose metabolic output rivals that of the liver in biochemical diversity. The relationship between gut microbiome composition and insulin resistance has been one of the most intensively investigated areas of metabolic medicine over the past two decades. While early enthusiasm sometimes outpaced

evidence, a clearer picture has emerged regarding what is confirmed, what is proposed, and what remains speculative.

Akkermansia muciniphila

Akkermansia muciniphila is a mucin-degrading bacterium that resides in the mucous layer of the intestinal epithelium. Multiple observational studies have confirmed that *Akkermansia* abundance is lower in individuals with insulin resistance, obesity, and type 2 diabetes compared to metabolically healthy controls. Intervention studies in murine models demonstrated that supplementation with *Akkermansia* improved metabolic parameters. The observational data in humans is confirmed and reproducible, though causality and the precise mechanisms through which *Akkermansia* modulates insulin sensitivity in humans remain under active investigation.

Butyrate-Producing Bacteria

Roseburia intestinalis and *Faecalibacterium prausnitzii* are among the best-characterised butyrate-producing species in the human gut. Both have been consistently associated with better metabolic profiles — lower insulin resistance, reduced systemic inflammation, and healthier lipid parameters — in observational human data. Whether these associations reflect a causal protective role or simply a shared metabolic environment that favours both butyrate producers and insulin sensitivity is an important unresolved question.

Indian Microbiome Data

The gut microbiome composition of Indian populations differs substantially from the Western cohorts in which much of the foundational microbiome-metabolism research was conducted. The landmark study by Dhakan DB et al. published in *GigaScience* in 2019 characterised the gut microbiome across geographically and dietary-diverse Indian populations. A consistent finding was the enrichment of *Prevotella* in plant-based dietary patterns — a bacterial genus associated with carbohydrate fermentation and SCFA production. This Indian-specific microbiome architecture has implications for interpreting metabolic risk in the context of traditionally high-carbohydrate dietary patterns prevalent across South Asia.

The LPS–TLR4–NF-κB Pathway

The most mechanistically proposed explanation for how gut dysbiosis drives systemic insulin resistance involves lipopolysaccharide (LPS) — a structural component of the outer membrane of gram-negative bacteria. When gut barrier integrity is disrupted — a condition termed intestinal hyperpermeability or "leaky gut" — small quantities of LPS translocate across the epithelium into the portal circulation. This low-grade systemic LPS exposure, termed metabolic endotoxaemia, was first described by Cani PD et al. in a seminal 2007 Diabetes paper.

The proposed mechanism is as follows: LPS activates Toll-Like Receptor 4 (TLR4) on hepatocytes, adipocytes, and skeletal muscle cells, triggering NF-κB activation and downstream pro-inflammatory cytokine production — including TNF-α, IL-6, and IL-1β. These cytokines impair insulin receptor signalling at the post-receptor level, specifically through serine phosphorylation of insulin receptor substrate-1 (IRS-1), effectively creating inflammation-mediated insulin resistance.

This is a well-supported and plausible mechanism with considerable experimental backing in animal models. However, it must be labelled accurately: the LPS–TLR4–NF-κB pathway is a proposed, well-supported mechanism. It has not been proven to be universally operative in all human subjects with insulin resistance, and the clinical significance of metabolic endotoxaemia in individual patients remains difficult to quantify without specialised assays not available in routine practice.

Evidence Grading Note: Akkermansia and butyrate associations — confirmed observational data. Indian Prevotella enrichment — confirmed. LPS–TLR4–NF-κB pathway — proposed, well-supported, not universally proven. Clinicians should communicate this hierarchy to patients and in academic presentations.

Section 5 — Short-Chain Fatty Acids: Metabolic Effectors from the Colon

Short-chain fatty acids (SCFAs) — principally butyrate, propionate, and acetate — are produced through anaerobic bacterial fermentation of dietary fibre in the colon. They represent the primary energy substrate for

colonocytes, and their systemic metabolic effects have attracted considerable research interest as potential mediators of the gut–muscle and gut–liver metabolic axes.

Butyrate, the most extensively studied SCFA, is proposed to activate AMP-activated protein kinase (AMPK) in skeletal muscle. AMPK is the master energy sensor of the cell — its activation promotes glucose uptake, fatty acid oxidation, and mitochondrial biogenesis. If butyrate reliably activates AMPK in skeletal muscle through portal or systemic delivery, it would represent a mechanistic link between gut microbial activity and peripheral insulin sensitivity.

Propionate is primarily transported to the liver via the portal vein, where it participates in gluconeogenic substrate supply and may have regulatory effects on hepatic lipid metabolism. Acetate enters the systemic circulation and has been proposed to modulate hypothalamic appetite regulation and adipose tissue metabolism.

SCFAs may also enhance GLUT4 expression and translocation in skeletal muscle, augmenting glucose uptake independently of insulin. This mechanism, if operative in humans at physiologically relevant SCFA concentrations, would constitute a significant insulin-sensitising effect arising from dietary fibre fermentation.

The evidence grade for SCFA-mediated metabolic effects must be clearly labelled: proposed mechanism, moderate evidence. The animal and in vitro data are compelling. Human intervention studies using dietary fibre supplementation to increase SCFA production have shown metabolic benefits, but confounding variables — dietary pattern, concurrent prebiotic effects, caloric intake — make it difficult to isolate SCFA effects specifically. The AMPK activation mechanism in skeletal muscle by butyrate, while mechanistically coherent, has not been established definitively in human in vivo studies at physiologically achievable gut SCFA concentrations.

In clinical practice, the actionable implication is straightforward: dietary patterns that promote SCFA production — high in diverse plant fibre, legumes, resistant starch, and fermented foods — are associated with better metabolic outcomes. Whether this reflects SCFA effects, microbiome changes, dietary displacement of refined carbohydrates, or a combination of

all three is less important than the clinical message: fibre-rich whole-food dietary patterns belong at the centre of the metabolic prescription for insulin resistance.

Section 6 — The Gut–Liver Axis and Non-Alcoholic Fatty Liver Disease

Non-alcoholic fatty liver disease (NAFLD) — now increasingly termed Metabolic-Associated Fatty Liver Disease (MAFLD) — represents one of the most prevalent manifestations of insulin resistance in the Indian population. The relationship between the gut and the liver in NAFLD pathogenesis is intimate and bidirectional, but must be understood within its appropriate causal hierarchy.

The gut is one contributor to NAFLD — it is not the sole cause. Hepatic de novo lipogenesis (DNL) — the conversion of excess carbohydrate substrate, principally fructose and glucose, into triglycerides within the hepatocyte — is a central and quantitatively significant driver of hepatic steatosis. Adipose tissue lipolysis, particularly from visceral adipose depots, delivers non-esterified fatty acids directly to the liver via the portal circulation and constitutes another major source of hepatic fat accumulation. The gut contributes through metabolic endotoxaemia, altered bile acid signalling, and dysbiosis-associated changes in gut permeability, but these operate in concert with — not instead of — the lipogenic and lipolytic pathways.

Boursier J et al., published in *Hepatology* 2016, demonstrated that specific gut microbiome signatures — particularly enrichment of *Ruminococcus* — were independently associated with more severe histological disease in NAFLD patients, suggesting that gut dysbiosis contributes to disease severity beyond baseline metabolic risk factors.

In clinical assessment of patients with suspected hepatic insulin resistance, the TyG Index serves as a validated operational marker. The threshold of TyG above 8.82 on the natural logarithm scale — derived from published Indian cohorts and reported in Nehru MS et al., *IJCPR* 2026 — provides a practical bedside tool for identifying patients with significant insulin resistance where hepatic mechanisms are likely to be operative. This must be computed on the natural logarithm scale: the formula $\ln(\text{fasting triglycerides in mg/dL} \times \text{fasting glucose in mg/dL} / 2)$, with the Indian cutoff of 8.82, not

the occasionally cited values of 4.x or 5.x that derive from different computational conventions.

Dr. Nehru Method Operational Threshold: TyG Index > 8.82 (natural log scale) as marker of hepatic and systemic insulin resistance — validated in Indian published cohorts (Nehru MS, IJCPR 2026). Always use natural logarithm computation.

HOMA-IR above 2.3 — the India-specific cutoff validated in published Indian research — complements TyG assessment and provides a direct estimate of insulin-mediated glucose disposal capacity. Together, these two indices offer a clinically deployable framework for metabolic risk stratification without requiring liver biopsy or advanced imaging in the majority of patients.

Section 7 — Micronutrient Absorption, Gut Disease, and the Metabolic Cascade

The gut is not only a site of energy substrate absorption — it is the primary interface through which micronutrients essential for metabolic health enter the body. Disruption of gut function therefore has profound consequences beyond macronutrient handling, with important implications for the insulin resistance–diabetes continuum.

Vitamin B12: Intrinsic Factor Dependency and Metformin Interference

Vitamin B12 absorption is a uniquely complex process governed by intrinsic factor — a glycoprotein secreted by parietal cells of the gastric fundus. Dietary B12, primarily from animal-derived foods, binds to intrinsic factor in the stomach and the complex is subsequently recognised by specific cubilin receptors in the terminal ileum, where absorption occurs. This anatomical specificity means that any pathology affecting the gastric mucosa, the terminal ileum, or the intrinsic factor molecule itself will impair B12 absorption — regardless of dietary intake.

Metformin — the most widely prescribed glucose-lowering agent globally and the cornerstone of therapy in type 2 diabetes — interferes with B12 absorption through a distinct mechanism. Metformin reduces calcium-

dependent binding of the intrinsic factor–B12 complex to ileal cubilin receptors. This effect is dose-dependent and cumulative, resulting in clinically significant B12 depletion in a substantial proportion of long-term metformin users. Monitoring of serum B12 — and ideally serum methylmalonic acid and homocysteine as functional markers — is essential in all patients on sustained metformin therapy.

In the Dr. Nehru Method, supplementation is always with methylcobalamin — the active neurological form of B12 that crosses the blood-brain barrier and is directly utilised by the nervous system without requiring hepatic conversion. Cyanocobalamin is not contraindicated and has an established pharmacological history, but methylcobalamin is the preferred form in clinical practice given its physiological superiority and direct neurological bioavailability.

Vitamin D: Bile Acid Dependency and Fat-Soluble Absorption

Vitamin D is a fat-soluble vitamin and prohormone whose intestinal absorption is dependent on the presence of bile acids. Bile acids, secreted by the liver and stored in the gallbladder, are essential for the formation of mixed micelles that solubilise dietary fat and fat-soluble vitamins in the intestinal lumen, enabling their uptake by enterocytes. Any condition that impairs bile acid availability — including cholestatic liver disease, surgical biliary diversion, or severe ileal disease causing bile acid malabsorption — will impair Vitamin D absorption regardless of sun exposure or oral supplementation.

In the Indian population, Vitamin D deficiency is endemic and multifactorial — driven by dietary insufficiency, limited sun exposure in urban settings, and dark skin melanin reducing cutaneous synthesis. When gut disease compounds this background deficiency, the resulting severe Vitamin D depletion acts as an additional upstream driver of insulin resistance through impairment of pancreatic beta cell function, immune dysregulation, and reduced insulin receptor expression.

SIBO and H. pylori: Incretin Secretion and Absorption Disruption

Small Intestinal Bacterial Overgrowth (SIBO) — defined as excessive bacterial colonisation of the proximal small intestine — and *Helicobacter pylori* infection represent two highly prevalent gut pathologies in the Indian population that intersect meaningfully with metabolic health.

SIBO disrupts the anatomical compartmentalisation that normally ensures nutrients contact L cells and K cells in the appropriate regions of the gut. Premature fermentation of carbohydrates in the proximal small intestine may alter nutrient sensing and incretin secretion patterns. SIBO is also associated with bile acid deconjugation, fat malabsorption, and SCFA overproduction in the small intestine — a combination that impairs fat-soluble micronutrient absorption and may promote gut permeability.

H. pylori infection, affecting an estimated 40–70% of adult Indians, causes chronic active gastritis and in a subset of patients, progressive gastric atrophy with loss of parietal cell mass. Loss of parietal cells directly reduces intrinsic factor secretion, impairing B12 absorption through the mechanism described above. Additionally, *H. pylori*-induced gastric acid suppression alters the upper gastrointestinal microenvironment in ways that may affect proximal incretin secretion. Eradication of *H. pylori* is therefore not merely a gastroenterological intervention — it is, in patients with concurrent metabolic disease, a metabolic intervention with measurable downstream effects.

Clinical Synthesis: Integrating the Gut–Liver–Pancreas Axis into the Dr. Nehru Method

The clinical practitioner managing a patient with insulin resistance or type 2 diabetes must resist two errors of cognitive simplification. The first is to ignore the gut–liver–pancreas axis entirely — to treat diabetes as purely a pancreatic failure requiring glucose-lowering agents without ever interrogating the upstream drivers. The second is to overstate gut causality — to frame every case of insulin resistance as a microbiome disorder requiring probiotic correction, obscuring the primary roles of visceral adiposity, hepatic lipogenesis, physical inactivity, and carbohydrate excess.

The truth lies in integration. In the Indian Health Constitution framework of the Dr. Nehru Method, the gut–liver–pancreas axis constitutes one of the eight major upstream triggers that converge on the final common pathway of HOMA-IR elevation above 2.3 and TyG Index elevation above 8.82. A patient presenting with these metabolic markers may have reached this threshold through visceral obesity, high glycaemic carbohydrate intake, B12 deficiency impairing neural metabolic regulation, sarcopenia reducing peripheral glucose disposal, or gut dysbiosis promoting metabolic endotoxaemia — or, most commonly, several of these acting simultaneously.

The therapeutic response must be correspondingly multidimensional: dietary fibre prescription to enhance SCFA production and diversify microbiome composition; B12 repletion with methylcobalamin in all metformin-treated patients; Vitamin D correction in deficient individuals; *H. pylori* eradication when present; GLP-1 receptor agonist therapy as a pharmacological amplification of the physiological incretin signal; and reduction of ultra-processed food intake to lower the LPS burden on the portal circulation.

No single intervention repairs the entire axis. But understanding the axis allows the clinician to prescribe with mechanistic clarity — to know why each element of the prescription is justified and how it connects to the pathophysiology of the individual patient sitting across the consultation table.

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CHAPTER 10

NAFLD, NASH and the Liver

Chapter 10 — NAFLD, NASH and the Liver in Metabolic Disease

10.1 — Introduction and Nomenclature Update

For decades, the accumulation of fat in the liver in the absence of significant alcohol consumption was termed Non-Alcoholic Fatty Liver Disease — NAFLD — with its inflammatory subtype known as Non-Alcoholic Steatohepatitis, NASH. This exclusionary nomenclature failed to capture the primary pathophysiological driver — systemic metabolic dysfunction. In a 2023 multisociety Delphi consensus, the nomenclature was updated. NAFLD was renamed Metabolic Dysfunction-Associated Steatotic Liver Disease — MASLD — and NASH was updated to Metabolic Dysfunction-Associated Steatohepatitis — MASH. This framework requires the presence of hepatic steatosis alongside at least one cardiometabolic risk factor, and retains steatohepatitis as a meaningful pathophysiological term.¹



Figure 10.1 — Progression from MASLD to MASH and fibrosis.

10.2 — Global Epidemiology

MASLD is a global pandemic advancing alongside the rising prevalence of obesity and type 2 diabetes. The foundational meta-analysis by Younossi and colleagues established the scale of the disease. Analysing 92 studies with a combined sample size of 9,361,716 participants across three decades, the global prevalence of NAFLD/MASLD was estimated at 30.05 percent, with 95 percent confidence interval 27.88 to 32.32 percent.² This makes MASLD the most common chronic liver condition worldwide, and its prevalence parallels the rising burden of insulin resistance and central obesity in populations transitioning to Western dietary patterns.

Nearly one in three adults globally now meets criteria for MASLD — a figure that should reframe fatty liver from an incidental ultrasound finding to a central marker of systemic metabolic risk.

India-specific large-scale prevalence data using consistent diagnostic criteria across multiple centres remains an area requiring further dedicated study. While regional Indian cohorts have reported high rates of hepatic steatosis among patients with central obesity and insulin resistance, readers should seek the most current published multicentre Indian epidemiological data before citing specific prevalence figures, as estimates vary considerably by region, diagnostic method, and population studied.

10.3 — Non-Invasive Diagnostics — Fibroscan and CAP Score

Liver biopsy remains the definitive standard for staging MASLD, but its invasive nature, cost, and sampling variability make it impractical for widespread screening. Non-invasive modalities are therefore central to clinical practice. Transient elastography — Fibroscan — is the cornerstone of non-invasive evaluation, providing two distinct metrics. Liver Stiffness Measurement uses shear wave velocity to estimate hepatic fibrosis stage. Controlled Attenuation Parameter measures the attenuation of ultrasound waves as they pass through liver parenchyma.

[IMAGE PLACEHOLDER — Figure 10.2]

Figure 10.2 — Fibroscan / Controlled Attenuation Parameter staging of hepatic steatosis.

CAP is a surrogate marker that correlates with histologic steatosis grade — it does not provide a literal measurement of the percentage of fat-laden hepatocytes, but rather an acoustic signal that has been validated against histological steatosis categories such as S1, mild steatosis, through S3, severe steatosis. Clinicians should understand CAP as a strong correlate of steatosis burden rather than a direct cellular count.³

10.4 — Pathophysiology — The Gut-Liver Axis

While insulin resistance remains the primary driver of MASLD, the gut-liver axis is increasingly recognised as an important contributory mechanism in the progression from simple steatosis to steatohepatitis. Intestinal dysbiosis can impair gut barrier integrity, permitting translocation of lipopolysaccharide — LPS — derived from Gram-negative bacterial cell walls. LPS entering the portal circulation binds Toll-like receptor 4 on hepatic Kupffer cells, triggering the NF-kB signalling cascade and the release of pro-inflammatory and pro-fibrotic cytokines including TNF-alpha and IL-6. This pathway is best understood as one significant contributory mechanism within a multifactorial disease process — not the sole or universally proven driver of steatohepatitis in every patient.⁴

10.5 — MASLD as a Cardiovascular Risk Marker

Patients with MASLD are more likely to die from cardiovascular disease than from liver-related complications. The relationship between the liver and the cardiovascular system extends beyond shared metabolic risk factors. Targher and colleagues established that NAFLD is associated with increased cardiovascular disease risk, with some analyses suggesting this association persists independent of traditional risk factors, though the degree of independence continues to be studied and refined in the literature.⁵

Fatty liver is not a bystander finding — it is a marker that the patient's cardiovascular risk profile deserves close attention, often before lipid panels or blood pressure reveal the same warning.

Surrogate markers for regional fat distribution, including DEXA-derived android fat measurement, have been proposed in the literature as plausible correlates for both hepatic steatosis and insulin resistance as measured by HOMA-IR. This relationship is biologically coherent given the shared origin of android and visceral fat depots, though clinicians should seek dedicated validation studies for android fat as a hepatic steatosis surrogate in their specific patient population before relying on it as a stand-alone diagnostic proxy.

10.6 — Pharmacological Interventions

10.6.1 — Saroglitazar

Saroglitazar is a dual PPAR-alpha and PPAR-gamma agonist developed and used in India, with an established indication for diabetic dyslipidaemia and hypertriglyceridaemia. The EVIDENCES II trial — a phase 2 study of 106 patients — demonstrated that the 4mg dose significantly improved ALT, liver fat content, and insulin resistance markers in patients with NAFLD/NASH. The PRESS V and PRESS VI trials established Saroglitazar's efficacy in lipid control, with PRESS V comparing Saroglitazar to pioglitazone and PRESS VI focusing on patients unresponsive to statin therapy. The exact current CDSCO-approved indication for Saroglitazar specifically in NASH, as distinct from its established dyslipidaemia indication, should be verified against the current official product label before being stated as definitive regulatory fact, as approval status and labelling can change and vary by formulation.⁶

10.6.2 — Resmetirom

In March 2024, the US FDA granted accelerated approval to Resmetirom, a thyroid hormone receptor-beta agonist, for MASH with moderate to advanced liver fibrosis, corresponding to fibrosis stages F2 to F3. In the MAESTRO-NASH phase 3 trial — Harrison SA et al., *New England Journal of Medicine*, 2024, N=966 — MASH resolution without worsening of fibrosis was achieved in 25.9 percent of patients on the 80mg dose and 29.9 percent on the 100mg dose, compared with 9.7 percent on placebo.⁷ As of 2026, Resmetirom is not yet widely available in India through standard commercial channels, and Indian clinicians should confirm current import and access protocols before discussing this option with patients.

10.6.3 — Tirzepatide and Incretin-Based Therapies

Incretin-based therapies offer substantial benefit for steatosis resolution. Tirzepatide, a dual GIP and GLP-1 receptor agonist, has shown notable efficacy in the SYNERGY-NASH trial — Loomba R et al., *New England Journal of Medicine*, 2024, N=190 patients with biopsy-proven MASH at fibrosis stage F2 or F3. MASH resolution without worsening of fibrosis was

achieved in 51.8 percent of patients on the 5mg dose, 62.8 percent on the 10mg dose, and 73.3 percent on the 15mg dose, compared with 13.2 percent on placebo. Fibrosis improvement of one stage or greater occurred in 59.1 percent, 53.3 percent, and 54.2 percent of patients across the three dose groups respectively, compared with 32.8 percent on placebo.⁸

Tirzepatide achieved MASH resolution in nearly three out of four patients at the highest dose — a result that may reshape the treatment hierarchy for steatohepatitis once broader access and India-specific cost data become available.

10.6.4 — SGLT2 Inhibitors and Liver Fat Reduction

Emerging data support the use of SGLT2 inhibitors, including dapagliflozin and empagliflozin, for liver fat reduction independent of weight loss. Trials evaluating empagliflozin added to standard of care in patients with type 2 diabetes and MASLD, assessed via MRI-derived proton density fat fraction, have shown statistically significant reductions in hepatic steatosis over treatment periods of approximately 24 weeks. Similar trials evaluating dapagliflozin using Fibroscan CAP scores have shown comparable reductions in steatosis, supporting SGLT2 inhibitors as a useful adjunctive class in the management of MASLD, particularly in patients who already require glucose-lowering therapy.⁹

10.7 — Practical Clinical Approach

For the Indian clinician managing a patient with suspected or confirmed MASLD, a structured approach should integrate metabolic staging with hepatic assessment. HOMA-IR above 2.3 and TyG above 8.82 on the natural log scale — the Dr. Nehru Method operational thresholds derived from published Indian cohorts — should prompt active screening for hepatic steatosis even in the absence of elevated transaminases, since a substantial proportion of patients with significant liver fat have normal liver enzymes.¹⁰¹¹ Fibroscan with CAP and LSM should be the first-line non-invasive assessment where available, with android fat on DEXA serving as a useful adjunctive marker. Pharmacological choice should be individualised — SGLT2 inhibitors and GLP-1 or dual incretin agonists are reasonable first considerations in patients with concurrent diabetes, while Saroglitazar remains a practical option within its established Indian regulatory indication.

Resmetirom represents an important option once broader Indian availability is established.

Clinical Summary — Chapter 10

- NAFLD renamed MASLD, NASH renamed MASH — 2023 multisociety Delphi consensus — requires steatosis plus one cardiometabolic risk factor.
- Global MASLD prevalence 30.05% — Younossi meta-analysis — 92 studies — N=9,361,716.
- CAP score correlates with histologic steatosis grade — not a literal fat cell count — surrogate marker only.
- Gut-liver axis via LPS-TLR4-NF-kB — important contributory mechanism — not sole proven driver in every patient.
- MASLD associated with increased cardiovascular risk — Targher et al. — degree of independence from traditional risk factors continues to be studied.
- Resmetirom — FDA approved March 2024 — MAESTRO-NASH N=966 — MASH resolution 25.9% to 29.9% vs 9.7% placebo — not yet widely available India.
- Tirzepatide — SYNERGY-NASH N=190 — MASH resolution up to 73.3% at highest dose vs 13.2% placebo — most effective agent currently reported.
- Saroglitazar — established India dyslipidaemia indication — EVIDENCES II N=106 — verify current exact NASH labelling before stating as regulatory fact.
- SGLT2 inhibitors reduce liver fat independent of weight loss — useful adjunct in diabetic patients with MASLD.

Multiple Choice Questions

1. NAFLD and NASH were renamed by the 2023 Delphi consensus to:
A. Fatty Liver Syndrome and Hepatic Steatosis Disorder
B. MASLD and MASH C. Metabolic Liver Disease and Steatofibrosis
D. Type 2 Hepatopathy
2. The global MASLD prevalence reported by Younossi et al. meta-analysis was approximately:

A. 15% B. 22% C. 30% D. 45%

3. Controlled Attenuation Parameter on Fibroscan is best described as:

- A. A direct count of fat-laden hepatocytes
- B. A surrogate marker correlating with histologic steatosis grade
- C. A measure of liver fibrosis only
- D. An invasive biopsy alternative requiring sedation

4. In the SYNERGY-NASH trial, the highest tirzepatide dose achieved MASH resolution in approximately:

A. 13.2% B. 51.8% C. 62.8% D. 73.3%

5. Resmetirom received FDA accelerated approval for MASH in:

A. March 2022 B. March 2023 C. March 2024 D. March 2025

Answers: 1. B 2. C 3. B 4. D 5. C

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CHAPTER 11

Atherogenic Dyslipidaemia

The Triglyceride-HDL Axis and the Indian Cardiovascular Paradox

Introduction

In the standard lipid report handed to an Indian patient, the number that receives the most attention is LDL cholesterol. Physicians adjust statin doses to drive it lower. Guidelines set targets. Patients ask whether their LDL has reached the goal. Yet across India's cardiology wards and coronary care units, a troubling pattern repeats itself daily: patients arrive with myocardial infarctions and strokes carrying LDL values that are entirely within the acceptable range. Their total cholesterol may even appear reassuring. What their reports consistently show, buried lower on the printout, is a triglyceride above 200 mg/dL and an HDL below 40 mg/dL. This is not a coincidence. It is a pattern, and it has a name: atherogenic dyslipidaemia.

Atherogenic dyslipidaemia is not a single lipid abnormality. It is a metabolic cluster — elevated triglycerides, reduced HDL cholesterol, a shift toward small dense LDL particles, and elevated remnant cholesterol — all arising from the same upstream driver: insulin resistance. In the Indian patient, this cluster is the dominant lipid phenotype associated with cardiovascular events. Yet it remains under-recognised, under-measured, and under-treated, in part because our clinical gaze remains fixed on LDL.



Figure 11.1 — Atherogenic dyslipidaemia: high triglycerides, low HDL, small dense LDL.

This chapter argues that the Indian lipid story is fundamentally a triglyceride-HDL story. It presents the evidence base for atherogenic dyslipidaemia, explains the mechanistic pathway through which insulin resistance generates this lipid phenotype, examines why standard LDL-centric therapy fails a substantial proportion of Indian patients, and proposes the role of the

Triglyceride-Glucose Index — the TyG Index — as a bedside marker that captures residual cardiovascular risk where the lipid panel falls short.

The Indian Cardiovascular Paradox

India presents an epidemiological paradox that has puzzled lipidologists for three decades. Indian populations consistently demonstrate lower mean LDL cholesterol compared to Western counterparts, yet carry cardiovascular disease burden that is not only higher in absolute terms but also occurs at a younger age. Indians experience their first myocardial infarction on average a full decade earlier than Europeans. Case-fatality rates are higher. Coronary artery disease is more diffuse rather than focal. The angiographic and autopsy pattern shows a generalised small-vessel disease rather than the dominant single-vessel occlusion more common in Western patients.

How does a population with lower LDL develop more cardiovascular disease? The paradox dissolves when the lipid focus shifts from LDL to the triglyceride-HDL axis. The landmark ICMR-INDIAB study by Joshi and colleagues, examining 16,607 individuals across India, documented that hypertriglyceridaemia and low HDL, not elevated LDL, were the dominant lipid abnormalities in the Indian population.¹ In urban areas, the prevalence of atherogenic dyslipidaemia exceeded 50 percent. Even in rural populations with low total cholesterol and low LDL, the triglyceride-HDL dysregulation was pervasive.

This finding reframes the Indian lipid narrative entirely. The Indian cardiovascular burden is not an LDL problem. It is a metabolic dyslipidaemia problem rooted in insulin resistance, visceral adiposity, and the high glycaemic dietary patterns that characterise the contemporary Indian diet.

Pathophysiology: How Insulin Resistance Generates Atherogenic Dyslipidaemia

The biochemical pathway connecting insulin resistance to atherogenic dyslipidaemia is well established. Under normal insulin signalling, the liver suppresses very low-density lipoprotein (VLDL) production. Adipose tissue, responsive to insulin, suppresses lipolysis and the release of free fatty acids into the portal circulation. Lipoprotein lipase activity in peripheral tissues is maintained, ensuring efficient triglyceride clearance from the bloodstream.

In the insulin-resistant state, each of these regulatory mechanisms fails. Hepatic insulin resistance impairs the suppression of VLDL synthesis; the liver overproduces large, triglyceride-enriched VLDL particles and releases them into the circulation in excess. Adipose tissue, resistant to insulin's antilipolytic effect, releases free fatty acids at an inappropriately high rate, further loading the liver with substrate for VLDL production. Lipoprotein lipase activity is reduced, impairing clearance of triglyceride-rich particles from the plasma.

The consequence is a sustained elevation in plasma triglycerides. These triglyceride-rich VLDL particles do not remain inert; they interact with HDL and LDL particles through the action of cholesterol ester transfer protein (CETP). CETP mediates an exchange: triglycerides move from VLDL into HDL and LDL particles, while cholesterol esters move in the reverse direction. The HDL particles, now triglyceride-enriched, become substrates for hepatic lipase, which hydrolyses them into smaller, denser, catabolically unstable forms. HDL cholesterol concentration falls. HDL particle function — specifically its capacity for reverse cholesterol transport and endothelial protection — is impaired.

The LDL particles undergo a parallel transformation. Triglyceride enrichment followed by lipase action produces small, dense LDL particles. These sdLDL particles are more atherogenic than large buoyant LDL for multiple reasons: they have lower affinity for the LDL receptor, resulting in longer plasma residence time; they penetrate the arterial intima more readily due to their smaller size; they are more susceptible to oxidative modification; and they are more pro-inflammatory in their interaction with vascular endothelial cells. Standard LDL measurement using the Friedewald equation or direct assay measures LDL cholesterol mass, not particle number or size. A patient with predominantly sdLDL may have a deceptively normal LDL-C value while carrying a substantially elevated atherogenic particle burden.

Taskinen and colleagues have provided a comprehensive mechanistic review of how this TG-HDL dyslipidaemia represents an integrated pathophysiological phenotype driven by insulin resistance, with each component amplifying the others in a self-reinforcing cycle.⁵

Remnant Cholesterol: The Missing Atherogenic Fraction

Beyond triglycerides and HDL lies a lipid fraction that has received increasing attention in recent years: remnant cholesterol. Remnant cholesterol represents the cholesterol carried in triglyceride-rich lipoprotein remnants — partially metabolised VLDL, intermediate-density lipoprotein (IDL), and chylomicron remnants. These particles are generated during the incomplete lipolysis of triglyceride-rich lipoproteins, a process that is particularly impaired in states of insulin resistance and high carbohydrate intake.

Remnant particles are directly atherogenic. Unlike LDL, which must undergo modification to enter the arterial wall, remnant particles can be taken up by macrophages directly, contributing to foam cell formation and plaque development. Their cholesterol content per particle is substantially higher than LDL, giving them disproportionate atherogenic potency relative to their circulating concentration.

The Copenhagen General Population Study, reported by Wadstrom and colleagues in 2024, analysed 93,461 individuals and provided robust population-level evidence that remnant cholesterol is an independent predictor of cardiovascular events beyond standard lipid parameters.⁶ This large-scale confirmation of remnant cholesterol's atherogenicity reinforces the mechanistic argument: in the insulin-resistant Indian patient with elevated triglycerides, the remnant cholesterol burden is high, and standard LDL measurement captures none of this risk.

The Failure of LDL-Centric Thinking: Lessons from PROMINENT

If atherogenic dyslipidaemia is real, mechanistically coherent, and epidemiologically prominent, why has the clinical community not moved beyond LDL? Part of the answer lies in the extraordinary success of statin trials in LDL reduction and cardiovascular event reduction. The intellectual infrastructure of lipidology was built on those trials, and LDL became the surrogate marker of cardiovascular risk management.

The Mendelian randomisation work of Ference and colleagues, analysing genetic data from 654,783 individuals, demonstrated convincingly that the magnitude of cardiovascular risk reduction from lipid-lowering interventions is proportional to the absolute reduction in apolipoprotein B — the structural protein present in all atherogenic lipoprotein particles, including VLDL, IDL, LDL, and remnants.² This finding carries a critical implication: triglyceride lowering that does not reduce apoB will not reduce cardiovascular events. Lowering triglycerides without reducing the number of atherogenic particles circulating in the plasma is biochemically insufficient.

[IMAGE PLACEHOLDER — Figure 11.2]

Figure 11.2 — ApoB particle burden as the driver of ASCVD risk.

The PROMINENT trial, reported by Das Pradhan and colleagues in the New England Journal of Medicine in 2022, tested pemafibrate — a selective peroxisome proliferator-activated receptor alpha modulator — in 10,497 patients with type 2 diabetes, mild-to-moderate hypertriglyceridaemia, and low HDL.³ The drug reduced triglycerides by 26 percent, raised HDL, and lowered remnant cholesterol. Yet it produced no reduction whatsoever in major adverse cardiovascular events. The trial was stopped early for futility.

PROMINENT's failure was intellectually clarifying rather than merely disappointing. Pemafibrate reduced triglycerides without meaningfully reducing apoB. The atherogenic particle number remained unchanged. Downstream lipid remodelling without upstream particle reduction delivered no clinical benefit. This mechanistic lesson must inform how we approach atherogenic dyslipidaemia in clinical practice: the therapeutic target is not triglyceride concentration in isolation. It is the insulin-resistant state that drives excess VLDL production and apoB particle burden.

This distinction is particularly relevant in the Indian context, where triglyceride elevation is almost always metabolic in origin — driven by insulin resistance, high glycaemic carbohydrate intake, and visceral adiposity — rather than a primary familial hypertriglyceridaemia. Addressing the metabolic root cause rather than the lipid number is the correct therapeutic logic.

HDL Efflux Capacity: Function Matters More Than Concentration

The relationship between HDL and cardiovascular risk has long been assumed to be simple: higher HDL is protective. Population studies consistently show an inverse relationship between HDL cholesterol and cardiovascular events. Yet multiple large trials attempting to raise HDL pharmacologically — using niacin and CETP inhibitors — failed to reduce cardiovascular events despite substantial increases in HDL concentration.

The resolution to this paradox came from the concept of HDL efflux capacity. HDL's protective role in atherosclerosis operates primarily through reverse cholesterol transport — the process by which HDL accepts cholesterol from lipid-laden macrophages in the arterial wall and carries it back to the liver for excretion. This function is captured by HDL efflux capacity: the ability of HDL particles to accept cholesterol from macrophages in an *ex vivo* assay.

Rohatgi and colleagues, in a landmark study of 2,924 adults published in the *New England Journal of Medicine* in 2014, demonstrated that HDL efflux capacity was a strong inverse predictor of incident cardiovascular disease and carotid intima-media thickness, independent of HDL cholesterol concentration.⁷ Critically, HDL efflux capacity was not strongly correlated with HDL-C. Patients with apparently adequate HDL cholesterol could have markedly impaired efflux capacity, particularly in the setting of insulin resistance, where dysfunctional HDL particles circulate in abundance.

In the Indian insulin-resistant patient, the HDL measured on a standard lipid panel is not merely low in concentration — it is dysfunctional. It has been remodelled through CETP-mediated triglyceride enrichment and lipase-mediated catabolism into particles with impaired efflux capacity. The clinical implication is that raising HDL-C pharmacologically without restoring HDL function will not reduce cardiovascular risk. Restoring insulin sensitivity, reducing VLDL overproduction, and normalising the metabolic milieu in which HDL is synthesised and remodelled is the appropriate intervention.

The TyG Index as a Marker of Atherogenic Dyslipidaemia

The Triglyceride-Glucose Index was originally developed as a surrogate marker of insulin resistance, derived from the product of fasting triglycerides and fasting glucose on a natural logarithmic scale. Its mathematical relationship to atherogenic dyslipidaemia is intrinsic: triglycerides are one of the two components of the index, and the metabolic state that elevates triglycerides simultaneously generates the full atherogenic dyslipidaemia phenotype through the pathways described above.

The India-specific TyG cutoff validated through clinical research at HOMA Healthcare Center is above 8.82 on the natural logarithm scale. This must be expressed exclusively on the natural log scale; the TyG Index is never expressed as a value in the 4.x or 5.x range, which would indicate incorrect calculation methodology. The range of clinical relevance extends from 7.5 at the lower boundary to 10.5 at the upper extreme of metabolic derangement.

Jin and colleagues, in a study of 4,321 individuals published in *Cardiovascular Diabetology* in 2018, demonstrated that the TyG Index was a stronger predictor of subclinical atherosclerosis — measured as carotid intima-media thickness and the presence of carotid plaques — than any individual lipid parameter including LDL, HDL, and total cholesterol.⁴ This finding positions the TyG Index not merely as an insulin resistance marker but as a cardiovascular risk marker in its own right.

Tirosh and colleagues, studying 13,936 young adults in the Israeli Defence Forces, demonstrated prospectively that the TyG Index predicted incident diabetes and metabolic syndrome years before clinical diagnosis, highlighting its utility in the pre-disease phase where intervention is most impactful.⁸ The predictive window offered by the TyG Index extends upstream of LDL dysregulation, capturing the metabolic trajectory toward cardiovascular disease at an earlier and more modifiable stage.

The proposed clinical staging framework applicable to this context, based on the author's experience across thousands of patients managed at HOMA Healthcare Center, suggests that TyG values between 8.82 and 9.5 represent moderate atherogenic risk, values between 9.5 and 10.0 represent high risk with likely established insulin resistance driving dyslipidaemia, and values above 10.0 represent extreme metabolic derangement requiring urgent

multimodal intervention. This staging is a proposed clinical framework based on observed clinical patterns and is not yet an internationally adopted consensus classification.

Remnant Cholesterol and Cardiovascular Outcomes: The Kanjilal Perspective

The Indian population provides a natural study context for atherogenic dyslipidaemia because the phenotype is so prevalent. Kanjilal and colleagues, examining a cohort of 1,321 Indian subjects, demonstrated that the triglyceride-HDL ratio — a proxy for insulin resistance and atherogenic dyslipidaemia — was significantly associated with coronary artery disease risk in Indians at lipid levels that would not trigger intervention under conventional LDL-focused guidelines.⁹ This study is particularly valuable because it is conducted in an Indian population and therefore avoids the extrapolation problem inherent in applying Western lipid thresholds to Indian patients.

The triglyceride-HDL ratio deserves specific attention as a bedside tool because it is calculable from the standard lipid panel without additional cost. A TG-HDL ratio above 3.0 in Indian patients is clinically significant and should prompt evaluation for insulin resistance, visceral adiposity, and metabolic syndrome. When this ratio is elevated in conjunction with a TyG Index above 8.82 and a HOMA-IR above 2.3, the metabolic risk profile is compelling irrespective of the LDL value.

Icosapentaenoic Acid and the REDUCE-IT Exception

The pharmacological management of atherogenic dyslipidaemia took an important turn with the publication of the REDUCE-IT trial. Bhatt and colleagues randomised 8,179 patients with elevated triglycerides despite statin therapy to icosapentaenoic acid (EPA) 4 grams daily or placebo.¹⁰ The EPA group demonstrated a 25 percent relative risk reduction in major adverse cardiovascular events. This was a striking result in a trial where conventional fibrate therapy had repeatedly failed.

The mechanism of benefit from high-dose EPA in REDUCE-IT is not simply triglyceride lowering. EPA has pleiotropic effects on membrane stabilisation, inflammation, platelet aggregation, and endothelial function. Importantly,

high-dose EPA does not raise LDL, unlike mixed omega-3 formulations containing both EPA and DHA. The REDUCE-IT result establishes that for patients on maximally tolerated statin therapy with persistent hypertriglyceridaemia and elevated cardiovascular risk, high-dose EPA is a proven intervention. This is distinct from conventional fibrate therapy and from the failed pemafibrate strategy tested in PROMINENT.

The clinical implication for Indian patients is cautiously optimistic: in those with established cardiovascular disease, statin therapy, and persistent triglyceride elevation as a marker of residual metabolic risk, high-dose EPA represents an evidence-based option. However, the primary intervention in the insulin-resistant Indian patient must remain metabolic — addressing the root cause of VLDL overproduction rather than managing its downstream lipid consequences.

The Dr. Nehru Method: Clinical Approach to Atherogenic Dyslipidaemia

The approach to atherogenic dyslipidaemia that has evolved through clinical practice at HOMA Healthcare Center begins with recognition. The first step is identifying the phenotype: elevated triglycerides, low HDL, and a TyG Index above 8.82 in any patient presenting with metabolic concerns, regardless of LDL value. These patients are not adequately characterised by their LDL; they need a metabolic risk assessment.

The second step is calculating the TG-HDL ratio and the TyG Index from the available fasting lipid profile and glucose. No additional testing is required. These calculations take less than a minute and provide far more prognostic information in the Indian patient than LDL alone.

The third step is assessing insulin resistance directly using the HOMA-IR calculation. A HOMA-IR above 2.3 — the India-specific cutoff validated in published research — confirms that the dyslipidaemia is metabolically driven. This is not a standalone lipid problem requiring a lipid-lowering drug. It is a metabolic problem requiring metabolic intervention.

Metabolic intervention in this context means carbohydrate quality modification before carbohydrate quantity restriction. The high glycaemic index carbohydrate intake that characterises the contemporary Indian diet — white rice consumed in excess, refined wheat flour, sugar-sweetened

beverages, processed snack foods — drives postprandial hyperglycaemia, hyperinsulinaemia, and VLDL overproduction through exactly the mechanism described above. Reducing glycaemic load reduces hepatic triglyceride synthesis, reduces VLDL output, and normalises the TG-HDL axis through metabolic correction rather than pharmacological suppression.

Physical activity, specifically resistance training to address the sarcopenia that coexists with atherogenic dyslipidaemia in many Indian patients, improves insulin sensitivity in skeletal muscle and reduces the hepatic lipid load. Visceral fat reduction through sustained caloric moderation reduces the free fatty acid flux to the liver that drives VLDL overproduction. These metabolic interventions address the apoB particle burden at its origin.

Pharmacological intervention, when required, should be guided by the apoB framework rather than LDL alone. Statins reduce cardiovascular risk primarily through LDL-C reduction, which is an imperfect proxy for apoB reduction. In patients with atherogenic dyslipidaemia where LDL-C is normal or near-normal despite high apoB particle burden, statin therapy may be underestimated in its necessity and underdosed. The TyG Index, used alongside HOMA-IR, helps identify this group.

Clinical Summary: Key Principles

1. Atherogenic dyslipidaemia — elevated TG, low HDL, small dense LDL, elevated remnant cholesterol — is the dominant lipid phenotype in insulin-resistant Indian patients.
2. The Indian cardiovascular paradox — lower LDL, higher CVD — is explained by this TG-HDL axis, not by LDL.
3. The PROMINENT trial confirms that TG lowering without apoB reduction provides no cardiovascular benefit. The metabolic driver must be addressed.
4. HDL function (efflux capacity) matters more than HDL concentration. Pharmacological HDL raising without metabolic correction fails.
5. The TyG Index above 8.82 (natural log scale) is a validated India-specific marker of insulin-resistance-driven atherogenic dyslipidaemia and predicts MACE beyond LDL.

6. High-dose EPA (REDUCE-IT) is a proven adjunct in statin-treated patients with persistent hypertriglyceridaemia, but is not a substitute for metabolic intervention.

7. The Dr. Nehru Method in atherogenic dyslipidaemia: measure TG-HDL ratio and TyG Index, confirm HOMA-IR above 2.3, address carbohydrate quality, reduce visceral fat, restore insulin sensitivity.

Multiple Choice Questions

1. Which lipid pattern characterises atherogenic dyslipidaemia in insulin-resistant Indian patients?

- A. Elevated LDL with normal triglycerides
- B. Elevated triglycerides, low HDL, and small dense LDL
- C. Elevated total cholesterol with normal HDL
- D. Isolated elevation of Lp(a)

2. The ICMR-INDIAB study by Joshi et al. (2014) involving 16,607 Indian subjects demonstrated which of the following?

- A. LDL elevation was the dominant lipid abnormality in urban Indians
- B. Hypertriglyceridaemia and low HDL were the dominant lipid phenotypes
- C. Total cholesterol above 240 mg/dL was the strongest cardiovascular predictor
- D. Statin therapy alone was sufficient for Indian cardiovascular prevention

3. The PROMINENT trial (Das Pradhan et al., NEJM 2022) tested pemafibrate in 10,497 patients with type 2 diabetes and hypertriglyceridaemia. The primary finding was:

- A. Significant reduction in major adverse cardiovascular events
- B. Significant reduction in heart failure hospitalisation
- C. No reduction in cardiovascular events despite triglyceride lowering
- D. Significant reduction in all-cause mortality

4. According to Ference et al. (JAMA 2019, N=654,783), the magnitude of cardiovascular risk reduction from lipid-lowering therapy is proportional to:

- A. The absolute reduction in LDL cholesterol
- B. The absolute reduction in total cholesterol
- C. The absolute reduction in apolipoprotein B

- D. The percentage reduction in triglycerides
5. Rohatgi et al. (NEJM 2014) demonstrated that which HDL parameter was the strongest predictor of cardiovascular events, independent of HDL-C concentration?
- A. HDL particle size
 - B. HDL efflux capacity
 - C. HDL triglyceride content
 - D. HDL apolipoprotein A-I concentration
6. The India-specific TyG Index cutoff for atherogenic risk, expressed on the natural logarithm scale, is:
- A. Above 4.5
 - B. Above 5.0
 - C. Above 8.82
 - D. Above 12.0
7. In the REDUCE-IT trial (Bhatt et al., NEJM 2019), the agent that reduced major adverse cardiovascular events by 25 percent relative risk in statin-treated patients with hypertriglyceridaemia was:
- A. Fenofibrate 145 mg daily
 - B. Pemafibrate 0.2 mg twice daily
 - C. Icosapentaenoic acid 4 grams daily
 - D. Omega-3 fatty acid mixed EPA/DHA 4 grams daily
8. Remnant cholesterol derives primarily from which circulating particles?
- A. LDL and HDL
 - B. VLDL remnants, IDL, and chylomicron remnants
 - C. Lp(a) and sdLDL
 - D. HDL2 and HDL3 subfractions
9. Which mechanism explains why small dense LDL is more atherogenic than large buoyant LDL?
- A. Small dense LDL has higher affinity for the LDL receptor and clears faster
 - B. Small dense LDL penetrates the arterial intima more readily and is more susceptible to oxidation
 - C. Small dense LDL is found only in familial hypercholesterolaemia
 - D. Small dense LDL carries more apolipoprotein E per particle

10. The triglyceride-HDL ratio above 3.0 in an Indian patient indicates which of the following?

- A. Familial combined hyperlipidaemia requiring immediate statin escalation
- B. A clinically significant atherogenic dyslipidaemia phenotype warranting metabolic evaluation
- C. Dietary fat excess requiring fat restriction
- D. Normal lipid variation within the Indian population reference range

Answers: 1. B 2. B 3. C 4. C 5. B 6. C 7. C 8. ?

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CHAPTER 12

Hormonal Drivers of Insulin Resistance

Chapter 12 — Hormonal Drivers of Insulin Resistance

Beyond Glucose — The Endocrine Perspective

Insulin resistance is best understood as a multisystem endocrine state in which sex steroids, thyroid hormones, glucocorticoids, growth hormone, and pituitary signals modulate glucose handling, lipid metabolism, and body fat distribution — rather than a simple defect in glucose uptake alone. In the Dr. Nehru Method for the Indian phenotype, HOMA-IR above 2.3 and TyG above 8.82 on the natural log scale are used as operational thresholds to flag clinically relevant insulin resistance. These are proposed cutoffs based on Indian cohort data and are not universal guidelines.¹²

[IMAGE PLACEHOLDER — Figure 12.1]

Figure 12.1 — Thyroid axis and cortisol contributions to insulin resistance.

12.1 — Insulin Resistance as an Endocrine State

Insulin resistance emerges at the intersection of pancreatic beta cell function, hepatic insulin signalling, adipose tissue biology, and a complex neuro-endocrine network involving sex steroids, the thyroid axis, the hypothalamic-pituitary-adrenal axis, growth hormone, and prolactin. Classical measures such as HOMA-IR and fasting glucose capture only part of this phenotype. Endocrine patterns — hypogonadism in men, androgen excess in women, subclinical hypothyroidism, subtle cortisol excess, and altered growth hormone signalling — often precede overt hyperglycaemia.

[IMAGE PLACEHOLDER — Figure 12.2]

Figure 12.2 — Hormonal drivers: the androgen axis in men and insulin resistance.

In the Dr. Nehru Method, HOMA-IR above 2.3 is taken as an operational threshold for insulin resistance in Indian adults, based on percentile-derived

cutoffs that overlap with published HOMA-IR values between 2.0 and 2.6 reported across South Asian and other populations. Similarly, a TyG index above 8.82 natural log is used as an India-specific operational marker of insulin resistance, building on global evidence that TyG strongly correlates with clamp-measured insulin sensitivity and predicts diabetes and cardiovascular events, particularly in high-carbohydrate, rice-predominant dietary settings. These thresholds are proposed for pragmatic risk stratification and should be viewed as staging tools within the Dr. Nehru framework rather than universal consensus guidelines.¹²

12.2 — The Androgen Axis in Men

Hypogonadism, Visceral Adiposity, and Insulin Resistance

Low testosterone in adult men is consistently associated with increased visceral adiposity, reduced lean mass, and a higher prevalence of metabolic syndrome and type 2 diabetes. Epidemiological and interventional data support a bidirectional relationship — visceral obesity promotes functional hypogonadism through increased aromatase activity and inflammatory signalling, while hypoandrogenism may itself worsen fat distribution and insulin sensitivity, creating a self-reinforcing cycle.

The Moscow Study programme led by Kalinchenko and colleagues examined testosterone undecanoate in hypogonadal men with metabolic syndrome. In a randomised, placebo-controlled trial of 184 men with metabolic syndrome and hypogonadism, testosterone undecanoate administration produced modest but significant reductions in body weight, BMI, and waist circumference, with progressive improvements in HOMA-IR over follow-up compared with placebo. These findings support the concept that androgen normalisation can favourably modify central obesity and surrogate markers of insulin resistance in selected hypogonadal men, although not all metabolic endpoints — lipids, HbA1c — improve uniformly across studies.³

A meta-analysis of 18 RCTs found testosterone replacement reduced HOMA-IR by an SMD of minus 1.94 — a substantial pooled effect — but this does not justify routine testosterone therapy for metabolic indications alone, without classical hypogonadism.

A meta-analysis of testosterone replacement therapy in men with hypogonadism and either type 2 diabetes or metabolic syndrome — 18 randomised controlled trials, N=1415 — showed a standardised mean difference in HOMA-IR of minus 1.94, with 95 percent confidence interval minus 2.65 to minus 1.23, indicating a substantial improvement in insulin resistance with testosterone replacement in pooled datasets. The same meta-analysis reported favourable effects on HbA1c, triglycerides, and LDL cholesterol, although erectile function did not consistently improve. These results support androgen therapy as a metabolic intervention in carefully selected hypogonadal men, but do not justify routine testosterone replacement solely for metabolic indications in men without classical hypogonadism.⁴

SHBG as a Hepatic Marker of Insulin Resistance

Sex hormone-binding globulin is produced by hepatocytes and reflects the integrated effect of hepatic insulin signalling, fatty liver status, and thyroid hormone action. Low plasma SHBG is strongly associated with obesity, insulin resistance, NAFLD, and type 2 diabetes, and has been proposed as a biomarker of hepatic insulin resistance. Hyperinsulinaemia suppresses hepatic SHBG gene expression through insulin-responsive pathways, providing a mechanistic link between high fasting insulin and low SHBG in insulin-resistant states. In clinical practice, low SHBG in men with central adiposity may be an additional clue to underlying insulin resistance, though it should be integrated with clinical context rather than used as a stand-alone diagnostic test.¹¹

12.3 — Oestrogen, Menopause, and Insulin Resistance

Menopausal Transition and Visceral Fat

The menopausal decline in oestradiol is associated with redistribution of body fat from gluteofemoral to visceral depots, increased central obesity, and adverse changes in insulin sensitivity and lipids. Observational cohorts demonstrate rising HOMA-IR and higher incidence of metabolic syndrome across the perimenopausal transition, independent of chronological ageing. These changes are particularly relevant in South Asian women, who often carry a high baseline burden of visceral adiposity and cardiometabolic risk.

Hormone Replacement Therapy and HOMA-IR

Salpeter and colleagues conducted a landmark meta-analysis of 107 hormone replacement therapy trials evaluating metabolic outcomes in women with and without diabetes. Pooled results showed that in women without diabetes, HRT reduced abdominal fat by minus 6.8 percent and HOMA-IR by minus 12.9 percent, with 95 percent confidence interval minus 17.1 to minus 8.6 percent. In women with diabetes, HRT decreased fasting glucose by minus 11.5 percent and HOMA-IR by minus 35.8 percent, with 95 percent confidence interval minus 51.7 to minus 19.8 percent. These data indicate that appropriately tailored HRT can improve insulin resistance and glucose profiles in many postmenopausal women, although individual risk-benefit assessment remains essential.⁵

Oral oestrogen formulations in this meta-analysis were associated with adverse changes in C-reactive protein and procoagulant factors, whereas transdermal oestrogen did not significantly affect CRP or protein S. Because transdermal delivery bypasses first-pass hepatic metabolism, it is often preferred in women at high cardiometabolic risk or with baseline hypertriglyceridaemia. The choice of formulation should be individualised and guided by guideline-based cardiovascular and thrombotic risk assessment rather than metabolic considerations alone.⁵

12.4 — PCOS, Androgen Excess, and Insulin Resistance in Women

Hyperandrogenism and Metabolic Risk

Polycystic ovary syndrome is characterised by ovulatory dysfunction, polycystic ovarian morphology, and hyperandrogenism, with pronounced insulin resistance even in non-obese women. Hyperinsulinaemia amplifies ovarian androgen production and suppresses SHBG, leading to higher free testosterone levels and a feed-forward cycle of androgen excess and insulin resistance. In metabolic staging frameworks, a total testosterone around 1.2 nmol/L may be used as a study-specific operational threshold to flag androgen excess in PCOS cohorts. This value should not be treated as a universal cutoff, and local assay characteristics and ethnic differences must be considered.

AMH Thresholds and Ethnic Caution

Chinese case-control data have explored anti-Mullerian hormone as a diagnostic biomarker for PCOS. In a cohort of 771 women — 653 with PCOS and 118 controls — the optimal AMH diagnostic thresholds for PCOS were 8.16 ng/mL in women aged 20 to 29 years and 5.89 ng/mL in those aged 30 to 39 years, with areas under the ROC curve of 0.846 and 0.865 respectively. Combined use of AMH, LH to FSH ratio, testosterone, and fasting insulin further improved diagnostic performance.⁶

AMH thresholds from Chinese cohorts should be applied cautiously in Indian patients — ovarian reserve markers, body size, assay platforms, and PCOS phenotype distributions differ meaningfully across ethnic groups.

These AMH thresholds originate from Chinese data and should be applied cautiously in Indian patients, because ovarian reserve markers, body size, assay platforms, and PCOS phenotype distributions differ across ethnic groups. In the Dr. Nehru staging system, AMH is interpreted as a supportive marker rather than a stand-alone diagnostic test. Operational AMH thresholds are used for internal consistency but are explicitly labelled as framework-specific, not international guidelines.⁶

12.5 — The Thyroid Axis and Insulin Resistance

Subclinical Hypothyroidism and HOMA-IR

Subclinical hypothyroidism is defined biochemically by elevated TSH with normal free T4, typically stratified into mildly elevated TSH 4.5 to 10 mIU/L and markedly elevated TSH at or above 10 mIU/L. Several case-control studies report higher HOMA-IR values in subclinical hypothyroidism compared with euthyroid controls, with positive correlations between TSH, fasting insulin, and HOMA-IR. Other cohorts show weaker or inconsistent relationships, illustrating that not all individuals with subclinical hypothyroidism develop measurable insulin resistance.⁹

Within the Dr. Nehru Method, a TSH above 4.5 mIU/L is treated as an operational threshold suggesting subclinical hypothyroidism-related metabolic stress, rather than a universal consensus cutoff. Clinicians should interpret HOMA-IR in the context of thyroid status, recognising that

subclinical hypothyroidism may exacerbate insulin resistance in some patients, particularly those with coexistent obesity or type 2 diabetes.

Levothyroxine, Metabolic Parameters, and B12

Levothyroxine replacement in subclinical hypothyroidism can improve lipids, weight, and surrogate markers of insulin resistance in selected patients, although randomised trials show variable metabolic responses. Initiation of levothyroxine in Indian patients with subclinical hypothyroidism and elevated HOMA-IR should be based on conventional thyroid guidelines — presence of symptoms, TSH level, antibody status, pregnancy plans — with metabolic benefit considered a potential additional advantage rather than the primary indication.

Metformin therapy, widely used in patients with concurrent type 2 diabetes or PCOS, is associated with reduced vitamin B12 absorption and biochemical B12 deficiency in many studies. In this framework, methylcobalamin is always preferred over cyanocobalamin for replacement, and periodic monitoring of B12 status is advised in metformin-treated thyroid patients to avoid compounding neuropathy and haematological risk. This preference reflects the Dr. Nehru Method clinical approach and does not imply universal consensus on B12 formulations.

12.6 — Cortisol, Subclinical Hypercortisolism, and Insulin Resistance

Glucocorticoid excess — whether overt Cushing syndrome or subtle autonomous cortisol secretion — promotes hepatic gluconeogenesis, antagonises peripheral insulin action, and contributes to central adiposity and sarcopenia. Cortisol increases lipolysis and free fatty acids, impairs muscle glucose uptake, and alters adipokine profiles, producing a characteristic phenotype of truncal obesity, proximal myopathy, and impaired glucose tolerance. In mild or subclinical hypercortisolism, these metabolic changes may be present without classic Cushingoid features, making biochemical screening important in selected high-risk individuals.⁷

Endocrine Society guidelines recommend the 1mg overnight dexamethasone suppression test, late-night salivary cortisol, or 24-hour urinary free cortisol as initial screening tests for endogenous Cushing syndrome. A post-DST

serum cortisol above 1.8 mcg/dL has been widely adopted as a threshold indicating inadequate suppression and prompting further evaluation for autonomous cortisol secretion. DST cutoffs vary between guidelines — ranging from 1.8 to 5 mcg/dL — and depend on assay methodology, dexamethasone pharmacokinetics, and clinical context.⁷

In this chapter, a cortisol above 1.8 mcg/dL after 1mg DST is treated as an operational threshold signalling possible subclinical hypercortisolism requiring confirmatory testing — repeat DST, urinary cortisol, imaging — rather than a definitive diagnostic boundary. Mechanistic and observational data link such mild cortisol excess to increased risk of diabetes, hypertension, and cardiovascular disease, but high-quality randomised trials of cortisol-targeted interventions on metabolic outcomes remain limited.

12.7 — Hyperprolactinaemia and Insulin Resistance

Hyperprolactinaemia has been associated with insulin resistance and low-grade inflammation, particularly in women with prolactinomas and coexistent obesity. In a descriptive clinical study of 15 patients with hyperprolactinaemia, baseline HOMA-IR and high-sensitivity CRP were significantly higher than in healthy controls, and short-term cabergoline therapy reduced serum prolactin, fasting insulin, HOMA-IR, and hsCRP over 12 weeks.⁸

The N=15 cabergoline study is pilot evidence only and cannot be generalised. Prolactin is normalised primarily for standard endocrine indications — metabolic improvement is a possible secondary benefit, not the reason for treatment.

The small sample size and non-randomised design mean these cabergoline data should be classified as pilot evidence only — findings that cannot be generalised to all patients with hyperprolactinaemia. In the Dr. Nehru staging system, hyperprolactinaemia is flagged as a potential contributor to insulin resistance, especially when accompanied by obesity and menstrual irregularity, but normalisation of prolactin with dopamine agonists is pursued primarily for standard endocrine indications — tumour control, symptom relief — with metabolic improvement considered a possible secondary benefit.⁸

12.8 — Growth Hormone Deficiency, Replacement, and Insulin Resistance

Adult growth hormone deficiency is characterised by increased visceral fat, reduced lean mass, dyslipidaemia, and reduced quality of life, with insulin resistance present in many but not all patients. Growth hormone has complex anti-insulin actions acutely, yet chronically promotes favourable body composition changes, making its net effect on glucose metabolism time-dependent.¹⁰

Several studies have evaluated growth hormone replacement in adult deficiency using HOMA-IR and oral glucose tolerance testing. In one twelve-month study of 24 adults, growth hormone therapy produced progressive increases in fasting insulin with deterioration in insulin sensitivity indices, despite significant reductions in visceral fat and waist-to-hip ratio. Another twelve-month study in 17 adults showed marked improvements in body fat percentage and lean mass with neutral effects on HOMA-IR. Long-term follow-up over five years in a cohort of 20 patients demonstrated an initial rise in HOMA-IR after year one, followed by partial improvement by year five as body fat decreased, with no cases of impaired fasting glucose.¹⁰

Taken together, these data suggest growth hormone replacement in adult deficiency may transiently worsen surrogate markers of insulin resistance in the early phase, while improving body composition and possibly attenuating long-term cardiometabolic risk. In the Dr. Nehru framework, growth hormone replacement is considered a specialised therapy in which HOMA-IR may worsen initially, so close monitoring of glucose and HbA1c is essential, though body composition improvements may justify continued therapy in appropriately selected patients.

12.9 — Integrating Hormonal Drivers in the Indian Phenotype

Populations with rice-predominant diets and high refined-carbohydrate intake often exhibit a phenotype of normal BMI but elevated visceral adiposity, high triglycerides, and subtle elevation of fasting glucose. In such settings, the TyG index offers a useful surrogate for insulin resistance that integrates fasting glucose and triglycerides and correlates with clamp-measured insulin

sensitivity and risk of diabetes and cardiovascular disease. In the Dr. Nehru Method, TyG above 8.82 natural log is used as a proposed operational cutoff for Indian adults, complementing HOMA-IR above 2.3 to stage insulin resistance. These thresholds reflect Indian data and literature from Asian cohorts but are explicitly described as staging tools rather than universal diagnostic criteria.¹²

From an endocrine perspective, staging insulin resistance in Indian patients involves a layered assessment across multiple axes simultaneously, rather than relying on glucose parameters alone.

- Pancreatic-hepatic axis — HOMA-IR, TyG, fasting insulin, SHBG as hepatic marker of insulin resistance.
- Gonadal axis — total and free testosterone, SHBG, and clinical features of hypogonadism in men; androgen excess and PCOS features in women.
- Thyroid axis — TSH and free T4, with TSH above 4.5 mIU/L as an operational marker of subclinical hypothyroidism-related metabolic stress.
- HPA axis — 1mg DST with cortisol above 1.8 mcg/dL as an operational threshold prompting further evaluation for autonomous cortisol secretion.
- Pituitary axis — prolactin levels, recognising that cabergoline-associated improvements in HOMA-IR arise from pilot data only.
- GH axis — clinical and biochemical assessment of adult deficiency, understanding that replacement may initially worsen HOMA-IR while improving body composition over time.

Methylcobalamin is preferred over cyanocobalamin for B12 replacement in this staging approach, especially in patients on long-term metformin, to minimise overlapping neuropathy and haematological risk. All hormonal cutoffs and staging thresholds described in this chapter are operational and intended for structured clinical reasoning within the Dr. Nehru Method — they do not represent universal guidelines or consensus positions of professional societies. Clinical decisions should continue to follow standard international and national endocrine guidelines, with this framework serving as an adjunctive lens for understanding the hormonal drivers of insulin resistance beyond glucose.

Clinical Summary — Chapter 12

- Testosterone replacement — 18 RCTs N=1415 — HOMA-IR SMD minus 1.94 — appropriate only in confirmed hypogonadism, not for metabolic indication alone.
- HRT — 107 trials — HOMA-IR falls 12.9% without diabetes, 35.8% with diabetes — transdermal preferred at high cardiometabolic risk.
- AMH PCOS cutoffs — 8.16 ng/mL ages 20-29, 5.89 ng/mL ages 30-39 — Chinese cohort N=771 — apply cautiously in Indian patients.
- TSH above 4.5 mIU/L — operational threshold for subclinical hypothyroidism-related metabolic stress, not universal cutoff.
- DST cortisol above 1.8 mcg/dL — operational threshold prompting further evaluation, not definitive Cushing diagnosis.
- Hyperprolactinaemia cabergoline data — N=15 — pilot evidence only — cannot be generalised.
- GH replacement — HOMA-IR worsens initially — body composition improves over 1 to 5 years — monitor glucose closely.
- All hormonal cutoffs in this chapter are Dr. Nehru Method operational tools — not international consensus guidelines.

Multiple Choice Questions

1. The meta-analysis of 18 RCTs on testosterone replacement reported a HOMA-IR SMD of:
A. Minus 0.54 B. Minus 1.94 C. Plus 1.20 D. Minus 3.58
2. In women with pre-existing type 2 diabetes, HRT reduced HOMA-IR by approximately:
A. 12.9% B. 20.5% C. 35.8% D. 42.0%
3. The AMH diagnostic threshold for PCOS in Chinese women aged 20-29 was:
A. 5.89 ng/mL B. 6.50 ng/mL C. 8.16 ng/mL D. 10.2 ng/mL
4. A post-DST cortisol above 1.8 mcg/dL indicates:
A. Normal suppression B. Inadequate suppression requiring further evaluation

C. Definitive Cushing syndrome D. Hypocortisolism

5. GH replacement in adult GH deficiency typically shows what early pattern?

A. Immediate HOMA-IR improvement B. No metabolic change

C. Initial HOMA-IR worsening with later body composition benefit D. Permanent insulin resistance

Answers: 1. B 2. C 3. C 4. B 5. C

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8. Influences of hyperprolactinaemia and obesity on cardiovascular risk markers — cabergoline therapy in 15 patients — pilot evidence.
9. Studies of insulin resistance in subclinical hypothyroidism — elevated HOMA-IR and TSH correlation.
10. Adult GH deficiency replacement trials — early HOMA-IR deterioration with later body composition improvement.
11. SHBG as hepatic marker of insulin resistance — low SHBG associated with obesity, NAFLD, diabetes.
12. Comparative analyses of TyG and insulin resistance biomarkers — associations with metabolic syndrome and cardiometabolic events.

CHAPTER 13

Precision Pharmacotherapy

Precision Pharmacotherapy in Metabolic Disease

Introduction

Metabolic disease is not a single condition but a spectrum of overlapping dysfunctions — hyperglycaemia, dyslipidaemia, insulin resistance, visceral adiposity, and cardiovascular risk — all converging on the same damaged endothelium. Treating each of these in isolation, with one drug per diagnosis, is the old model. Precision pharmacotherapy, as proposed in the Dr. Nehru Method, treats the underlying metabolic biology first and allows the clinical labels — diabetes, hypertension, hyperlipidaemia — to follow the correction of the root cause.

This chapter presents the key pharmacological agents available in India in 2026, organised by their mechanism of action within the metabolic cascade. The staging framework used throughout this text — HOMA-IR greater than 2.3 as the threshold for insulin resistance in the Indian population, validated by Nehru MS et al. (International Journal of Medicine 2026;8(2):977–983), and TyG Index greater than 8.82 on the natural logarithm scale, validated by Nehru MS et al. (IJCPR 2026;18(2):826–833) — determines which drugs are indicated at which stage. A patient at Stage 0 (HOMA-IR less than 1.0) requires lifestyle correction. A patient at Stage 4 or Stage 5 (HOMA-IR greater than 5.0) requires combination pharmacotherapy targeting insulin resistance, atherogenic dyslipidaemia, and beta cell exhaustion simultaneously.



Figure 13.1 — Precision pharmacotherapy mapped to HOMA-IR stages.

Throughout this chapter, drug benefit is expressed using Number Needed to Treat (NNT) from TheNNT.com, not manufacturer promotional data. This distinction matters enormously in clinical practice and patient counselling.

[IMAGE PLACEHOLDER — Figure 13.2]

Figure 13.2 — Number needed to treat (NNT) framing for statin therapy.

1. Metformin — The Indispensable First Line

Metformin remains the cornerstone of pharmacotherapy for type 2 diabetes and insulin resistance. Its mechanism is primarily hepatic — it suppresses gluconeogenesis through AMPK activation, reducing fasting glucose without causing hypoglycaemia. It modestly reduces HOMA-IR, reduces circulating insulin levels, and has a favourable effect on body weight. The UKPDS demonstrated long-term cardiovascular benefit independent of glycaemic control, an effect that has been called the "legacy effect."

The mandatory companion: methylcobalamin. Metformin consistently depletes vitamin B12 through impairment of ileal calcium-dependent absorption. The depletion is dose-dependent and duration-dependent. Every patient on metformin must receive methylcobalamin supplementation — not cyanocobalamin. Methylcobalamin is the neurologically active form, crosses the blood-brain barrier, and is directly usable in myelin synthesis without requiring hepatic conversion. Cyanocobalamin requires multiple conversion steps and is significantly less effective in metformin-treated patients. This is not optional. It is a mandatory co-prescription in the Dr. Nehru Method.

The clinical consequence of untreated B12 deficiency in metformin-treated patients includes peripheral neuropathy, which is indistinguishable on examination from diabetic neuropathy. Thousands of Indian patients have received gabapentin or pregabalin for a neuropathy that was actually metformin-induced B12 deficiency — a preventable and reversible condition. Annual B12 monitoring and routine methylcobalamin supplementation at 500 mcg twice daily are recommended.

Metformin is effective up to HOMA-IR Stage 3. Beyond Stage 3, it is insufficient as monotherapy and must be combined with agents that directly address beta cell protection and insulin resistance amplification.

2. Thiazolidinediones — Targeting Insulin Resistance Directly

Pioglitazone acts as a PPARgamma agonist, improving insulin sensitivity in peripheral tissues — muscle, fat, and liver. It directly reduces HOMA-IR and has demonstrated benefit in non-alcoholic fatty liver disease (NAFLD), which is increasingly recognised as the hepatic manifestation of the same metabolic cascade described in the Indian Health Constitution framework. Pioglitazone also has modest anti-atherogenic properties, reducing intimal media thickness and inflammatory markers.

Clinical drawbacks include fluid retention, weight gain, and the well-established association with bladder cancer with long-term use at high doses. In Indian patients with sarcopenia, the weight gain associated with pioglitazone must be monitored carefully, as it may disproportionately favour fat gain over muscle preservation.

Saroglitazar — the Indian option. Saroglitazar is a dual PPARalpha/gamma agonist developed and approved in India. It combines the insulin-sensitising effect of a PPARgamma agonist with the lipid-lowering properties of a PPARalpha agonist, making it particularly useful in patients with concurrent atherogenic dyslipidaemia (elevated triglycerides, reduced HDL). The PRESS V and PRESS VI trials (PMC4287720) validated its efficacy in Indian patients. Saroglitazar reduces triglycerides by 35 to 45 percent, reduces LDL, improves HDL, and reduces HOMA-IR. It is available at a fraction of the cost of novel agents and should be considered particularly in Indian patients with diabetic dyslipidaemia and NAFLD where the metabolic phenotype favours a dual PPARagonist approach.

3. SGLT2 Inhibitors — Cardiorenal Protection Beyond Glycaemia

Sodium-glucose cotransporter 2 inhibitors — empagliflozin, dapagliflozin, and canagliflozin — represent a genuine paradigm shift in metabolic pharmacotherapy. Their glycaemic mechanism (blocking renal glucose reabsorption, producing glucosuria) is secondary to their organ-protective effects. The cardiovascular and renal outcomes trials have transformed these drugs from antidiabetic agents into cardiorenal protective agents.

The EMPA-REG OUTCOME trial demonstrated that empagliflozin reduced cardiovascular death by 38 percent and hospitalisation for heart failure by 35 percent in patients with established cardiovascular disease. The DAPA-HF trial extended dapagliflozin benefit to patients with heart failure with reduced ejection fraction regardless of diabetes status. The CREDENCE and DAPA-CKD trials confirmed renal protective effects, with dapagliflozin reducing the composite of renal decline and death.

In the Dr. Nehru staging framework, SGLT2 inhibitors are indicated from Stage 2 onwards (HOMA-IR 2.3 to 3.5) in patients with established cardiovascular disease, heart failure, or chronic kidney disease, and from Stage 3 onwards (HOMA-IR 3.5 to 5.0) in patients with high cardiovascular risk. The glucose-lowering effect is HbA1c dependent; the cardiorenal benefit is not.

Practical considerations in India: empagliflozin and dapagliflozin are both available as generics at approximately Rs 15 to 30 per tablet. Patients must be counselled regarding adequate hydration, genital mycotic infections, and the rare but serious risk of euglycaemic diabetic ketoacidosis (particularly in patients who fast or undergo surgery). SGLT2 inhibitors should be withheld 72 hours before any surgical procedure.

4. GLP-1 Receptor Agonists — Beta Cell Protection and Weight Reduction

Glucagon-like peptide-1 receptor agonists — semaglutide, dulaglutide, liraglutide — act on the incretin axis to stimulate glucose-dependent insulin secretion, suppress glucagon, slow gastric emptying, and reduce appetite. Their weight reduction effect is central to their metabolic benefit: in patients with HOMA-IR greater than 2.3, sustained weight reduction directly lowers insulin resistance and reduces TyG Index.

Semaglutide (oral and subcutaneous) has demonstrated HbA1c reductions of 1.5 to 2.0 percent and weight reduction of 4 to 6 kg in the SUSTAIN and PIONEER trials. The LEADER trial (liraglutide) and REWIND trial (dulaglutide) demonstrated significant cardiovascular mortality reduction in patients with established cardiovascular disease.

Triglyceride reduction. GLP-1 receptor agonists reduce triglycerides by 20 to 30 percent through reduced hepatic VLDL synthesis and enhanced

lipoprotein lipase activity. In the Dr. Nehru Method, this TG-lowering effect contributes directly to reduction in TyG Index — a dual benefit that amplifies the metabolic impact beyond glycaemic control alone.

Beta cell protection is a critical and underappreciated benefit. GLP-1 receptor agonists reduce beta cell apoptosis in animal models and appear to preserve beta cell function in early-stage disease. Initiating GLP-1 receptor agonists at HOMA-IR Stage 2 (HOMA-IR 2.3 to 3.5, before clinical diabetes) may represent the optimal window for beta cell preservation.

Dulaglutide (0.75 mg and 1.5 mg weekly injection) is the most cost-accessible option in India at approximately Rs 1,200 to 2,500 per month. Once-weekly injection supports adherence in the Indian outpatient setting.

5. Tirzepatide — Dual GIP/GLP-1 Agonism

Tirzepatide is a dual glucose-dependent insulinotropic polypeptide (GIP) and GLP-1 receptor agonist. The SURMOUNT and SURPASS trial programmes have demonstrated weight reduction of 15 to 22 percent — substantially greater than any previous pharmacological intervention — with HbA1c reductions exceeding 2.0 percent. The SURMOUNT-1 trial in non-diabetic obese individuals demonstrated that 72 weeks of tirzepatide at the 15 mg dose produced mean weight loss of 20.9 percent.

In the context of the Dr. Nehru Method, tirzepatide is the most powerful single agent available to reduce HOMA-IR and TyG Index simultaneously. Its dual mechanism — GIP agonism promoting insulin sensitivity in adipose tissue combined with GLP-1 mediated appetite suppression and beta cell protection — addresses multiple nodes of the metabolic cascade in a single injection.

India cost reality. Tirzepatide (Mounjaro, Eli Lilly) is currently priced at approximately Rs 25,000 to Rs 40,000 per month in India (verified approximate; verify before advising patients). This places it beyond the reach of the majority of Indian patients. It is appropriate to discuss tirzepatide with patients who have Stage 4 or Stage 5 metabolic disease (HOMA-IR greater than 5.0) with established cardiovascular disease or severe obesity, particularly those with the financial capacity to sustain treatment. Biosimilar development is expected to reduce costs significantly within the next three to five years.

6. Statins — The NNT Conversation

Statins are the most prescribed class of drugs in cardiovascular medicine globally, yet the gap between public perception of their benefit and the evidence base deserves honest discussion in clinical practice.

NNT in primary prevention. Per TheNNT.com analysis of primary prevention trials, the NNT to prevent one major adverse cardiovascular event over 5 years is approximately 70 to 100. This means that 70 to 100 patients must be treated for 5 years for one patient to benefit. Absolute risk reduction in primary prevention is approximately 1 percent. This must be disclosed to patients making informed decisions about long-term statin therapy.

NNT in secondary prevention. In patients with established cardiovascular disease, the NNT drops to approximately 30 to 40 over 5 years — a clinically significant and defensible benefit. All patients with established coronary artery disease, prior myocardial infarction, stroke, or peripheral arterial disease should be on high-intensity statin therapy unless contraindicated.

In Indian patients, several considerations modify statin choice. Indians have higher rates of statin-induced myopathy at standard Western doses, partly due to pharmacogenomic differences in CYP2C9 and SLCO1B1 polymorphisms. Rosuvastatin and atorvastatin remain the evidence-based choices for LDL reduction. Pitavastatin deserves particular mention: it is relatively glucose-neutral compared to atorvastatin and rosuvastatin, which have documented dose-dependent diabetogenic effects. For patients at the pre-diabetes stage (HOMA-IR 2.3 to 3.5 with borderline fasting glucose), pitavastatin represents a more metabolically appropriate statin choice.

7. Ezetimibe and Bempedoic Acid — Non-Statin LDL Reduction

Ezetimibe. Ezetimibe inhibits intestinal cholesterol absorption via the NPC1L1 transporter, reducing LDL by 15 to 20 percent as monotherapy and an additional 15 to 20 percent when added to statin therapy. The IMPROVE-IT trial demonstrated modest but statistically significant cardiovascular benefit when ezetimibe was added to simvastatin in post-ACS patients (NNT approximately 50 over 7 years). Ezetimibe is well tolerated, relatively inexpensive in India (Rs 10 to 20 per day), and should be considered in all

patients who cannot tolerate high-dose statins or who require further LDL reduction beyond statin monotherapy.

Bempedoic acid. Bempedoic acid inhibits ATP-citrate lyase, an enzyme upstream of HMG-CoA reductase in the cholesterol synthesis pathway, reducing LDL by 15 to 25 percent. Its critical advantage is that it is not activated in muscle tissue, making it an option for statin-intolerant patients. The CLEAR Outcomes trial (Nissen SE et al., NEJM 2023) demonstrated a 13 percent relative reduction in major adverse cardiovascular events (MACE) — the first non-statin LDL-lowering agent to demonstrate primary cardiovascular event reduction in statin-intolerant patients. Bempedoic acid availability in India is limited in 2026 but is expected to expand as global generic manufacturing scales up.

8. PCSK9 Inhibitors — Evolocumab Biosimilars in India

PCSK9 inhibitors — evolocumab and alirocumab — reduce LDL by 50 to 60 percent through prevention of LDL receptor degradation, resulting in dramatically enhanced LDL clearance. The FOURIER trial (evolocumab) demonstrated a 15 percent reduction in cardiovascular events in patients with established cardiovascular disease on maximally tolerated statin therapy. The ODYSSEY OUTCOMES trial confirmed similar benefits with alirocumab.

Biosimilar access in India. The Drugs Controller General of India (DCGI) has approved evolocumab biosimilars manufactured by Zydus and Lupin, with pricing in the range of Rs 5,000 to Rs 8,000 per injection (approximate; verify before advising patients). This represents a significant reduction from the originator product cost but remains a substantial outpatient expenditure. In the Indian context, evolocumab biosimilars are appropriate for patients with familial hypercholesterolaemia, established cardiovascular disease with LDL above 70 mg/dL despite maximal statin plus ezetimibe therapy, or statin-intolerant patients with very high cardiovascular risk.

9. Fenofibrate and Omega-3 — Targeting Atherogenic Dyslipidaemia

Fenofibrate. Fenofibrate is a PPAR α agonist that reduces triglycerides by 30 to 50 percent and raises HDL by 10 to 20 percent. It is the cornerstone of treatment for severe hypertriglyceridaemia (TG greater than 500 mg/dL) and

has a role in mixed dyslipidaemia when TG and non-HDL cholesterol remain elevated despite statin therapy. In the Dr. Nehru framework, fenofibrate directly lowers the TyG Index by reducing the triglyceride component. Patients with TyG Index greater than 8.82 and fasting triglycerides greater than 300 mg/dL are candidates for fenofibrate in addition to lifestyle modification. A critical interaction: fenofibrate combined with statin increases the risk of myopathy; the combination should be used only when the clinical benefit clearly outweighs the risk, and creatine kinase should be monitored.

Omega-3 fatty acids — REDUCE-IT. The REDUCE-IT trial (Bhatt DL et al., NEJM 2019; N=8,179) demonstrated that icosapentaenoic acid (EPA) at 4g/day (Vascepa/icosapent ethyl) reduced major cardiovascular events by 25 percent in statin-treated patients with elevated triglycerides (150 to 499 mg/dL) and established cardiovascular disease or diabetes. This is a robust result with a relatively low NNT of approximately 21 over 4.9 years for cardiovascular death, MI, stroke, coronary revascularisation, or unstable angina. The mechanism goes beyond simple TG reduction and likely involves membrane stabilisation, anti-inflammatory effects, and plaque stabilisation. Standard over-the-counter omega-3 supplements contain mixed EPA and DHA and do not replicate the REDUCE-IT results; the trial used pharmaceutical-grade pure EPA. Icosapent ethyl availability in India is limited; standard omega-3 1–4g/day may be used as an adjunct with appropriate patient counselling about the limitations of the evidence.

10. Rational Combination Strategy by HOMA-IR Stage

The Dr. Nehru Method does not prescribe by HbA1c alone. The pharmacological approach is staged according to HOMA-IR and TyG Index:

Stage 0 (HOMA-IR less than 1.0): Lifestyle only. No pharmacotherapy required. Reassess annually.

Stage 1 (HOMA-IR 1.0 to 2.3): Lifestyle intensification. Consider methylcobalamin and vitamin D supplementation if deficient. No hypoglycaemic agent indicated.

Stage 2 (HOMA-IR 2.3 to 3.5): Metformin with mandatory methylcobalamin. Add SGLT2 inhibitor if cardiovascular disease or heart

failure present. Consider GLP-1 receptor agonist if obesity (BMI greater than 25 kg/m² Indian cutoff) and high cardiovascular risk.

Stage 3 (HOMA-IR 3.5 to 5.0): Metformin plus SGLT2 inhibitor plus GLP-1 receptor agonist. Add saroglitazar if TyG Index remains above 8.82 and TG elevated. Statin if LDL above target. Ezetimibe if LDL target not achieved on statin alone.

Stage 4 (HOMA-IR 5.0 to 10.0): Combination of three or four agents targeting different nodes. Consider tirzepatide if patient can sustain cost. PCSK9 inhibitor biosimilar if established cardiovascular disease with residual LDL burden.

Stage 5 (HOMA-IR greater than 10.0): Maximum combination therapy. Consider endocrinology or bariatric surgery referral. Methylcobalamin, vitamin D, and iron correction are mandatory as micronutrient deficiencies at this stage amplify metabolic dysfunction.

All drug costs cited in this chapter are approximate figures for 2026 and are subject to market variation. The author recommends verifying current pricing before advising patients. Costs of biosimilars and generics in India are expected to continue declining as the patent landscape evolves.

Multiple Choice Questions

1. A 48-year-old man with type 2 diabetes has been on metformin 1000 mg twice daily for 4 years. He presents with tingling in both feet. His HbA1c is 7.1%. What is the most important investigation to order first?
 - A. Nerve conduction velocity
 - B. Serum vitamin B12 level
 - C. Fasting blood glucose
 - D. Doppler study of lower limb vessels
2. Which of the following statements about NNT for statins in primary prevention is most accurate according to evidence from TheNNT.com?
 - A. NNT is approximately 10 over 5 years
 - B. NNT is approximately 30 to 40 over 5 years
 - C. NNT is approximately 70 to 100 over 5 years
 - D. NNT is less than 5 over 5 years

3. A patient with HOMA-IR of 4.2 and TyG Index of 9.1 (natural log scale) has fasting triglycerides of 380 mg/dL and HDL of 32 mg/dL. Which Indian drug would best address both the insulin resistance and the atherogenic dyslipidaemia in a single agent?
- A. Pioglitazone
 - B. Saroglitazar
 - C. Fenofibrate
 - D. Ezetimibe
4. The REDUCE-IT trial demonstrated cardiovascular benefit with high-dose EPA (4g/day) in patients with elevated triglycerides on statins. What was the approximate NNT for the primary composite endpoint over 4.9 years?
- A. 5
 - B. 21
 - C. 50
 - D. 100
5. Which statin is considered relatively glucose-neutral and is a preferred choice for a patient with pre-diabetes (HOMA-IR 2.6, FPG 108 mg/dL) who requires LDL lowering?
- A. Atorvastatin 40 mg
 - B. Rosuvastatin 20 mg
 - C. Pitavastatin 2 mg
 - D. Simvastatin 40 mg
6. Bempedoic acid acts by inhibiting ATP-citrate lyase. What is its major clinical advantage over statins for patients with statin-associated muscle symptoms?
- A. It works only in the liver and is not activated in muscle tissue
 - B. It has stronger LDL reduction than any statin
 - C. It also reduces blood glucose and HbA1c
 - D. It can be given once weekly as an injection
7. According to the Dr. Nehru India-specific cutoffs, the TyG Index threshold for metabolic risk in the Indian population is:
- A. Greater than 4.5 on natural log scale
 - B. Greater than 8.82 on natural log scale
 - C. Greater than 10.5 on natural log scale
 - D. Greater than 5.0 on natural log scale

8. A 55-year-old woman with established coronary artery disease, LDL 110 mg/dL on maximum-dose rosuvastatin plus ezetimibe, cannot tolerate further lipid therapy due to myalgia. Which of the following is the most appropriate next step?

- A. Add fenofibrate
- B. Consider evolocumab biosimilar
- C. Add omega-3 supplements
- D. Switch to pitavastatin monotherapy and stop ezetimibe

Answers: 1. B 2. C 3. B 4. B 5. C 6. A 7. B 8. B

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CHAPTER 14

Pipeline Drugs and Emerging Therapies

Pipeline Drugs and Emerging Therapies in Metabolic Medicine

14.1 Introduction: The New Frontier of Metabolic Medicine

The landscape of metabolic medicine is undergoing a seismic transformation. Decades of incremental progress are now giving way to a wave of mechanistically novel compounds that act on pathways once considered undruggable. For the clinician managing type 2 diabetes, obesity, non-alcoholic fatty liver disease (NAFLD/MASH), chronic kidney disease, and cardiovascular risk in the Indian population, this chapter offers an honest appraisal of what is coming — and what remains investigational.

In the Dr. Nehru Method framework, the eight metabolic triggers converge on a final common pathway: progressive insulin resistance quantified by HOMA-IR (India-validated cutoff >2.3) [1] and the Triglyceride–Glucose Index (TyG, natural log scale, India cutoff >8.82) [2]. Pipeline drugs do not replace this foundational assessment — they extend our therapeutic reach once lifestyle modification and available agents have been optimised.



Figure 14.1 — Pipeline drugs and emerging metabolic therapies — mechanism overview.

This chapter presents each emerging therapy under a structured format: mechanism, key trial evidence, current approval status, India availability, and clinical relevance. Drugs are honestly labelled as approved, approved but not yet available in India, or investigational. No benefit is overstated.

14.2 Triple Receptor Agonists: Pushing the Frontier of Weight Reduction

14.2.1 Retatrutide (GIP / GLP-1 / Glucagon Triple Agonist)

Retatrutide is a once-weekly injectable peptide that simultaneously activates three receptors: glucose-dependent insulintropic polypeptide (GIP), glucagon-like peptide-1 (GLP-1), and the glucagon receptor. The triple agonism creates a synergistic metabolic effect — the GLP-1 arm suppresses appetite, the GIP arm potentiates insulin secretion and modulates adipose lipolysis, and the glucagon arm drives hepatic fat oxidation and increases energy expenditure. This mechanistic breadth distinguishes it from tirzepatide, which targets only GIP and GLP-1.

In a landmark Phase 2 trial by Jastreboff et al. published in the New England Journal of Medicine in 2023, 338 adults with obesity were randomised to various doses of retatrutide or placebo [3]. At 24 weeks, the highest dose cohort demonstrated mean body weight reduction of 17.5%, with 48-week data showing 24.2% weight loss — the highest figure ever reported for a once-weekly subcutaneous agent at the time of publication. Preliminary data released subsequently suggest weight loss approaching 30.3% at 104 weeks in the most responsive cohorts, a magnitude that blurs the line between pharmacotherapy and bariatric surgery outcomes.

The Phase 3 TRIUMPH programme is currently ongoing and will determine whether these extraordinary results replicate across broader populations, including those with type 2 diabetes and established cardiovascular disease. Safety data from Phase 2 were consistent with the GLP-1 class profile: nausea, vomiting, and diarrhoea, predominantly mild to moderate and dose-dependent.

Status: Investigational. Not approved anywhere as of 2025. Phase 3 TRIUMPH trials ongoing. Not available in India.

14.2.2 Oral Semaglutide 50 mg (OASIS-1)

The development of oral GLP-1 receptor agonists has historically been constrained by the poor oral bioavailability of peptides. Semaglutide 14 mg

oral was the first approved oral GLP-1 agonist for glycaemic control. The OASIS-1 trial published in *The Lancet* in 2023 by Knop et al. evaluated a higher oral dose of 50 mg in 667 adults with overweight or obesity without diabetes [4]. After 68 weeks, participants achieved a mean body weight reduction of 15.1% — comparable in magnitude to injectable semaglutide 2.4 mg — with an acceptable gastrointestinal side-effect profile.

For Indian patients, the prospect of effective oral weight management without injections is highly relevant given the significant cultural and practical barriers to injectable therapy in both urban and rural settings. This dose of oral semaglutide awaits regulatory approval in India, though the approved 14 mg formulation (Rybelsus) is available.

Status: Oral semaglutide 50 mg — regulatory submission pending in major markets. Not yet approved in India. The 14 mg dose (Rybelsus) is approved in India for glycaemic control only.

14.3 Therapies for Metabolic Dysfunction-Associated Steatohepatitis (MASH)

14.3.1 Resmetirom (Thyroid Hormone Receptor-Beta Agonist)

Resmetirom is a liver-directed, thyroid hormone receptor-beta (THR- β) selective agonist. Systemic thyroid hormone action is broad and includes cardiac and bone effects; hepatic THR- β specificity allows resmetirom to harness thyroid hormone's lipid-lowering and metabolic benefits within the liver without systemic thyrotoxicity. It reduces hepatic triglyceride synthesis, increases mitochondrial beta-oxidation, and reduces LDL cholesterol through upregulation of hepatic LDL receptors.

The pivotal MAESTRO-NASH trial reported by Harrison et al. in the *New England Journal of Medicine* in 2024, involving 966 patients with biopsy-proven MASH and fibrosis stage F1B to F3, demonstrated that resmetirom 100 mg achieved NASH resolution (without worsening fibrosis) in 29.9% of patients and fibrosis improvement of at least one stage in 25.9% — both meeting pre-specified primary endpoints versus placebo [5].

On the basis of this trial, the United States Food and Drug Administration (FDA) granted resmetirom (brand name: Rezdiffra) accelerated approval in March 2024 — making it the first approved pharmacological therapy for MASH. This is a historic milestone. India has no approved MASH-specific drug at the time of this writing.

Status: FDA approved (USA) March 2024 for MASH with hepatic fibrosis. Not yet available in India. Clinicians should monitor CDSCO submissions.

14.3.2 Tirzepatide for MASH (SYNERGY-NASH)

Beyond its approved indications in type 2 diabetes and obesity, tirzepatide has demonstrated striking hepatic efficacy. The SYNERGY-NASH Phase 2 trial published in the New England Journal of Medicine in 2024 by Loomba et al. enrolled 190 patients with biopsy-proven MASH and fibrosis [6]. Tirzepatide 10 mg weekly achieved MASH resolution in 55% of patients, and the 15 mg dose achieved MASH resolution in 62% — far exceeding the resmetirom Phase 3 results, although cross-trial comparisons must be interpreted cautiously given different populations and design.

The implication for India is direct: tirzepatide (Mounjaro) is available in India, though at significant cost. Clinicians managing patients with MASH who also have type 2 diabetes or obesity have an evidence-based reason to prefer tirzepatide over older agents when hepatic disease is the primary concern, pending formal approval of this indication.

Status: Tirzepatide approved in India for T2DM and obesity. MASH indication is Phase 2 only — not approved for this specific indication anywhere. Phase 3 trials ongoing.

14.4 Addressing the Sarcopenia Problem: Bimagrumab

14.4.1 Bimagrumab (Anti-Myostatin / ActRII Antibody)

One of the most persistent limitations of GLP-1 receptor agonist therapy — and of caloric restriction more broadly — is the concurrent loss of lean muscle mass alongside fat. In the Indian context, where sarcopenic obesity and the TOFI (Thin Outside, Fat Inside) phenotype are disproportionately

prevalent, preserving or augmenting skeletal muscle index (SMI) is as clinically important as reducing fat mass.

Bimagrumab is a fully human monoclonal antibody that blocks activin receptor type IIA and IIB (ActRIIA/IIB), thereby inhibiting myostatin and other negative regulators of muscle growth. By releasing the physiological brake on muscle protein synthesis, bimagrumab promotes genuine skeletal muscle hypertrophy — not merely preservation.

In a randomised Phase 2 trial by Heymsfield et al. published in JAMA Network Open in 2021, 75 adults with type 2 diabetes and obesity received bimagrumab or placebo over 48 weeks [7]. The results were unprecedented in metabolic pharmacotherapy: lean body mass increased by 3.6 percentage points (approximately 1.7 kg absolute), while fat mass declined by 7.5 kg in the bimagrumab group compared to only 0.2 kg in the placebo group. HbA1c improved by 0.76 percentage points. This unique body composition profile — simultaneously building muscle while reducing fat — is not achievable with any currently approved metabolic drug. The mechanism is entirely distinct from GLP-1-based therapies.

This is a promising signal, but it must be stated clearly: Phase 2 trials establish proof-of-concept, not regulatory efficacy. Larger Phase 3 trials are needed to confirm safety and sustained benefit before clinical use can be recommended.

Status: Investigational. Not approved anywhere. Not available in India. Phase 3 development pathway under evaluation. For research interest only.

14.5 Cardiorenal Protection: Extending the Evidence Base

14.5.1 Semaglutide in Chronic Kidney Disease (FLOW Trial)

The FLOW trial, reported by Perkovic et al. in the New England Journal of Medicine in 2024, enrolled 3,533 patients with type 2 diabetes and chronic kidney disease (CKD stages 2 to 4, urine albumin-to-creatinine ratio ≥ 300 mg/g) [8]. Participants were randomised to semaglutide 1.0 mg weekly or

placebo in addition to standard of care including renin-angiotensin system blockade and SGLT-2 inhibitors (where indicated).

The trial was stopped early by the Data Safety Monitoring Board on grounds of clear efficacy. Semaglutide reduced the composite primary endpoint — major kidney events, kidney failure, or death from kidney or cardiovascular causes — by 24% compared to placebo. This was a landmark finding, extending the nephroprotective class beyond SGLT-2 inhibitors.

For Indian clinicians, diabetic CKD represents an enormous unmet burden. Semaglutide (Ozempic, Rybelsus) is available in India, and the FLOW data provide a compelling evidence basis for its use in patients with diabetic CKD, even in the absence of a formal CKD-specific approval in India at present.

Status: Semaglutide approved in India for T2DM and cardiovascular risk reduction. CKD indication approval pending regulatory submissions based on FLOW data. Clinically relevant now.

14.5.2 Finerenone (Non-Steroidal MRA for Diabetic CKD)

Finerenone is a non-steroidal mineralocorticoid receptor antagonist (MRA) mechanistically distinct from spironolactone and eplerenone. Its selectivity profile offers cardiorenal protection without the anti-androgenic side effects of spironolactone and with a lower risk of hyperkalaemia compared to steroidal MRAs. In the FIDELIO-DKD trial reported by Bakris et al. in the New England Journal of Medicine in 2020, 5,734 patients with diabetic CKD on standard ACE inhibitor or ARB therapy were randomised [9]. Finerenone significantly reduced the composite of kidney disease progression and kidney failure, while the companion FIGARO-DKD trial confirmed cardiovascular benefits.

Importantly, finerenone is approved in India and is the one drug in this chapter that is both novel in mechanism and currently accessible to Indian patients with diabetic CKD. It should be considered an essential component of the Dr. Nehru Method's cardiorenal protection protocol alongside SGLT-2 inhibitors and RAS blockade.

Status: Approved in India for diabetic CKD. Available now.

14.6 Lipoprotein(a) — The South Asian Time Bomb

Lipoprotein(a) [Lp(a)] is a genetically determined atherogenic particle whose plasma concentration is set almost entirely by the LPA gene locus and is resistant to all conventional lipid-lowering therapies including statins, ezetimibe, and even PCSK9 inhibitors. Epidemiological data consistently demonstrate that South Asians have higher mean Lp(a) levels than European populations, contributing to the disproportionate premature cardiovascular mortality observed in Indians even at levels of LDL cholesterol considered acceptable by conventional targets.

[IMAGE PLACEHOLDER — Figure 14.2]

Figure 14.2 — Emerging Lp(a)-lowering agents: antisense and siRNA approaches.

Two mechanistically distinct investigational agents are approaching Phase 3 completion with dramatic Lp(a)-lowering capabilities.

14.6.1 Pelacarsen (Antisense Oligonucleotide)

Pelacarsen is an antisense oligonucleotide (ASO) that targets hepatic apolipoprotein(a) mRNA synthesis, the rate-limiting step in Lp(a) particle assembly. In a Phase 2 trial by Tsimikas et al. published in the New England Journal of Medicine in 2020, 286 patients with established cardiovascular disease and elevated Lp(a) (>60 mg/dL) received pelacarsen subcutaneously [10]. The agent achieved Lp(a) reductions of approximately 80% at the highest dose — a magnitude of reduction not achievable with any available therapy. The Phase 3 Lp(a) HORIZON trial, specifically powered to demonstrate cardiovascular event reduction, has enrolled over 7,000 patients and results are anticipated. Given the high prevalence of elevated Lp(a) in South Asians, this trial is especially relevant to our population.

Status: Investigational. Phase 3 HORIZON trial ongoing. Not available in India.

14.6.2 Olpasiran (siRNA Therapy)

Olpasiran takes a different molecular approach — it is a small interfering RNA (siRNA) packaged in GalNAc conjugate for hepatocyte delivery, silencing apolipoprotein(a) gene expression at the post-transcriptional level.

Its durability advantage is remarkable: a single subcutaneous injection produces Lp(a) suppression lasting approximately three months. In the OCEAN(a)-DOSE trial by O'Donoghue et al. published in the New England Journal of Medicine in 2022, 281 patients with established atherosclerotic cardiovascular disease and Lp(a) >150 nmol/L received olpasiran at various doses [11]. The highest dose cohort demonstrated Lp(a) reductions exceeding 95% — a near-complete suppression. The Phase 3 OCEAN(a)-Outcomes trial is ongoing.

Status: Investigational. Not approved anywhere. Not available in India.

14.7 Targeting Residual Inflammatory Risk: Ziltivekimab

Even after optimal lipid and glycaemic control, a substantial proportion of patients with established cardiometabolic disease remain at residual risk driven by chronic low-grade inflammation — measurable as elevated high-sensitivity C-reactive protein (hsCRP) and interleukin-6 (IL-6). The RESCUE trial evaluated ziltivekimab, a monoclonal antibody targeting IL-6 ligand, in 264 patients with CKD and elevated hsCRP, reported by Ridker et al. in The Lancet in 2021 [12]. Ziltivekimab achieved reductions in hsCRP of 80 to 90% in a dose-dependent manner, along with significant reductions in fibrinogen, serum amyloid A, and other inflammatory biomarkers.

The agent is being evaluated in the ZEUS cardiovascular outcomes trial in patients with CKD at high inflammatory risk. In the Indian context, where insulin resistance per se is a driver of chronic inflammation (IL-6 is upregulated in visceral adiposity), an IL-6–targeting agent could theoretically complement the eight-trigger framework by directly addressing the inflammatory limb of beta cell toxicity.

Status: Investigational. ZEUS outcomes trial ongoing. Not available in India.

14.8 Summary Table: Pipeline Agents at a Glance

14.9 Integration with the Dr. Nehru Method Framework

The eight metabolic triggers — high glycaemic index carbohydrate load, visceral obesity and the TOFI phenotype, vitamin D deficiency, methylcobalamin deficiency, iron deficiency, sarcopenia, the MTHFR

C677T polymorphism, and ultra-processed food exposure — converge on progressive beta cell exhaustion measurable through HOMA-IR (>2.3, India-validated) and TyG Index (>8.82 on natural log scale, India-validated) [1,2]. Pipeline drugs extend the reach of therapy at different points in this cascade:

Triple agonists (retatrutide) and dual agonists (tirzepatide, when used for MASH) act upstream at the adipose-hepatic axis, directly reducing the visceral fat trigger.

Bimagrumab addresses sarcopenia — the sixth trigger — through a unique anabolic mechanism currently unavailable in any approved therapy.

Resmetirom targets end-organ damage (steatohepatitis) arising from chronic hepatic lipid accumulation driven by insulin resistance.

Semaglutide in CKD and finerenone protect the kidney against the cardiorenal consequences of prolonged HOMA-IR elevation.

Lp(a)-targeting agents (pelacarsen, olpasiran) address a genetically fixed risk amplifier that is disproportionately prevalent in South Asian patients.

Ziltivekimab addresses the chronic inflammatory milieu — IL-6 excess — that links visceral adiposity to both beta cell toxicity and vascular damage.

No drug in this pipeline replaces the foundational work: correcting methylcobalamin deficiency, restoring vitamin D, eliminating ultra-processed foods, and achieving sustained reductions in HOMA-IR and TyG through lifestyle modification. Pipeline drugs are the extended toolkit, not the foundation.

14.10 Conclusion

We stand at an inflection point in metabolic medicine. The next five years will likely bring approval of retatrutide, resmetirom availability in India, Phase 3 data on Lp(a)-targeted agents, and clearer positioning of bimagrumab in sarcopenic obesity. For the Indian clinician, two imperatives emerge: first, identify and treat the currently approved options aggressively (tirzepatide, semaglutide, finerenone) while monitoring pipeline progress; second, educate patients that emerging therapies are not available yet and not substitutes for the disciplined metabolic work that the Dr. Nehru Method demands.

The future belongs to precision metabolic medicine — matching the right mechanistic agent to the patient's dominant trigger profile as measured by validated biomarkers. That future is arriving faster than many physicians appreciate.

Multiple Choice Questions

Q1. Retatrutide is best described as:

- a) A GLP-1 receptor agonist identical in mechanism to semaglutide
- b) A dual GIP/GLP-1 receptor agonist with hepatic selectivity
- c) A triple receptor agonist acting on GIP, GLP-1, and glucagon receptors
- d) An antisense oligonucleotide targeting apolipoprotein B

Q2. In the OASIS-1 trial (Knop et al., Lancet 2023), oral semaglutide 50 mg achieved what mean body weight reduction over 68 weeks?

- a) 8.2%
- b) 10.5%
- c) 15.1%
- d) 22.4%

Q3. The unique body composition finding with bimagrumab in the Heymsfield et al. JAMA Network Open 2021 trial was:

- a) Loss of fat mass only, with no change in lean mass
- b) Simultaneous gain in lean mass (+1.7 kg) and loss of fat mass (−7.5 kg)
- c) Reduction in visceral fat alone without change in subcutaneous fat
- d) Improvement in HbA1c with no change in body composition

Q4. Resmetirom received FDA approval in March 2024 as the first approved therapy for:

- a) Non-alcoholic fatty liver disease (NAFLD) without fibrosis
- b) Cirrhosis secondary to alcohol-related liver disease
- c) Metabolic dysfunction-associated steatohepatitis (MASH) with hepatic fibrosis
- d) Primary biliary cholangitis

Q5. The FLOW trial (Perkovic et al., NEJM 2024) demonstrated that semaglutide:

- a) Reduced major kidney events by 24% in patients with T2DM and CKD
- b) Was inferior to empagliflozin for kidney protection
- c) Improved eGFR without reducing proteinuria
- d) Was effective only in CKD stages 4–5

Q6. Which of the following drugs is currently approved and available in India for diabetic chronic kidney disease?

- a) Bimagrumab
- b) Resmetirom
- c) Finerenone
- d) Pelacarsen

Q7. Pelacarsen reduces Lp(a) by which mechanism?

- a) Inhibiting PCSK9-mediated LDL receptor degradation
- b) Blocking apolipoprotein(a) mRNA using antisense oligonucleotide technology
- c) Activating LPL-mediated triglyceride hydrolysis
- d) Upregulating hepatic LDL receptors via THR-beta activation

Q8. Olpasiran differs from pelacarsen in that it:

- a) Acts at the protein level by blocking Lp(a) receptor binding
- b) Is an siRNA agent delivered via GalNAC conjugate achieving >95% Lp(a) reduction
- c) Is an oral medication requiring daily dosing
- d) Has completed Phase 3 trials and is now FDA approved

Q9. In the Dr. Nehru Method framework, bimagrumab would address which of the eight metabolic triggers?

- a) High glycaemic index carbohydrate load
- b) MTHFR C677T polymorphism
- c) Sarcopenia
- d) Vitamin D deficiency

Q10. The India-validated cutoff for HOMA-IR indicating insulin resistance, as published by Nehru MS et al. in Int J Med 2026, is:

- a) >1.5
- b) >2.0
- c) >2.3
- d) >3.0

Answers: 1. C 2. C 3. B 4. C 5. A 6. C 7. B 8. B 9. C 10. C

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